NarrativeWiki: Probing the Trade-offs of Decomposition in Graph-Built Character Wikis at Reader-Selected Narrative Cutoffs

Anonymous ACL submission

Abstract

A character wiki that reveals only what a reader has reached invites decomposing the novel into a graph of claims filtered by volume. We instrument this design as NARRATIVEWIKI and test it against six systems at every cutoff of a five-volume series (111 gold facts; LLM-finalized labels; two human reviewers relabeled 554 original-text items). A renamed and paraphrased copy (the treated text) tests reliance on recognition: closed-book recall falls from 0.683 to 0.003, while systems given text lose at most 0.070. The design trades context for accumulation. NARRATIVEWIKI's pages grow with the story: on treated text it best covers the most-documented characters (recall 0.929; long context 0.794; original: 0.894, 0.837), loses one fact across cutoffs per text (others: 4–24), and on the original text publishes 3.8 times long context's distinct assertions. But claims lose context: NARRATIVEWIKI turns a prediction into a marriage (treated text), groups entities unlike prefix-built indexes (0.220 of first-cutoff pages, original text) and misplaces true statements. Prompting with only the text read discloses no assessed later fact; an instruction to withhold later volumes discloses 17/88 later facts on treated text (original: 8/88). Code, gold facts and labels: https://anonymous.4open.science/r/NarrativeWiki-Anon-341E/.

1 Introduction

A reader consulting a character wiki midway through a series may encounter a later identity, marriage or death while seeking a reminder of events already read. A reader who has finished *Anne of Avonlea*, the second book of L. M. Montgomery's Anne series, may learn from Anne Shirley's whole-series page about her marriage to Gilbert Blythe in book four (Figure 2).[1] A useful reference must respect the reader's progress: at cutoff t, every statement should be established by the first t volumes.

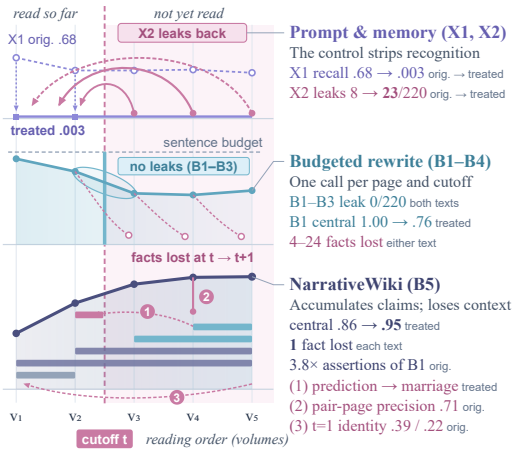


Figure 1: Three ways the cutoff behaves over volumes $v_1..v_5$; every number names its text (orig.: original; treated: renamed and paraphrased). **Top:** the control removes closed-book recall (X1) and raises disclosures under a cutoff instruction (X2). **Middle:** pages rewritten in one call per cutoff (B1–B3) disclose no assessed later fact but lose facts between cutoffs. **Bottom:** NARRATIVEWIKI's accumulated claims keep central-character recall rising, but lose context: (1) a volume-2 prediction printed as the volume-4 marriage, (2) true lines placed on the wrong pair page, (3) first-cutoff pages grouped unlike a prefix-built index.

The natural engineering answer is structural. Decompose the text into a graph of claims, date each claim by the volume of its evidence, and generate each page from the claims dated no later than the cutoff, in the manner of graph-based retrieval-augmented generation (Edge et al., 2024; Guo et al., 2025). The boundary is then enforced by data, not by an instruction to the generator, and one build serves every cutoff. This paper asks what that decomposition gains and what it costs.

NARRATIVEWIKI instruments this design as an *analytical probe*, not an advocate: verbatim-evidence gates, a claim verifier and logged model

[1] Examples in this paper reveal plot points of the series.

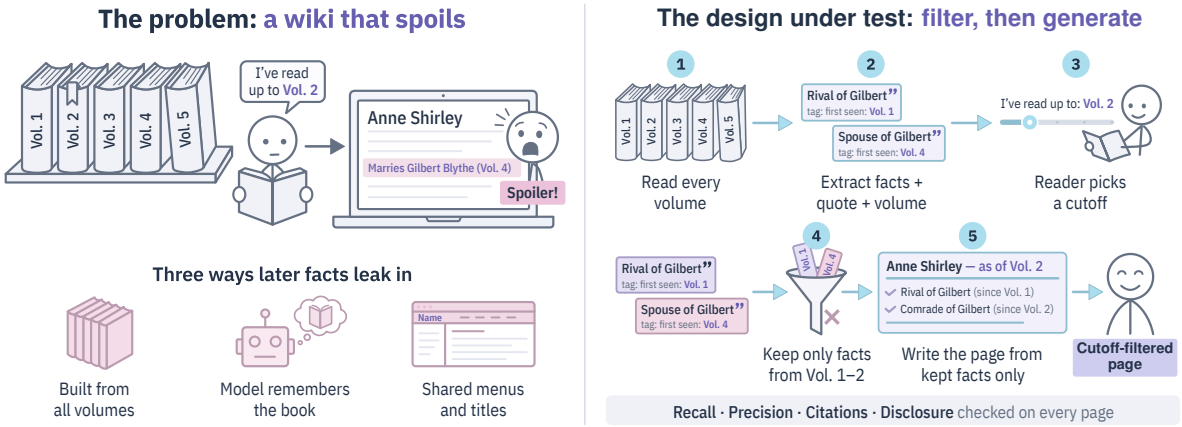


Figure 2: Overview. Left: a whole-series wiki reveals Anne's marriage to Gilbert (volume 4) to a reader at volume 2. Later information can enter through a full-series index, generator memory or surfaces shared across cutoffs. Right: the filter-before-generate design that NARRATIVEWIKI implements and this paper probes.

calls trace each error to its stage, and it is compared with systems that differ in what the generator sees and how pages are written (Figure 3).

Such a comparison is interpretable only if scores reflect reading: given only a title and a character name, the generator recovers 0.683 of the eligible gold facts. We therefore pair every measurement with a *renamed-text control*: a *treated* copy of the series renames characters and places and paraphrases nearly every paragraph, keeping paragraph identifiers so that gold facts and citations stay aligned. Closed-book recall falls to 0.003, while systems given text largely keep theirs (Section 6.1).

Headline means rank NARRATIVEWIKI below prompting with the text read so far (Table 1). The probe asks instead *where* decomposition gains and *where* a volume filter fails, through five research questions on recognition, cutoff enforcement, accumulation, page context and construction scope (RQ1–RQ5), each answered in the subsection of Section 6 with its number (Table 2).

Prior work bounds dialogue or answers in time (Ahn et al., 2024; Huang et al., 2025). We study a linked reference artifact at every cutoff:

- **A probe study of structural decomposition** under a reading cutoff, separating its gains from failures that a volume filter cannot detect (Sections 4, 6).
- **A paired renamed-text control** as evaluation methodology: seven systems rebuilt on aligned treated text with unchanged code, prompts and gold facts (Section 6.1).
- **A task, threat model and artifact-level protocol** for cutoff-bounded reference generation, with 111 gold facts dated by reveal volume (Sections 3, 5; Appendix C).

2 Related Work

Spoilers, characters and temporal bounds. Spoiler detection labels review sentences without conditioning on a reader's position (Wan et al., 2019); Jones et al. (2018) serve fan-wiki revisions archived before an episode aired, and e-readers offer position-bounded answers (Amazon Staff, 2026), although we are unaware of a published evaluation. CHIRON builds validated character sheets (Gurung and Lapata, 2024), BookWorm pairs books with character descriptions (Papoudakis et al., 2024), and EvolvTrip graphs characters' mental states over time (Yang et al., 2025). TimeChara measures point-in-time hallucination in role-play (Ahn et al., 2024); Living the Novel gates conversation to the narrative present (Huang et al., 2025), the filter-before-generate principle we adopt; NS-ST-GraphRAG answers from temporally valid subgraphs (Lin, 2026); temporal knowledge graphs attach validity intervals to facts (Dasgupta et al., 2018), ours indexed by reading volume rather than story time. These works bound answers or dialogue in time; none evaluates a complete linked reference artifact at every cutoff (Table 3). LitBank annotates identities using the whole text, including revelations unavailable at an earlier mention (Bamman et al., 2020).

Grounded article and wiki generation. STORM outlines Wikipedia-like articles (Shao et al., 2024); GraphRAG summarizes a corpus entity graph (Edge et al., 2024); LightRAG, a

baseline here, combines graph structure with lightweight retrieval (Guo et al., 2025); SearchWiki builds linked wikis for agents (Singh et al., 2026); book-length benchmarks evaluate question answering and summarization (Wang et al., 2025; Chang et al., 2024). Our reference also depends on the reader's position.

Factuality, attribution and contamination. We evaluate atomic support (Min et al., 2023), attribution to identified sources (Rashkin et al., 2023) and citation quality (Gao et al., 2023) across the whole artifact and in page context. Automatic raters of book-length text can disagree with knowledgeable readers (Kim et al., 2024); on STORYSUMM, MiniCheck's Flan-T5 checker (Tang et al., 2024) scored below chance (Subbiah et al., 2024), so we use evidence-reading annotators. Published novels may be known to the generator (Chang et al., 2023; Cheng et al., 2025); renaming entities is an established control (Gupta, 2025; Zhang et al., 2024), and rewriting can damage the narrative it protects (Sun et al., 2025); we add aligned paraphrase and audit the treatment.

3 Task and Threat Model

Cutoffs and the delivered artifact. A series comprises ordered volumes $v_1, \ldots, v_N$. At cutoff t, the reader has read $D_{\leq t} = v_1 \cdots v_t$; the cutoff follows reading order, not story time. The delivered artifact W_t includes page text, links and backlinks, titles and addresses, shared navigation and any search index, group and place pages, and source views opened by citations. All these surfaces belong to the evaluation scope. Unrequested later information is in scope; deliberately opening a later cutoff is outside the threat model.

Facts and assertions. A gold fact g is a selected character statement with reveal volume $r(g)$, the earliest volume whose text suffices to establish it. Eligible gold facts are $G_t = \{g : r(g) \leq t\}$; assessed later facts are $F_t = \{g : t < r(g) \leq N\}$. The latter are a finite probe set, not an exhaustive spoiler inventory. Story-level truth, support from eligible text and citation sufficiency are distinct properties: a true assertion supported only by later text violates the cutoff. Support must preserve attribution and modality: *A believes B is her father* does not establish *B is A's father*.

Threats. We distinguish three channels. *Construction dependence*: processing volumes beyond t shapes the early artifact through alias merges, canonical names or pages created for later characters, even when rendered claims are filtered. *Published disclosure*: a delivered surface directly exposes later content, such as a name in shared navigation. *Parametric contribution*: the generator supplies memorized information about the work. An observed disclosure alone cannot identify its channel; experiments therefore vary what each system can access (Figure 3).

4 The Probe: NARRATIVEWIKI

NARRATIVEWIKI constructs a graph of evidence-bearing claims with volume intervals and renders one wiki per cutoff (Figure 3). Claims are filtered to evidence volumes $\leq t$ before page assembly and prose generation. Filtering precedes generation because removing explicit later facts from generated text can leave implications intact.

Ingestion and entities (stages 1–2). Volumes become ordered paragraphs with stable citation identifiers such as `v01:c06:p0003`. Lexical mining identifies capitalized n-grams, honorifics and vocatives; the model types them as characters, groups, places, events or objects. Surface forms are clustered by exact match, fuzzy match and whole-word subset, followed by model adjudication of ambiguous pairs using passages containing both names. A bare name is not automatically merged with a fuller name first appearing later. Shared first names are assigned volume by volume to the locally dominant full name. Entities use the fullest title occurring in their first volume, and surface forms retain their first-occurrence volume.

Claims with evidence (stage 3). Three extraction passes share a schema: subject, predicate, object or value, qualifier, first-evidence volume (`first_vol`), verbatim quotes and paragraph identifiers. A chapter pass covers every paragraph in overlapping spans and extracts scenes, participants and summaries backed by up to four paragraphs. A mention pass reads windows around each character's mentions; a long-context pass reads all their paragraphs in a volume for attributes and background. Claims are retained only when the quote occurs verbatim in its cited paragraph, the subject is named in or beside that paragraph (and, for twelve two-person relations, the object in it), and relation objects resolve to known entities.

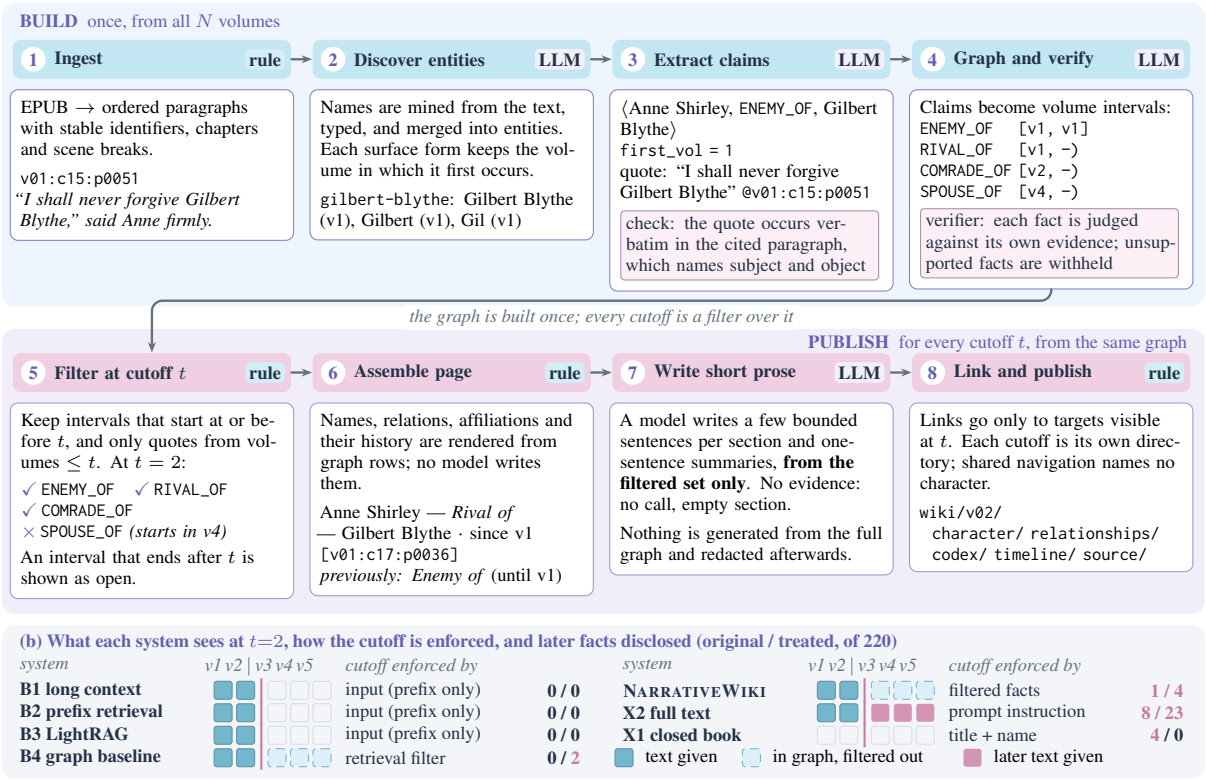


Figure 3: (a) The NARRATIVEWIKI pipeline on a stored fact. BUILD turns all volumes into paragraphs, entities and verifier-accepted claim intervals; PUBLISH assembles each cutoff's wiki from eligible intervals. LLM stages call the generator; the rest are deterministic. (b) What each system sees at cutoff 2, how it enforces the cutoff, and the later facts it disclosed over $t = 1..4$ (original, treated). X1, X2: diagnostics.

Intervals and verification (stage 4). Claims become volume intervals. Conflicting values of a single-valued attribute within one volume are arbitrated as extraction conflicts; values in different volumes are treated as story changes, closing the earlier interval when the later one opens. Near-duplicate values are clustered with BGE-M3 embeddings (Chen et al., 2024). A verifier judges each assembled fact against its cited evidence in a separate call; rejected facts (21.2% of 2,535 checked) are withheld. On 81 constructed minimal-edit items the verifier reaches 0.989 balanced accuracy (MiniCheck: 0.718), and 0.750 on 12 harder items of embedded speech (Appendix K).

Cutoff assembly and publication (stages 5–8). Deterministic assembly selects intervals opening by t (Algorithm 1, Appendix D). Each quote is filtered by its own volume; intervals closing after t appear current to conceal future changes; and qualifiers must be supported no later than the interval's start. Names, attributes, affiliations and relations are rendered directly from graph rows. The model writes only short prose from filtered facts, with sentence limits per section (Appendix D), and is instructed to identify the input lines supporting it. Line keys select source paragraphs; returned paragraph IDs are normalized during citation resolution. Sections without evidence remain empty without a model call. String matching adds links using surface forms revealed by t. Cutoffs occupy separate directories, shared navigation names no characters, and no site-wide search index is shipped because it would span cutoffs.

Build scope. One graph is built from all volumes. Additional cutoffs require no extraction, structured content is rendered without model calls, and prose is generated once per distinct eligible fact set. This efficiency retains construction dependence (Section 3), which we assess against prefix-only builds (Section 6.5). The filter applies to published claims, not to build inputs: extraction prompts list each entity's aliases from the whole-series index, so volume-1 prompts can name later aliases (Appendix Q, D3). Every model stage, including verification, uses the generator specified in Section 5.

5 Experimental Setup

Corpus and gold facts. We evaluate the first five novels in publication order of L. M. Montgomery's Anne series, from *Anne of Green Gables* to *Rainbow Valley*, using Project Gutenberg editions: 428,122 words, 8,489 paragraphs and 184 chapters (Appendix C). The gold inventory contains 111 facts about 43 characters, each with a reveal volume and an evidence paragraph. It was drafted from the source text without access to system outputs; eligible counts at each cutoff appear in Table 8. For 85 of the 111 facts the cited paragraph suffices alone; the other 26 (23 established in part, three not) need text beyond that paragraph and are retained unchanged (Appendix B). Recall measures coverage of this selected inventory rather than all facts in the novels.

Systems. All systems use Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). Comparison systems share a character-page template, requested sentence limits and citation format. A cast character receives a page at t if an eligible gold fact concerns them or their full name occurs in $D_{\leq t}$ (14, 21, 25, 33 and 43 pages at $t = 1..5$; Table 27). **B1** (long context) supplies the complete prefix $D_{\leq t}$ to the generator. **B2** (prefix retrieval) combines BM25 and dense retrieval from a prefix-only index. **B3** uses prefix-built LightRAG (Guo et al., 2025) as the shared generator's retriever. **B4**, the graph baseline, retrieves from the full five-volume NARRATIVEWIKI graph, filtering facts at retrieval time; it controls for the graph while changing page generation and publication scope. **B5** is NARRATIVEWIKI: it discovers its own entities and also publishes other character, relationship, codex and timeline pages. We evaluate its full artifact and separately examine the cast character pages shared with the baselines. **X1** receives only the title and character name (closed book). **X2** receives all five volumes with an instruction to reveal nothing after t. These two diagnostics are excluded from best-system comparisons. Baseline settings were fixed on a separate development novel (Appendix E).

Paired text conditions. Every system is built twice, on the original and on the treated text (Appendix N). A reviewed map of 239 entries renames characters, families and local places, and the generator paraphrases each renamed paragraph, preserving facts, order and speakers. Paragraph identifiers and the mapped gold facts carry over, so rules, code and prompts are unchanged.

Metrics. We report four separate outcomes (Figure 6; Appendix B). *Recall* is the fraction of eligible gold facts stated of the correct character anywhere in W_t. *Precision* measures full support in $D_{\leq t}$ for 40 uniformly sampled atomic assertions per system and cutoff, drawn from an inventory of all delivered surfaces (Min et al., 2023). Assertions are judged in their page context: a true scene line on a relationship page is supported there only if both characters participate in a shared episode of that scene (the *scene rule*); we also record whether each assertion is true as a statement. *Citation sufficiency* is the fraction of sampled assertions fully established by their cited paragraphs alone; uncited assertions count as insufficient. *Disclosure* searches every page for every assessed later fact using three lexical routes. Retrieved candidates are labeled *stated*, *implied*, *uncertain* or absent; stated and implied count as disclosure. There are 220 fact–cutoff assessments per system at $t = 1..4$; disclosure is undefined at $t = 5$. We measure the candidate search's recall on seeded disclosures and add two complementary checks that do not use it: later-only support among sampled assertions, and names in W_t that occur in the full entity index but not in $D_{\leq t}$.

Annotation. Labels come from LLM coding agents that read the evidence under written rules (Appendix B): OpenAI Codex and Claude Opus 5.5 produced first-pass labels, and Claude Opus 5.5 finalized every reported label, including the gold facts. The pair verdict is the judgment that moves precision most: first-pass and final verdicts agree on 90 of 99 (Cohen's $\kappa = 0.79$), and the first-pass verdicts would give NARRATIVEWIKI a precision of 0.753 instead of 0.789. Reported values incorporate consistency audits and a review-triggered recall recheck (Appendix B). Two human reviewers independently relabeled 554 original-text items, all gold facts and 10% samples of the recall, precision and citation labels, without seeing the AI answers (κ from 0.39 on precision to 0.83 on recall; Table 5).

Statistics. Means are unweighted over cutoffs; each cell is built once, so standard deviations describe variation across cutoffs (medians, ranges, paired differences: Tables 14, 15).

System	Recall ↑ treated	Recall ↑ orig.	Recall ↑ diff.	Precision ↑ treated	Precision ↑ orig.	Citation suff. ↑ treated	Citation suff. ↑ orig.	Later facts /220 ↓ orig.	Later facts /220 ↓ treated	Distinct assertions	Tokens in / out (M)
B1 Long context	**0.845**$_{\pm 0.078}$	**0.848**	−0.003	0.920$_{\pm 0.060}$	0.929	0.530$_{\pm 0.239}$	0.658	0	0	1,821	69.23 / 0.37
B2 Prefix retrieval	0.784$_{\pm 0.052}$	0.834	−0.050	0.950$_{\pm 0.025}$	0.960	0.885$_{\pm 0.045}$	0.774	0	0	1,718	**3.42 / 0.37**
B3 LightRAG	0.470$_{\pm 0.074}$	0.536	−0.067	**0.960**$_{\pm 0.038}$	0.950	0.794$_{\pm 0.108}$	0.765	0	0	1,404	5.08 / 1.01
B4 Graph baseline	0.704$_{\pm 0.057}$	0.761	−0.057	0.940$_{\pm 0.034}$	**0.965**	**0.895**$_{\pm 0.037}$	**0.915**	0	**2**	1,701	+1.11 / 0.36
B5 NarrativeWiki	0.843$_{\pm 0.017}$	0.802	**+0.041**	0.770$_{\pm 0.065}$	0.789	0.625$_{\pm 0.087}$	0.512	**1**	**4**	**6,970**	33.13 / 1.86
X1 Closed book	0.003$_{\pm 0.007}$	0.683	**−0.680**	0.145$_{\pm 0.037}$	0.814	–	–	**4**	0	1,526	0.06 / 0.15
X2 Full text	0.744$_{\pm 0.063}$	0.814	−0.070	0.845$_{\pm 0.045}$	0.910	0.475$_{\pm 0.088}$	0.625	**8**	**23**	1,829	98.67 / 0.37

Table 1: Main results on the original and the treated text (renamed, paraphrased), with unchanged code, prompts and gold facts. Rates: unweighted means over $t = 1..5$, treated rates ± sample SD (original: Table 14). Diff.: treated minus original. Later facts: stated or implied disclosures of 220 fact–cutoff pairs, $t = 1..4$. Distinct assertions: mean per cutoff, original text; tokens: all cutoffs (B4: added to NarrativeWiki's build). X1, X2: diagnostics. Cell colours in every table: cyan, best; magenta, failure, disclosure or worst; violet or lavender, size, cost or inventory.

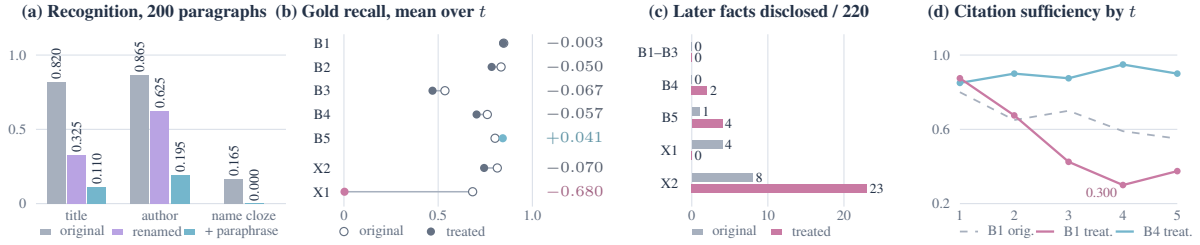


Figure 4: The renamed-and-paraphrased control (RQ1, RQ2). (a) Renaming alone leaves the work recognizable; paraphrase reduces recognition further without removing it. (b) Recall without text collapses; recall from text survives. (c) The instruction to withhold later volumes (X2) fails more often once the work is unrecognizable. (d) Long-context citations lose their volume on treated text; B4 on NarrativeWiki's graph does not.

6 Results

6.1 RQ1. Without the text, memory writes the wiki; without memory, recall needs reading

Title and name alone recover two-thirds of the gold facts. The closed-book generator (0.826 at $t = 1$) also states 4 of 220 assessed later facts and reaches sampled precision 0.814 against text it never received.

Renaming leaves the author recognizable; paraphrase reduces recognition. Title identification falls from 0.820 to 0.325 after renaming alone and to 0.110 after paraphrase (Figure 4a). Renaming alone, a common control (Gupta, 2025; Zhang et al., 2024), leaves the author identifiable in 0.625 of paragraphs, and paraphrase in 0.195: the prose carries the work, not only its names.

Recall without text collapses; recall from text survives. Closed-book recall falls from 0.683 to 0.003; its single remaining match (1 of 335) is a death that also occurs in the generator's invented story. Every system given text changes far less (Figure 4b): B1 by −0.003, NarrativeWiki by +0.041, at most 0.070 (X2). The closed-book generator still writes full, invented pages (Appendix N).

Under the control, instruction leaks nearly triple and long-context citations lose their volume. X2 discloses 23 of 220 assessed later facts on treated text against 8 of 220 on the original (unique facts: 17/88 against 8/88; Figure 4c), all but one at $t \leq 3$; two treated-text assessments come from a namesake created by the name map (Appendix N). Original-text compliance therefore overstates the instruction's reliability in this comparison. B1's precision holds (0.920 versus 0.929), but its citation sufficiency falls from 0.875 at $t = 1$ to 0.300 at $t = 4$ (Figure 4d), as citations keep a paragraph number and lose its volume.

6.2 RQ2. Volume filters block future evidence, but modality collapse still spoils

Prefix-bounded prompting discloses no assessed later fact. B1, B2 and B3 each have no stated or implied disclosure in 220 fact–cutoff assessments on either text (Figure 3; Table 1); B4 has none on the original.

A prediction becomes a marriage. On treated text, NARRATIVEWIKI lists the two leads as spouses at $t = 2$ and $t = 3$, a book-four fact (4 of 220: reciprocal facts at two cutoffs from one mechanism; Table 24). The cited evidence is a neighbour's prediction in book two. Extraction stored a presumed marriage; the relation line marks it, but the infobox's family field prints the spouse unmarked, and the other lead's page has an asserted spouse line citing a paragraph without a marriage. B4, on the same graph, writes "anticipated future marriage" without the speaker (2 of 220, implied). The original build did not extract this relation; whether paraphrase strengthened the prediction was not checked. Once stored, the claim keeps a coarse *presumed* flag but not who spoke or that it was a prediction: a volume filter checks *when* evidence appears, not *what* it licenses (Proposition 2, Appendix I). On the original text NARRATIVEWIKI has one implied disclosure: at $t = 3$ a summary says Phil Gordon *later serves as a bridesmaid for Anne*, where the eligible passage makes Anne Phil's bridesmaid.

Zero counts are bounded, not certified. The candidate search misses some paraphrases and most pronoun references (Table 17; uncertain verdicts: Appendix G); no sampled assertion at $t \leq 4$ in NARRATIVEWIKI or B1–B3 is established only by later volumes (NARRATIVEWIKI: 0 of 159).

6.3 RQ3. Accumulated claims grow with the protagonists; budgeted rewrites forget

Central-character recall rises as budgeted pages decay. Comparison systems rewrite every page at every cutoff in one call under a shared sentence budget (Appendix E); NARRATIVEWIKI renders every eligible claim, so Anne Shirley's page grows from 1,813 words to 3,439, while B1's never exceeds 1,157 (Figure 5b). In an exploratory split, chosen after the results were known, on the three characters with the most gold assessments, NARRATIVEWIKI has the highest recall of any system on treated text (0.929, against B1's 0.794) and, by a smaller margin, on the original (0.894 against 0.837), where the lead comes from Anne Shirley alone (Appendix H). On treated text its recall on them rises from 12/14 at $t = 1$ to 36/38 at $t = 5$, while B1's falls from 14/14 to 29/38 (Figure 5a).

Stored claims are not forgotten. A fact conveyed at t and lost at $t + 1$ occurs once for NARRATIVEWIKI on each text; every other system loses 4–19 on the original and 11–24 on treated text (Figure 5c). B4 reads the same graph and loses 12 and 14: the comparison points to rendering stored claims as a source of stability (Proposition 1).

One build publishes 3.8 times long context's distinct assertions. On the original text, one build publishes a mean of 6,970 textually distinct assertions per cutoff, against 1,821 for long context, the most among B1–B4 (Table 29); reciprocal phrasings and alias variants count separately.

6.4 RQ4. Context, not truth, costs precision

Coverage trails on minor characters. On the original text NARRATIVEWIKI trails long context (0.802 versus 0.848) and prefix retrieval (paired difference -0.032; Table 15), which uses a tenth of the input tokens; the deficit lies with the other 40 characters (0.737, against 0.845 and 0.871; Table 28).

Placement, not invention, lowers precision. Whole-artifact precision is 0.789 for NARRATIVEWIKI (0.628 under the stricter line rule), yet 0.934 of its sampled assertions are true as statements. On cast character pages, the matched population, NARRATIVEWIKI reaches 0.886 (31/35), versus B1's pooled 0.930 (185/199). The gap concentrates on relationship pages (46% of its assertions): 0.971 (99/102) are true as statements but 0.706 (72/102) are supported in page context, because scene lines naming both characters reach the pair's page even when the two share no episode (Figure 8a, Appendix G; Proposition 3).

The same graph supports precise, well-cited pages. Citation sufficiency is 0.512 for NARRATIVEWIKI against 0.658 (B1) and 0.774 (B2), and 0.534 (102/191) against 0.668 (131/196) and 0.802 (154/192) on sampled assertions that carry a citation, so omission does not explain the gap. B4, which writes cast pages from facts retrieved from NARRATIVEWIKI's graph, has lower recall (0.761) but the highest original-text precision and citation sufficiency (0.965, 0.915; its citation labels have no human check): the comparison points to selection, publication and citation attachment as additional sources of error. The reported NARRATIVEWIKI is the second build, revised after the first was labeled; the first had citation sufficiency 0.112 (Table 25).

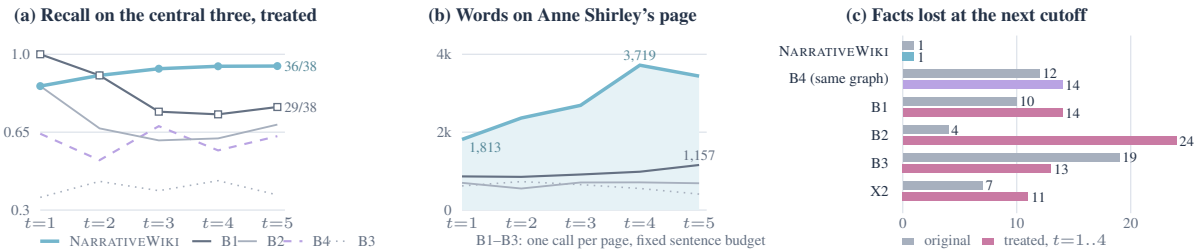


Figure 5: Where accumulation helps (RQ3). (a) Exploratory split, treated text: on the three most-documented characters NARRATIVEWIKI's recall rises with the prefix while budgeted rewrites fall. (b) NARRATIVEWIKI's page for Anne grows with the story (original text). (c) Facts conveyed at t and lost at $t+1$: B4 reads the same graph but rewrites each page, which points to rendering stored claims as a source of stability.

Question	What the probe found	What limits it
RQ1 (§6.1) Recognition	Closed-book recall 0.683 (orig.) falls to 0.003 (treated); systems given text lose at most 0.070.	Treatment reduces recognition without removing it (title 0.110, author 0.195).
RQ2 (§6.2) Cutoff	No assessed later fact from prefix-bounded prompting; unique later facts disclosed: X2 8/88 (orig.), 17/88 (treated); NARRATIVEWIKI 1/88, 2/88.	NARRATIVEWIKI: one mechanism at two cutoffs; the search misses most pronoun-mediated disclosures (Table 17).
RQ3 (§6.3) Accumulation	Central-three recall 0.929 against long context's 0.794 (treated); one fact lost across cutoffs per text.	Exploratory split, chosen after the results; comparison pages are budgeted, not length-matched.
RQ4 (§6.4) Context	0.934 of NARRATIVEWIKI's assertions are true as statements, 0.789 supported on their page; B4 on the same graph 0.965 (orig.).	Page scope differs from B1–B4; 40 assertions per cell; lowest human agreement ($\kappa = 0.39$).
RQ5 (§6.5) Scope	0.390 of $t = 1$ pages differ from a prefix-built index (0.220 regrouped or absent); prefix-index wikis match pooled recall (orig.).	One build per index, so model variance is inside the differences.

Table 2: The paper at a glance: each research question, its answer and what limits it. Cyan: the gain the design delivers; magenta: the failure a volume filter cannot detect.

6.5 RQ5. Later volumes shape early pages

For the reported build, 0.390 of character pages at $t = 1$ and 0.089 at $t = 4$ have no prefix-built character with exactly their set of names observed by t; excluding alias-only differences, 0.220 and 0.052 are regrouped or have no prefix counterpart (Figure 8b; Tables 21, 20). Wikis built on those prefix indexes match the full build's pooled recall (180 of 224 each, $t = 1..4$) and state no later fact, though both keep the implied bridesmaid disclosure at $t = 3$ (Table 22): prefix-built indexes remove the dependence at no cost in pooled recall, at one build per cutoff.

7 Discussion

One account of both sides of the trade-off. The pipeline replaces passages with claims (Table 2). As claims accumulate, pages can grow with the story and keep facts; budgeted rewrites in these comparisons kept fewer (Proposition 1, Appendix I). Volume filtering checks the date of the evidence, while the representation weakly preserves speaker, modality, namesake identity and scene context (Propositions 2–3). The observed failures include a prediction rendered as a marriage, identities shaped by later volumes and true statements misplaced on a page. B4 kept a hedge that rendering dropped, but lost more facts between cutoffs on the same graph. Extraction and publication both warrant scrutiny; stage ablations could separate them.

Where accumulation may matter. Accumulation may pay where rewriting each page is costly: series longer than one context window, and characters whose record outgrows a page budget. Untested repairs that keep it: carry speaker and modality in the claim schema, validate entity groupings against the prefix, and check each assertion in its page context (Appendix J).

8 Conclusion

In our builds, decomposing a narrative into dated claims filtered at the reader's cutoff trades context for accumulation. Our gold facts, labels and renaming pipeline make both sides testable on longer works.

Limitations

One series, one generator, one build per cell. The evaluation covers five volumes of one English public-domain series and one generator, so run-to-run variance is unknown, and cutoffs share characters and evidence rather than being independent samples. Precision and citation sufficiency rest on 40 sampled assertions per cell. All five volumes fit in the generator's context window, which favors direct prompting; longer series, other genres, languages and generators would test where accumulation pays (Section 7).

The Anne series informed development and is not a held-out test set. The reported NARRATIVEWIKI is the second of two builds, revised after the first was labeled; Table 25 reports both, with only their relationship lines re-read under one rule, and the revision moved metrics in both directions. The baselines, configured on another novel, were not retuned. The prefix-index wikis of Section 6.5 were built on the original text only and labeled in one pass.

An unblinded LLM finalized the labels; human checks cover original-text subsets. The finalizing annotator, Claude Opus 5.5, knew which system produced each output and assisted in building the system. Two human reviewers relabeled original-text gold, recall, precision and citation items (Table 5); disclosure and treated-text labels have no human check, nor do the citation labels of B3, B4 and X2, added after the human sample was drawn. Where the human check disagreed with the reported labels, only six recall facts were re-read; precision and citation labels were left unchanged. Agreement is lowest for precision and citation sufficiency. Some gold evidence paragraphs establish their facts only in part (Section 5).

Lexical candidate retrieval under-samples pronoun-mediated disclosures. Disclosure is assessed for a selected inventory of later facts through candidate retrieval, validated on seeded fixtures written by the same LLM annotator (Table 17). The line between an implied disclosure and an eligible prediction rests on the annotator. A zero count therefore does not certify spoiler-free output, and NARRATIVEWIKI's larger output leaves more room for missed disclosures.

Accumulation gains are measured against sentence-budgeted, not length-matched, pages. The comparison systems write each page in one call under a sentence budget, while NARRATIVEWIKI renders every eligible claim; an append-only or length-matched baseline would separate graph representation from page length, and was not run. The central/other split was chosen after the results were known, and facts lost across cutoffs also count labeling inconsistencies between adjacent cutoffs. Identity-dependence rates compare single builds. B4 differs from NARRATIVEWIKI in page scope and generation as well as citation attachment, and neither the account of Section 7 nor the proposed repairs have been tested by ablation.

Renaming and paraphrase reduce recognition without removing it. The treated text was not proofread against its source: some paragraphs keep inflected source names, the name map creates namesakes, and one paraphrase changes a fact (Appendix N). Its fidelity to the gold facts was checked by an unblinded LLM. A work published after the generator's training cutoff would offer a complementary test (Karpinska et al., 2024).

The representation is coarse and readers were not studied. Cutoffs are whole volumes; identity links are not dated; attribute intervals track explicit replacements rather than implicit aging; a value that changes within one volume is arbitrated as an extraction conflict (Appendix Q); and cross-claim inference is limited (Appendix O). The reported extraction and verifier prompts contain implementation notes in place of semantic definitions, with unmeasured effect (Appendix Q). We do not measure readers' ability to find information or their experience of spoilers. External search engines and deliberately opened later cutoffs are outside the audited artifact.

Ethical Considerations

The evaluation uses public-domain Project Gutenberg editions. The accompanying materials include code, identifiers, gold facts, AI labels and human-review summaries, and exclude copyrighted development texts; no reported measurement comes from those development series. No reader-study participants were recruited. Claude Opus 5.5 finalized the AI labels, with human review of selected original-text items as described in Section 5 and Appendix B.

AI coding assistants contributed to implementation, evaluation tools, drafting and the data figures,

which are drawn from stored data and measured values. The overview illustration (Figure 2) was generated with an image model and is included as a vector graphic. The authors reviewed these materials and remain responsible for their content. Cutoff-bounded references are intended to assist readers, although generated summaries can substitute for reading and can inadvertently disclose later events. Released pages concern public-domain works. The paper itself contains plot spoilers for the Anne series.

References

Jaewoo Ahn, Taehyun Lee, Junyoung Lim, Jin-Hwa Kim, Sangdoo Yun, Hwaran Lee, and Gunhee Kim. 2024. TimeChara: Evaluating point-in-time character hallucination of role-playing large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3291–3325, Bangkok, Thailand. Association for Computational Linguistics.

Amazon Staff. 2026. How Kindle helps you stay immersed in every book you read. About Amazon. Published 2 April 2025, updated 2 July 2026. `https://www.aboutamazon.com/news/books-and-authors/kindle-recaps-feature-ebook-series-refreshers`. Accessed 11 October 2026.

David Bamman, Olivia Lewke, and Anya Mansoor. 2020. An annotated dataset of coreference in English literature. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 44–54, Marseille, France. European Language Resources Association.

Kent K. Chang, Mackenzie Cramer, Sandeep Soni, and David Bamman. 2023. Speak, memory: An archaeology of books known to ChatGPT/GPT-4. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7312–7327, Singapore. Association for Computational Linguistics.

Yapei Chang, Kyle Lo, Tanya Goyal, and Mohit Iyyer. 2024. BooookScore: A systematic exploration of book-length summarization in the era of LLMs. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, Bangkok, Thailand. Association for Computational Linguistics.

Yuxing Cheng, Yi Chang, and Yuan Wu. 2025. A survey on data contamination for large language models. *arXiv preprint arXiv:2502.14425*.

Shib Sankar Dasgupta, Swayambhu Nath Ray, and Partha Talukdar. 2018. HyTE: Hyperplane-based temporally aware knowledge graph embedding. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2001–2011, Brussels, Belgium. Association for Computational Linguistics.

Jasper Dekoninck, Mark Niklas Müller, and Martin Vechev. 2024. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 92420–92464. Curran Associates, Inc.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. 2024. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, Singapore. Association for Computational Linguistics.

Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2025. LightRAG: Simple and fast retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 10746–10761, Suzhou, China. Association for Computational Linguistics.

Kshitij Gupta. 2025. WHODUNIT: Evaluation benchmark for culprit detection in mystery stories. *arXiv preprint arXiv:2502.07747*.

Alexander Gurung and Mirella Lapata. 2024. CHIRON: Rich character representations in long-form narratives. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 8523–8547, Miami, Florida, USA. Association for Computational Linguistics.

Yifei Huang, Tianyu Yan, Sitong Gong, Xiwei Gao, Caixin Kang, Ruicong Liu, Huchuan Lu, and Bo Zheng. 2025. Living the novel: A system for generating self-training timeline-aware conversational agents from novels. *arXiv preprint arXiv:2512.07474*.

Yeonseok Jeong, Minsoo Kim, Seung-won Hwang, and Byung-Hak Kim. 2025. Agent-as-judge for factual summarization of long narratives. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 23591–23608, Suzhou, China. Association for Computational Linguistics.

Shawn M. Jones, Michael L. Nelson, and Herbert Van de Sompel. 2018. Avoiding spoilers: Wiki time travel with Sheldon Cooper. *International Journal on Digital Libraries*, 19(1):77–93.

Marzena Karpinska, Katherine Thai, Kyle Lo, Tanya Goyal, and Mohit Iyyer. 2024. One thousand and one pairs: A "novel" challenge for long-context language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17048–17085, Miami, Florida, USA. Association for Computational Linguistics.

Yekyung Kim, Yapei Chang, Marzena Karpinska, Aparna Garimella, Varun Manjunatha, Kyle Lo, Tanya Goyal, and Mohit Iyyer. 2024. FABLES: Evaluating faithfulness and content selection in book-length summarization. In *First Conference on Language Modeling (COLM)*.

Klaus Krippendorff. 2011. Computing Krippendorff's alpha-reliability. Departmental Papers (ASC), Annenberg School for Communication, University of Pennsylvania.

Zheng Kui Lin. 2026. NS-ST-GraphRAG: Neuro-symbolic spatio-temporal GraphRAG for literary knowledge processing. *arXiv preprint arXiv:2609.05139*.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.

Argyrios Papoudakis, Mirella Lapata, and Frank Keller. 2024. BookWorm: A dataset for character description and analysis. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4471–4500, Miami, Florida, USA. Association for Computational Linguistics.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and David Reitter. 2023. Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4):777–840.

Yijia Shao, Yucheng Jiang, Theodore A. Kanell, Peter Xu, Omar Khattab, and Monica S. Lam. 2024. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6252–6278, Mexico City, Mexico. Association for Computational Linguistics.

Guransh Singh, Vishwajeet Kumar, Arkadeep Acharya, Adnan Qidwai, Jaydeep Sen, and Sachindra Joshi. 2026. SearchWiki: Learning to build and navigate knowledge wikis for active information seeking. *arXiv preprint arXiv:2608.29953*.

Melanie Subbiah, Faisal Ladhak, Akankshya Mishra, Griffin Thomas Adams, Lydia B. Chilton, and Kathleen McKeown. 2024. STORYSUMM: Evaluating faithfulness in story summarization. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9988–10005, Miami, Florida, USA. Association for Computational Linguistics.

Yifan Sun, Han Wang, Dongbai Li, Gang Wang, and Huan Zhang. 2025. The emperor's new clothes in benchmarking? A rigorous examination of mitigation strategies for LLM benchmark data contamination. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 57728–57753. PMLR.

Liyan Tang, Philippe Laban, and Greg Durrett. 2024. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8818–8847, Miami, Florida, USA. Association for Computational Linguistics.

Mengting Wan, Rishabh Misra, Ndapa Nakashole, and Julian McAuley. 2019. Fine-grained spoiler detection from large-scale review corpora. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2605–2610, Florence, Italy. Association for Computational Linguistics.

Cunxiang Wang, Ruoxi Ning, Boqi Pan, Tonghui Wu, Qipeng Guo, Cheng Deng, Guangsheng Bao, Xiangkun Hu, Zheng Zhang, Qian Wang, and Yue Zhang. 2025. NovelQA: Benchmarking question answering on documents exceeding 200K tokens. In *The Thirteenth International Conference on Learning Representations (ICLR)*.

Bohao Yang, Hainiu Xu, Jinhua Du, Ze Li, Yulan He, and Chenghua Lin. 2025. EvolvTrip: Enhancing literary character understanding with temporal theory-of-mind graphs. *arXiv preprint arXiv:2506.13641*.

Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junhao Chen, Moo Khai Hao, Xu Han, Zhen Leng Thai, Shuo Wang, Zhiyuan Liu, and Maosong Sun. 2024. ∞Bench: Extending long context evaluation beyond 100K tokens. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15262–15277, Bangkok, Thailand. Association for Computational Linguistics.

Appendix Contents

The appendices provide annotation rules, implementation details, complete results, a formal account of the trade-off, reporting recommendations and the prompt and defect record.

A Comparison with Prior Work 12
B Annotation Rules and Metric Definitions 12
C Corpus and Evaluation Dataset 17
D Implementation Details 18
E Comparison Systems 20
F Results at Every Cutoff 20
G Additional Results 21
H Where Accumulation Helps 29
I A Formal Account of the Trade-off 31
J Reporting Recommendations 32
K Validation of Automatic Judges 32
L Disclosure Casebook 32
M Qualitative Example: One Page, Two Cutoffs 32
N Contamination and the Treated Text 33
O Development Observations 40
P Reproducibility and Release 41
Q Prompts and Model Exchanges 43

A Comparison with Prior Work

Table 3 compares NARRATIVEWIKI with closely related systems and evaluations. The entries follow each cited paper's description of its task, output, and evaluation. Among these works, none combines a published reference artifact, a reader-selected volume cutoff, and evaluation of atomic assertions across the delivered pages.

B Annotation Rules and Metric Definitions

Annotators and workflow. Labels were produced by LLM coding agents working interactively in the project repository under the written rules of this appendix, not by a fixed prompt: for each item the agent read the fact or assertion with the eligible source paragraphs and recorded a label in a released label file. First-pass labels came from two agents, OpenAI Codex and Claude Opus 5.5; Claude Opus 5.5 finalized every reported label. The model version behind the Codex agent was not recorded. The prefix-index builds of Table 22 received a single pass.

How each metric is computed. Let A_t be the atomic assertions of W_t, $S_t \subset A_t$ the sampled atoms (40 drawn uniformly, with enumeration frame errors removed), $c(a)$ the set of paragraphs cited for atom a, $c_t(a) = c(a) \cap D_{\le t}$ its eligible part, and $\mathbf{1}[\cdot]$ the indicator. For an assessed later fact $g \in F_t$, $d_t(g)$ is the verdict of reading every candidate sentence of W_t for it.

$$
\begin{aligned}
\mathrm{M4}_t &= |P_t| \,/\, |F_t| \qquad (t \le 4),\\
P_t &= \{g \in F_t : d_t(g) \in \{\text{stated}, \text{implied}\}\},\\
\mathrm{M8}_t &= \frac{1}{|S_t|} \sum_{a \in S_t} \mathbf{1}\big[\sigma_t(a)\big],\\
\sigma_t(a) &\equiv c_t(a) \neq \varnothing \wedge c_t(a) \models a,\\
\mathrm{CITE}_t &= \frac{|\{a \in A_t : c(a) \neq \varnothing\}|}{|A_t|},\\
\mathrm{M5} &= \frac{|L|}{\sum_{t \le 4} |S_t|},\\
L &= \{(t,a) : 1 \le t \le 4,\ a \in S_t,\\
&\qquad D_{\le t} \not\models a,\ D_{\le 5} \models a\},
\end{aligned}
$$

where $c_t(a) \models a$ means the eligible cited paragraphs, read alone and jointly, establish every element of a (*sufficient* below); cited paragraphs after the cutoff are ignored, so an atom citing only later paragraphs is insufficient. The disclosure count of Table 11 is $\sum_{t=1}^{4} |P_t|$; the inclusive count adds the facts whose verdict is *uncertain*. A system's value is the unweighted mean over the cutoffs measured, $\bar{M} = \frac{1}{|T|} \sum_{t \in T} M_t$. A paired difference between systems A and B is $\Delta = \frac{1}{|T|} \sum_{t \in T} \left(M_t^A - M_t^B\right)$, each system scored on its whole artifact against the same gold facts; its standard deviation, median, lowest and highest value are those of the per-cutoff differences. The recall gap relative to the closed-book diagnostic is $\bar{\mathrm{M}}1 - \bar{\mathrm{M}}1^{X1}$; it does not isolate the contribution of reading from memory. The seeded search recall is the share of seeded disclosure sentences returned as candidates. M5 assesses a uniform assertion sample, whereas M4 uses fact-directed searches across the delivered artifact; their denominators and detection coverage differ.

Unit. An atomic assertion states one predicate about one subject, and an object for relations. Its interpretation includes polarity, modality, and attribution to the speaker or believer when the narrative does not establish it as true.

Support labels (M3). *Supported*: the text through the cutoff establishes the assertion with the stated attribution and modality. *Partial*: part of it is established, or a transient state is presented as a lasting trait. *Unsupported*: nothing through

Work	Task	Output artifact	Cutoff	Construction scope	Evaluation unit
Jones et al. (2018)	spoiler-free wiki browsing	archived fan-wiki revision	episode air date	human-written revisions saved before the cutoff	revision selected for a requested date
CHIRON (Gurung and Lapata, 2024)	character representation	character sheet	story snippet	story so far	generated statement; masked-character prediction
TimeChara (Ahn et al., 2024)	point-in-time role-play	dialogue answers	story moment	not applicable	response to one interview question
Living the Novel (Huang et al., 2025)	conversational agent	dialogue	narrative present	whole novel, retrieval gated at t	character dialogue; user study
STORM (Shao et al., 2024)	article writing	article	none	web search, target article excluded	outline headings; article sentence citations
NarrativeFact-Score (Jeong et al., 2025)	summary evaluation	factuality score and feedback	none	whole narrative	atomic fact of a summary
STORYSUMM (Subbiah et al., 2024)	faithfulness benchmark	labels	none	not applicable	summary sentence
NS-ST-GraphRAG (Lin, 2026)	question answering	answers	temporal scope of a query	whole work, subgraph selected at query time	answer to a question
SearchWiki (Singh et al., 2026)	information seeking	linked wiki for an agent	none	whole corpus	downstream answer
NARRATIVEWIKI (this work)	reference generation	linked wiki per cutoff	reading volume	all volumes, rendered from facts filtered at t; compared with prefix-only builds	atomic assertion of a delivered page; later gold fact against every page

Table 3: Comparison with prior work by task, artifact, cutoff, construction scope, and evaluation unit.

the cutoff establishes it. *Uncertain*: the annotator cannot decide from the eligible text. The annotator may read surrounding eligible context but never text after the cutoff. Only *supported* counts. Each atom is judged in the context of the page on which it appears; its factuality in isolation is also recorded (*true as a statement*). An error introduced by the enumeration tool, such as assigning a quotation to the wrong speaker, is a frame error and is excluded from the denominator (at most 1 per cell).

The pair rule. Relationship pages display lines from scene summaries as shared scenes. A factually true line is supported *on that page* only if both characters are present and participate in at least one episode of the scene summarized. Being mentioned, quoted, written to, remembered, dead, or watching from a distance does not count. This *scene rule* is the standing one. Under the stricter *line rule*, reported as a sensitivity value, the pair must take part in the episode that the sampled line itself describes. An earlier rule that accepted a pair present anywhere in the chapter was retired because it conflated shared scenes with shared chapters. The first-pass labels of the first and the reported build had used different pair rules, so Claude Opus 5.5 re-read every sampled relationship line from the first build, reported build, and two prefix builds against the chapter text. The annotator recorded verdicts under both the scene rule and the line rule. These contextual verdicts are combined with each build's factuality labels to compute the page-context precision reported here.

Reveal volume. The earliest t whose prefix $D_{\leq t}$ establishes the fact without later text; foreshadowing that a first-time reader could not resolve is not sufficient. Each later fact's reveal volume was checked against the earliest statement in the text by a keyword scan of every earlier volume followed by a read; the check moved 2 of 90 drafted facts to an earlier volume.

Conveyed (M1). A gold fact is conveyed at cutoff t when any page of that cutoff's wiki (a character, relationship, codex or timeline page) states its core proposition of the right character (the same subject, the same event, relation or attribute, and the same object) in any wording. A conjunction of two propositions needs both (*a newspaper man and a grandson of the schoolmaster*). An incidental qualifier may be missing (*four years ago*, *at Christmas*, a color or a material); a wrong or missing core ele-

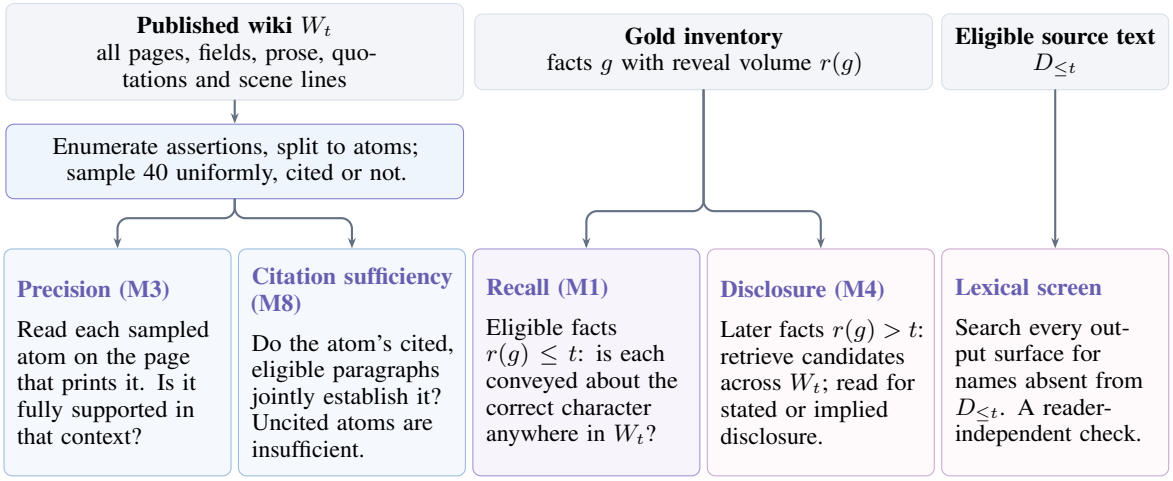


Figure 6: The evaluation protocol, applied identically to every system's wiki at every cutoff. Cyan: support for the wiki's assertions. Lavender: coverage of information available to the reader. Pink: checks for later information reaching the reader.

ID	Definition
M1	Recall: eligible gold facts conveyed about the correct character anywhere in W_t, divided by $\|G_t\|$.
M3	Precision: fully supported sampled atoms divided by sampled atoms across all surfaces, cited or uncited; enumeration frame errors are excluded.
M4	Disclosure: assessed later facts stated or implied anywhere in W_t, divided by $\|F_t\|$; undefined when $F_t = \varnothing$.
M5	Later-volume content: sampled atoms not established through the cutoff but established by the full five-volume text, divided by sampled atoms at $t = 1, \ldots, 4$.
M8	Citation sufficiency: sampled atoms established by their cited, eligible paragraphs, divided by all sampled atoms; uncited atoms are insufficient.
M12	Verifier errors: unsupported candidates admitted and supported candidates rejected on a challenge set.
M13	Cost: model calls, input tokens, and output tokens of the reported build.

Table 4: Metrics reported in this paper. IDs retain the historical measurement-ledger numbering to match archived results and labels; measures outside the reported set leave gaps.

ment may not: who did it, to whom, or the number when the number is the fact (*Una is ten*). The same proposition asserted of a different character does not count (where the statement stands does not matter, whom it is about does), and an engagement does not convey a marriage. We formalized this rule across all systems after an audit of the first-pass recall labels found it inconsistently applied. Labels had been carried forward across cutoffs when similar wording persisted; a fact marked absent early was therefore not reassessed when a later wiki paraphrased it. The audit rechecked every gold fact in every system using one search pattern per fact and changed 108 (96 to conveyed, 12 to not conveyed) labels; the reported build of NARRATIVEWIKI and the graph baseline on it were labeled afterward with the same patterns.

Disclosed (M4). *Stated*: a page asserts the later fact. *Implied*: a page asserts something from which a reader would infer it. *Uncertain*: a page asserts something adjacent that is available at the cutoff and that a reader could, but need not, read toward the later fact. *Absent*: no candidate sentence was found, or every candidate read was about something else. Stated and implied count as disclosure. Candidates are found by three lexical routes: vocabulary that first occurs after t, overlap with the content words of the fact, and marriage, death and birth words near the character's name.

Citation sufficiency (M8). For a sampled atom that the page cites, the annotator opens every paragraph in the citation list of that line or section (only paragraphs at or before the cutoff count) and asks whether they establish the atom without any other text. *Sufficient*: every element of the atom is in the cited paragraphs, jointly if need be; a relation is sufficient when the paragraph states it of a person the reader can identify, even if the page's canonical surname is given elsewhere. *Partial*: the core is there and a stated element is not (an added qualifier, a trait inferred from behavior). *Insufficient*: the supporting paragraph is not in the list, the list points to another volume or person, or the atom is itself unsupported. An assertion with no paragraph cita-

tion is insufficient; a link to a whole chapter is not a paragraph citation. Only *sufficient* counts. One correction was made after labeling. The tool that enumerates assertions did not attach to a timeline line of the reported build the *Sources* line printed under it, so the 22 sampled timeline atoms had been labeled as uncited. Their citations were read from the delivered pages and judged by the rule above: 11 are sufficient. Counting them as uncited would give NARRATIVEWIKI a citation sufficiency of 0.457 instead of 0.512, and a share of assertions with a paragraph citation of 0.840 instead of 0.930.

The precision audit. Citation assessment for M8 revealed comparison-system atoms previously labeled unsupported or uncertain that were established by the text, sometimes by the page's own citations. The first pass searched for a character's name near the fact, missing pronoun references, and examined only the first six cited passages. All 119 comparison-system atoms not labeled supported were re-read against the best-matching paragraphs of the complete eligible text, and 34 were corrected to supported. The atom-level negatives of NARRATIVEWIKI's first build were re-read the same way and none changed. Labels already marked supported were not re-audited for any system.

Two-reviewer agreement. Two human reviewers each labeled the complete review packet (554 original-text items: every gold fact and the samples of recall, precision and citation labels listed below), giving two separate sets of final labels. Table 5 recomputes their agreement on the original-text validation tasks. Cohen's κ is reported separately for gold evidence, recall, precision and citation sufficiency, since these tasks have different questions and label spaces. Across the 554 reviewed items, all assessed decisions agree on 458; at least one decision differs on 96. These values describe concordance between the reviewers' final judgments, rather than agreement with the AI annotations or correctness of the annotations. Each reviewer labeled every item from scratch, without seeing the AI answers, so these judgments are not anchored on them.

Table 6 compares each reviewer's label set with the accepted AI-assisted review snapshot finalized by Claude. Its mapping to the accepted source sheet is verified; it is not independently verified as unchanged blind first-pass AI labels. Each comparison uses all 554 reviewed items. Pair placement is evaluated separately on 14 relationship-page items within the precision sample. All applicable decisions agree on 473 items for reviewer 1, with at least one disagreement on 81; for reviewer 2, the corresponding counts are 457 and 97. These comparisons measure reviewer–AI concordance, not annotation accuracy.

Task	Items	Agreement	κ
Gold evidence	111	82.88%	0.55
Recall (M1)	235	92.77%	0.83
Precision (M3)	148	70.95%	0.39
Citation (M8)	60	71.67%	0.57
Pair placement	14	92.86%	0.86

Table 5: Agreement between the two human reviewers' final labels. Cohen's κ is reported per task; no mixed-task coefficient is pooled. Pair placement is a precision subset, not additional items.

Reviewer provenance. Both final label sets were produced entirely by human reviewers; no model generated or proposed these decisions. Both CSVs cover the complete review packet, and their labels match the accompanying response records. A released aligned audit maps each decision to its original item and marks remaining disagreements as unadjudicated.

Joint agreement across the three label sets. Table 7 recomputes nominal Krippendorff's α for the same accepted AI-assisted reference and the two reviewers' final-label sets. All 554 items have complete labels from all three. For each task, $\alpha = 1 - D_o/D_e$ uses nominal distance (zero for identical labels, one otherwise), observed within-item coincidences, and expected disagreement from pooled category counts with the $n(n-1)$ finite-sample correction, where n is the number of pairable label values (Krippendorff, 2011, pp. 1–5). It is calculated jointly, rather than by averaging pairwise kappas. Pair placement is again assessed separately within the precision sample. The coefficient measures concordance among the AI labels and two human label sets; it is not an all-human reliability estimate or an accuracy measure. The recomputed coefficients match the summary distributed with the label files. This calculation reproduces decisions; it does not adjudicate disagreements or change system scores.

Earlier human review. Claude Opus 5.5 finalized all reported AI labels. Before the two-reviewer

Task	Items	Reviewer 1 vs. AI		Reviewer 2 vs. AI	
		Agreement	κ	Agreement	κ
Gold evidence	111	85.59%	0.58	80.18%	0.51
Recall (M1)	235	94.47%	0.87	89.36%	0.75
Precision (M3)	148	77.70%	0.48	80.41%	0.50
Citation sufficiency (M8)	60	71.67%	0.57	68.33%	0.49
Pair placement	14	71.43%	0.44	64.29%	0.29

Table 6: Agreement recomputed for each human reviewer's label set against the accepted AI-assisted reference snapshot. Each comparison uses all 554 reviewed items. Pair placement is reported separately for a subset of the precision sample; its items are not added to the total. No mixed-task coefficient is pooled.

Task	Items	α	Mean pairwise agreement	Unanimous items
Gold evidence	111	0.54	82.88%	83
Recall (M1)	235	0.82	92.20%	209
Precision (M3)	148	0.45	76.35%	97
Citation sufficiency (M8)	60	0.54	70.56%	35
Pair placement	14	0.54	76.19%	9

Table 7: Nominal Krippendorff's α recomputed jointly over the accepted AI-assisted reference and the two human reviewers' final-label sets. All 554 items have complete labels from all three. Mean pairwise agreement averages the three label-set pairs; unanimous items have identical labels in all three sets. Pair placement is a subset of precision, not additional items. No mixed-task α is pooled.

check, one human reviewer examined and accepted an AI-checked sheet with answers and reasons visible; this pass yields no agreement estimate. It assessed the labels against the source text and the rules above. Sets of at most 500 items are checked in full; larger sets use a uniform random 10% sample. Each set is checked separately:

- gold facts, 111 items (at most 500): every fact and its evidence paragraph;
- recall labels, 2,345 items (over 500): a sample of 235;
- precision atoms, 1,480 items (over 500): a sample of 148;
- citation-sufficiency labels, 600 items (over 500): a sample of 60;
- disclosure verdicts, 1,540 items, and the 12,689 candidate sentences behind them: not checked by a human; these are LLM labels only.

In all, 554 labels were reviewed. The sampling frame was fixed before the last labels were added: its precision atoms are those of every original-text cell then labeled, including prefix-build cells, and its citation labels are those of B1, B2 and NARRATIVEWIKI; the citation labels of B3, B4 and X2 were added later and are outside it. All of them are labels of the original text: the labels of the treated text (Appendix N) were finalized afterward by Claude and are not part of the check. The accepted answers agree with the paper's original labels on 204 of 235 recall items, 120 of 148 precision atoms and 46 of 60 citation labels. Of the gold facts, 85, 23 and 3 are established by their evidence paragraph in full, in part and not at all; 12 of those established in part are dated by a paragraph that announces the event, which is completed within the same volume, so no reveal volume moves. For the three not established, the fact holds in the volume it is dated to, but the paragraph recorded as its evidence does not carry it alone (a greeting for a household arrangement, a plan for a residence, an introduction that does not yet tie a surname to the man). In recall and precision the accepted answers are stricter; citation-sufficiency differences occur in both directions.

The recall differences had one cause: in some cells a page sentence weaker than the gold fact had been credited ("fascinated by insects" for a boy who collects toads, bugs and frogs; a fight for a hatred). The written rule already asked for the same event, relation or attribute. We therefore re-read six gold facts in every cell of every system, on both texts, with one test per fact, and changed 64 (48 to not conveyed, 16 to conveyed) recall labels. The re-read moved no system's mean recall by more than 0.021 and is in every value reported here; after it the accepted answers match 215 of 235 recall

t	Book	Par.	Words	Ch.	$\lvert G_t \rvert$	$\lvert F_t \rvert$
1	*Anne of Green Gables*	1,788	101,965	38	23	88
2	*Anne of Avonlea*	1,705	90,685	30	46	65
3	*Anne of the Island*	1,759	76,061	41	66	45
4	*Anne's House of Dreams*	1,606	78,931	40	89	22
5	*Rainbow Valley*	1,631	80,480	35	111	0
	Total	**8,489**	**428,122**	**184**		

Table 8: The evaluated corpus. Paragraphs, words and chapters are per volume; $|G_t|$ is the number of gold facts eligible at cutoff t and $|F_t|$ the number of later facts assessed for disclosure at t.

items. It covers the six facts the check surfaced, not all 111. Precision and citation labels were not changed on the sampled items alone, since that would correct a tenth of the labels under a reading the other nine tenths did not get.

Zero counts. A zero disclosure count is printed as $0/n$. Under an idealized model of k independent, representative character-level Bernoulli trials with zero events, where k is the number of characters with an assessed later fact at t (40 at $t = 1$, 12 at $t = 4$), the exact one-sided 95% upper bound is $1 - 0.05^{1/k}$: 0.072 at $t = 1$ and 0.221 at $t = 4$. Grouping by character avoids treating correlated assertions as independent. However, characters also interact, the cast is purposively selected, and candidate retrieval is incomplete. These values illustrate the effect of the sampling unit; they are not calibrated population bounds on this system's spoiler risk.

C Corpus and Evaluation Dataset

Editions. The Anne series is read from Project Gutenberg eBooks 45 (*Anne of Green Gables*), 47 (*Anne of Avonlea*), 51 (*Anne of the Island*), 544 (*Anne's House of Dreams*) and 5343 (*Rainbow Valley*), in that order (Table 8). The corpus follows the publication order of the first five Anne books (1908–1919), excluding two later-written books set between them. Throughout the paper, a volume is one book; word counts use whitespace-delimited tokens in the parsed text.

Development history. The pipeline was developed on other series, which contribute no reported results. The Anne books were processed while the pipeline was still being revised, so this corpus is not a held-out evaluation. Baseline settings were chosen on George Eliot's *Middlemarch* (Gutenberg eBook 145, Books I–V) and not on the Anne series (Appendix E).

Primary-matrix unit	Original	Treated
System–cutoff artifacts	35	35
Selected gold records	111	111
Cast characters	43	43
Recall judgments	2,345	2,345
Disclosure judgments	1,540	1,540
Sampled assertion records	1,400	1,400
Excluded frame errors	4	2
Evaluated precision atoms	1,396	1,398

Table 9: Size of the primary evaluation matrix, excluding earlier and prefix-build diagnostics. Counts describe records and judgments, not independent trials. Both conditions reuse the same gold inventory. Citation sufficiency uses the sampled assertion records; X1 supplies no citations.

Cast. The evaluated cast contains the 43 characters with at least one gold fact. A comparison system generates a character page at cutoff t if a gold fact about that character is evidenced by t or their full name occurs in $D_{\leq t}$. Characters without a page remain in the evaluation denominators. This rule replaces an earlier rule that matched any single word of a name and generated baseline pages for characters not yet introduced (Appendix L).

Dataset size and units. The source corpus contains 428,122 whitespace-delimited words in 8,489 paragraphs and 184 chapters. The evaluated inventory contains 111 selected facts about 43 characters. Table 9 distinguishes these unique records from repeated judgments. Each original or treated condition contains the same seven systems at five cutoffs; renaming and paraphrasing reuse the inventory and reveal dates rather than adding an independent corpus. Recall labels judge eligible fact–cutoff pairs; disclosure labels judge later fact–cutoff pairs. Assertions are a separate system-generated population sampled for precision and citation sufficiency. Repeated cutoffs share facts and characters.

What a record contains. The gold YAML groups facts by character. Each fact records a claim and a paragraph identifier; the identifier encodes its evidence volume, chapter and paragraph. Optional keyword lists assist disclosure search, while reader and note fields record provenance. The stored YAML retains historical later-volume entries, but this evaluation selects evidence volumes one to five only. Below is a minimal excerpt from an existing record, with provenance metadata omitted:

```
series: anne
characters:
  Matthew Cuthbert:
    facts:
      - claim: "Matthew is the shyest man alive."
        evidence: v01:c01:p0004
```

Easier example: a direct statement. The cited paragraph describes Matthew as "the shyest man alive" and says that he "hated to have to go among strangers". The human-accepted gold check (Appendix B) labels this record fully supported. The character and description occur together, so matching the proposition requires neither a cross-paragraph identity link nor a conjunction. Its evidence volume is one: it is eligible at every evaluated cutoff. This is the narrator's characterization of Matthew, rather than a claim about his current life status.

Harder example: conjunction and local coreference. Owen Ford's record reads: "Owen Ford is a newspaper man and a grandson of the schoolmaster who built the house", with evidence `v04:c22:p0019`. Miss Cornelia's report includes "His name is Owen Ford", "a newspaper man" and "a grandson of the schoolmaster who built this house". The human-accepted gold check labels the record fully supported. Interpreting the pronouns requires linking them to Owen; recall requires both the occupation and the family/house relation. A generated statement that supplies only his occupation does not convey the full record. At cutoff three the fact is a disclosure probe; at cutoff four it becomes eligible. The example illustrates the existing conjunction rule, not a measured difficulty grade or a newly assigned annotation.

An evidence boundary case. The gold claim "Anne marries Gilbert Blythe" points to `v04:c04:p0000`, which begins "Anne wakened on the morning of her wedding day". That paragraph alone neither names Gilbert nor establishes the completed marriage. The human-accepted gold check therefore marks it partial, even though the fact is assigned to the correct volume. This example distinguishes volume-level support from paragraph-level evidence sufficiency; the recorded evidence defect remains part of the inventory, as reported in Appendix B. These illustrative cases do not constitute new labels, a measured easy/hard split, or a repair of the gold evidence.

Review population and release scope. The two-reviewer check covers 554 original-text items: 111 gold-evidence, 235 recall, 148 precision and 60 citation checks, sampled from the labels that existed when the sample was drawn (Appendix B). Original and treated gold, per-system judgments, sampled assertions, candidate witnesses, the two reviewers' final label files and recomputed agreement form the evaluation materials. The reviewer files, original response records and input hashes are released; the human review is described in Appendix B.

D Implementation Details

Claim schema. A claim is a record with a subject (an entity identifier, never a raw surface form), a predicate, a kind (attribute, relation or trait), an object entity or a scalar value, an optional short qualifier, a polarity (asserted, denied or presumed), the volume and chapter of its first evidence, and one or more evidence items, each a paragraph identifier with a verbatim quote of at most 240 characters. *Presumed* is meant to keep a belief apart from a fact: a character thought dead in one volume is recorded as presumed dead there, not as dead. The renderer marks a presumed value in attribute and relation lines, but the infobox's family field and the list of related characters print it unmarked (Section 6.2). Polarity and qualifier are part of a claim's identity, so two observations that disagree on either never merge. A claim whose subject or object cannot be resolved to a known entity is dropped.

Ontology. Entities are typed as character, group, place, event or object; only characters get their own pages, and the other types share codex pages. There are nine attributes (age, gender, status, rank, title, nickname, origin, occupation, appearance), five trait keys (personality, motivation, skill, fear, background) and twenty relation predicates covering kinship (parent, child, sibling, relative, spouse), affect (romantic, friend, rival, enemy), roles (mentor, student, commands, serves under, comrade, affiliated with) and events (origin, participated in, pilots, killed, saved). The ontology is shared by all series and was not adapted to the evaluated work.

Extraction settings. The chapter pass reads spans of about 4,000 tokens with 500 tokens of overlap and records, for each scene, a summary, the characters who take part and up to four paragraphs that support the summary. The mention pass reads a window of four paragraphs before and after a mention, never across a hard scene break,

Algorithm 1 Assembling the page of entity e at cutoff t.

Require: graph of verifier-accepted intervals; entity e; cutoff t
$I \leftarrow$ copies of $\{\, i \in \textsc{Intervals}(e) : i.start \le t \,\}$ ▷ a local view; the graph is not modified
if $I = \varnothing$ **then return** no page ▷ an entity with no interval by t has no page and is no link target
end if
for $i \in I$ **do**
  $i.quotes \leftarrow \{\, q \in i.quotes : \mathrm{vol}(q) \le t \,\}$ ▷ each quote filtered by its own volume
  if $i.end > t$ **then**
    $i.end \leftarrow \bot$ ▷ do not reveal that, or when, it ends
  end if
end for
$I \leftarrow \{\, i \in I : \textsc{Verified}(i) \,\}$ ▷ facts the verifier rejected are withheld
$\mathit{fields} \leftarrow \textsc{Render}(I)$ ▷ deterministic: names, attributes, relations, history
$\mathit{prose} \leftarrow \textsc{Generate}(I)$ ▷ model sees only I; no evidence ⇒ no call, empty section
$\mathit{page} \leftarrow \textsc{Link}(\mathit{fields}, \mathit{prose}, \{\, s : \mathrm{vol}(s) \le t \,\})$ ▷ surface forms and targets revealed by t
return *page* with cutoff-t title and navigation

with at most 40 windows per entity per volume, one window per call and at most 6,000 tokens per window (a denser cluster of mentions is split into consecutive windows). The long-context pass reads all of a character's paragraphs in a volume (up to 150,000 tokens) for gender, age and origin, and separately for background. A claim needs a self-reported confidence of at least 0.5 to be kept and 0.7 to fill a structured field. An evidence-adequacy gate rejects quotes shorter than six words, quotes in which the subject is not named within one paragraph, and quotes carrying irrealis markers ("would have", "might have").

Intervals. Single-valued attributes supersede: a value's interval closes when the next distinct value first appears. Relations in a declared conflict group supersede the same way; every other relation and multi-valued attribute is an open interval that opens at its earliest evidence, with its qualifier taken from evidence no later than that volume. Free-text values whose BGE-M3 embeddings have cosine similarity of at least 0.86 are clustered before intervals are built.

Generation. Every model call uses Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). The intended character-page template specifies appearance 2–4, personality 2–6, and history 3–12 sentences; shared-page summaries request one sentence. The archived NARRATIVEWIKI appearance prompt instead requests 3–4 sentences (Appendix Q). These are prompt requirements, not a measurement of output-length enforcement. Generation prompts request keys identifying the input lines supporting the text, which map back to source paragraphs. The parser also normalizes returned paragraph IDs during citation resolution, as in the codex example below. A shared-page summary is retained only if a second call judges every statement to be established by its input passages. A rejected summary is regenerated once and omitted if it fails again. Responses are cached by provider, model, prompt, and parameters, so rebuilding an unchanged configuration requires no new calls. All prompts are released verbatim with the code. Prompts for NARRATIVEWIKI and text-reading comparison systems do not state the work's title or author, but mined names can carry it: the series title is an entity in NARRATIVEWIKI's extraction prompts, and the typing model's exhibited output names the novel (Appendix Q.2). The closed-book diagnostic receives the title by design. A character with no STATUS claim is printed as *alive (assumed)*: a configured default with no evidence behind it, marked as assumed on the page, since the absence of a death claim does not establish that a character is alive.

Prompt and response defects. The extraction prompt of the reported runs exposes internal revision labels, module paths, and implementation notes in the STATUS and APPEARANCE descriptions. The prompt builder passes the same canonicalization note to the verifier as the *definition* of APPEARANCE, which the verifier is told to judge against; a rule for deduplicating values does not define a physical characteristic. The exhibited call accepts its appearance claim; the effect on other decisions is not measured. The codex response returns a paragraph ID where the prompt requests an input-line key; the parser maps it to its paragraph during citation resolution, without checking that the paragraph supports the sentence. All reported results use this configuration; a repaired prompt builder would require new runs, which this paper

does not report.

Changes between builds. The reported build differs from the first build (Section 6.4, Table 25) by one batch of changes, made after the first build had been labeled and applied to every series alike. *Entity resolution:* a bare name and a fuller name first seen in a later volume go to the model instead of merging for free; a first name shared by several people is assigned per volume; fuzzy matching may not exchange a whole word; the cap on model-judged name pairs rose from 1,000 to 5,000; hyphenated names and titles of the form "X of the Y" are mined whole. *Ingestion:* publisher's back matter is dropped. *Scenes:* the participants of a scene are those who take part, not those mentioned, and each summary names up to four supporting paragraphs. *Citations:* generated text names the input lines it rests on, and scene lines print their paragraphs on timeline, pair and codex pages; the first build cited a chapter for a scene line and nothing for most prose. *Gates:* a prose answer that says it found nothing is discarded, and summaries on shared pages pass the check described above. *Windows:* the 6,000-token ceiling on a mention window, which the first build did not enforce on a dense cluster. One side effect: the first build held two entities for several main characters, and each entity received its own windows, so the same text was read twice; with the entities merged it is read once, and the claims stored for two such characters fell from 162 and 66 to 118 and 46 on as many windows. We did not isolate this change's contribution to the reported build's lower recall.

	Adaptation and settings
B1	Every paragraph of volumes $1, \ldots, t$, labeled with its identifier, precedes the task so that pages at a cutoff share a cacheable prefix. Prompts exceeding the model's context limit would be marked unsupported rather than truncated; none exceeded it.
B2	Prefix-only index of consecutive paragraphs within a chapter (150 words). Okapi BM25 (k_1=1.5, b=0.75) and BGE-M3 cosine rankings for the character name alone and with seven facet words (appearance, personality, family, relations, goals, past, speech), fused by reciprocal rank (k=60); chunks added in fused order up to 20,000 tokens of retrieved text, then restored to reading order. Budgets count retrieved text only, estimated as characters divided by four, so a call's provider-counted input, which adds the prompt, can exceed them. Chunk size (150 or 300 words) and budget (6,000, 12,000 or 20,000 tokens) were chosen at t=2 of the development novel.
B3	LightRAG 1.5.7, default chunking (1,200 tokens) and its default extraction prompt and limits (at most 100 rows and 40 entities per chunk), mode `mix`, context capped at the same 20,000 tokens, reranking off. The index at cutoff t is the index at $t-1$ plus volume t, so no index sees a later volume. LightRAG supplies retrieval only: its retrieved entities, relations and chunks are passed to the shared generator, and its own extraction calls use the same model.
B4	The NARRATIVEWIKI graph built from all five volumes. For each character, the facts visible at t and their evidence paragraphs of volumes $\leq t$, up to the same budget, are passed to the shared generator in place of NARRATIVEWIKI's structured rendering (graph `anne@v2`).
X1	The shared prompt with the work's title, the character's name and no passages, with an instruction to write what is known of the character up to volume t; citations are empty by construction.
X2	Every paragraph of volumes 1–5 regardless of t, with the instruction to write for a reader of volume t and reveal nothing later.

Table 10: Adaptations of the comparison systems.

E Comparison Systems

All comparison systems (B1–B4, X1, X2) use the same cast list, character-page template, generator, and intended character-page budget: the shared sections and sentence limits (appearance 2–4, personality 2–6, history 3–12), at most five quotes, and a shared citation format (Table 10). For systems supplied with passages, citations to paragraphs outside their input are removed, and characters absent from the eligible text receive no page but remain in the evaluation denominators. Settings were chosen on the development novel (*Middlemarch*) and fixed before the Anne runs. NARRATIVEWIKI itself is not given the cast: it discovers entities and also publishes pages for other characters, relationships, codex entries, and a timeline. The shared template therefore controls the evaluated cast's character pages, while the complete artifacts differ in scope (Table 18). The exhibited comparison prompts request 2–4 appearance sentences; the archived NARRATIVEWIKI appearance prompt requests 3–4 (Appendix Q). The template budget therefore does not imply identical prompt wording or verified enforcement of output length.

F Results at Every Cutoff

Tables 12 (original text) and 13 (treated text) report the individual cells underlying the means in Table 11; Figure 7 plots them. Long-context recall is approximately stable across cutoffs. NARRATIVEWIKI has its highest recall on the first book, where it matches the closed-book diagnostic, and its lowest at $t = 2$. Closed-book recall declines from the first book to the fifth. LightRAG's low

System (cutoff enforced by)	Recall ↑	Precision ↑	Citation sufficiency ↑	Later facts disclosed ↓	Distinct assertions	Tokens in / out (M)
B1 Long context (input)	**0.848**$_{\pm 0.022}$	0.929$_{\pm 0.028}$	0.658$_{\pm 0.098}$	0 of 220	1,821	69.23 / 0.37
B2 Prefix retrieval (input)	0.834$_{\pm 0.063}$	0.960$_{\pm 0.029}$	0.774$_{\pm 0.088}$	0 of 220	1,718	**3.42 / 0.37**
B3 LightRAG (input)	0.536$_{\pm 0.133}$	0.950$_{\pm 0.018}$	0.765$_{\pm 0.118}$	0 of 220	1,404	5.08 / 1.01
B4 Graph baseline	0.761$_{\pm 0.049}$	**0.965**$_{\pm 0.014}$	**0.915**$_{\pm 0.029}$	0 of 220	1,701	+1.11 / 0.36
B5 NarrativeWiki (data)	0.802$_{\pm 0.026}$	0.789$_{\pm 0.087}$	0.512$_{\pm 0.035}$	**1 of 220**	**6,970**	33.13 / 1.86
X1 Closed book (none)	0.683$_{\pm 0.082}$	0.814$_{\pm 0.071}$	–	**4 of 220**	1,526	0.06 / 0.15
X2 Full text (instruction)	0.814$_{\pm 0.043}$	0.910$_{\pm 0.058}$	0.625$_{\pm 0.064}$	**8 of 220**	1,829	98.67 / 0.37

Table 11: Main comparison on Anne: rates are mean ± sample standard deviation across $t = 1..5$. Recall uses eligible gold facts; precision and citation sufficiency use 40 sampled atoms per cutoff, judged in page context. Disclosure counts stated or implied later facts across $t = 1..4$. Distinct assertions are means per cutoff; tokens (millions, input / output) cover the reported build. X1 and X2 are diagnostics; X1 cites nothing. Bold identifies rate and assertion-count maxima and minimum measured cost among B1–B5, plus non-zero disclosures. Cyan highlights performance maxima, magenta disclosures, and violet inventory and minimum cost. B4 reuses B5's graph; its cell is the page generation added to B5's build. Distributions are in Table 14.

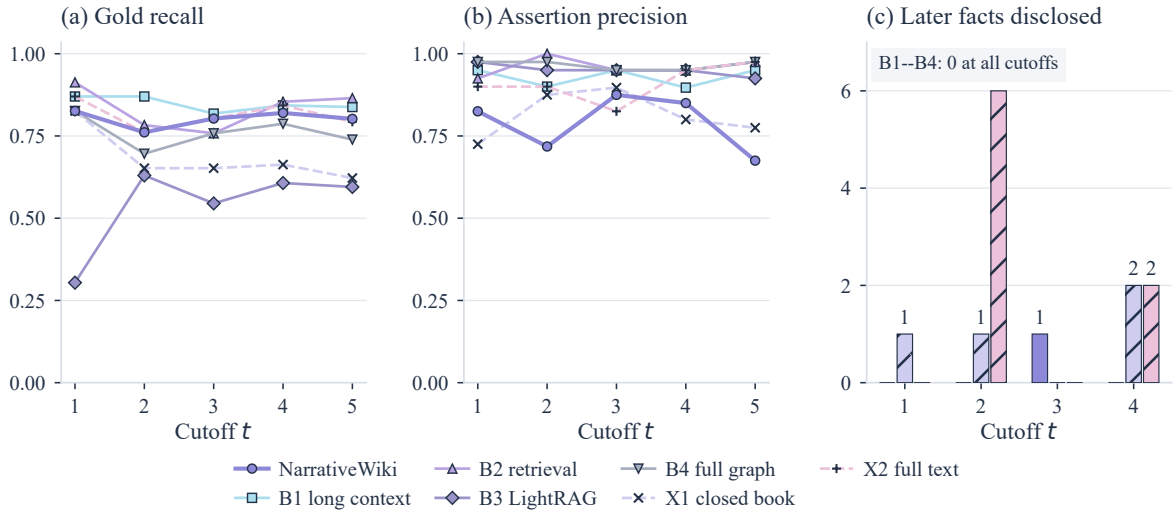


Figure 7: Original text: recall and precision at each cutoff (lines), and counts of assessed later facts disclosed (bars). Dashed lines denote the X1 and X2 diagnostics. B1–B4 have zero observed disclosures at every cutoff. Disclosure denominators are $|F_t| = 88, 65, 45, 22$ at $t = 1, \ldots, 4$; disclosure is undefined at $t = 5$.

recall at $t = 1$ accompanies retrieved context concentrated on the opening chapters. With only 40 sampled atoms per cell, the precision results do not establish a trend across cutoffs. Every reported model call used the shared generator. Table 14 gives the mean, standard deviation, median, minimum, and maximum across cutoffs; Table 15 reports the same summaries for the paired differences discussed in Section 6.

G Additional Results

Coverage of the evaluation. Table 16 lists the available builds and labels. B4's aggregate results are included in Table 11, with per-cutoff values below. The first build of NarrativeWiki and its prefix variants support the analyses in Sections 6.4 and 6.5.

What the disclosure search retrieves. Disclosure judgments are limited to sentences returned by candidate search. We therefore test retrieval on seeded sentences with known disclosures (Table 17). Search retrieves every verbatim claim, about 85% of named paraphrases and about 70% of named implications; pronouns on another character's page reduce retrieval to roughly one third. The corresponding rates are 220 of 220 for verbatim claims, 0.845 (186/220) and 0.768 (169/220) for paraphrases, and 0.714 (157/220) and 0.623 (137/220) for implications. Search produced 5,702 candidate sentences for NarrativeWiki, 520 for B1 and 12,689 across systems and builds. A lexical-

t	System	Recall (M1)	Precision (M3)	Citation sufficiency (M8)	Disclosed (M4)	Assertions
1	B1	0.870 (20/23)	0.950 (38/40)	0.800 (32/40)	0/88	837
	B2	**0.913 (21/23)**	0.925 (37/40)	**0.850 (34/40)**	0/88	848
	B3	0.304 (7/23)	**0.975 (39/40)**	0.575 (23/40)	0/88	622
	B4	0.826 (19/23)	0.975 (39/40)	0.925 (37/40)	0/88	784
	NARRATIVEWIKI	0.826 (19/23)	0.825 (33/40)	0.575 (23/40)	0/88	**4,606**
	X1	0.826 (19/23)	0.725 (29/40)	–	**1/88**	807
	X2	0.870 (20/23)	0.900 (36/40)	0.700 (28/40)	0/88	862
2	B1	**0.870 (40/46)**	0.900 (36/40)	0.650 (26/40)	0/65	1,350
	B2	0.783 (36/46)	**1.000 (40/40)**	0.650 (26/40)	0/65	1,369
	B3	0.630 (29/46)	0.950 (38/40)	**0.800 (32/40)**	0/65	1,102
	B4	0.696 (32/46)	0.975 (39/40)	0.875 (35/40)	0/65	1,284
	NARRATIVEWIKI	0.761 (35/46)	0.718 (28/39)	0.487 (19/39)	0/65	**9,695**
	X1	0.652 (30/46)	0.875 (35/40)	–	**1/65**	1,280
	X2	0.761 (35/46)	0.900 (36/40)	0.675 (27/40)	**6/65**	1,432
3	B1	**0.818 (54/66)**	**0.950 (38/40)**	0.700 (28/40)	0/45	1,823
	B2	0.758 (50/66)	**0.950 (38/40)**	**0.800 (32/40)**	0/45	1,730
	B3	0.545 (36/66)	**0.950 (38/40)**	0.775 (31/40)	0/45	1,331
	B4	0.758 (50/66)	0.950 (38/40)	0.950 (38/40)	0/45	1,704
	NARRATIVEWIKI	0.803 (53/66)	0.875 (35/40)	0.500 (20/40)	**1/45**	**13,947**
	X1	0.652 (43/66)	0.897 (35/39)	–	0/45	1,511
	X2	0.803 (53/66)	0.825 (33/40)	0.575 (23/40)	0/45	1,740
4	B1	0.843 (75/89)	0.897 (35/39)	0.590 (23/39)	0/22	2,345
	B2	**0.854 (76/89)**	0.949 (37/39)	0.718 (28/39)	0/22	2,187
	B3	0.607 (54/89)	**0.950 (38/40)**	**0.775 (31/40)**	0/22	1,837
	B4	0.787 (70/89)	0.950 (38/40)	0.900 (36/40)	0/22	2,227
	NARRATIVEWIKI	0.820 (73/89)	0.850 (34/40)	0.500 (20/40)	0/22	**17,853**
	X1	0.663 (59/89)	0.800 (32/40)	–	**2/22**	2,016
	X2	0.843 (75/89)	0.950 (38/40)	0.625 (25/40)	**2/22**	2,421
5	B1	0.838 (93/111)	0.950 (38/40)	0.550 (22/40)	–	3,234
	B2	**0.865 (96/111)**	**0.975 (39/40)**	0.850 (34/40)	–	2,892
	B3	0.595 (66/111)	0.925 (37/40)	**0.900 (36/40)**	–	2,440
	B4	0.739 (82/111)	0.975 (39/40)	0.925 (37/40)	–	2,981
	NARRATIVEWIKI	0.802 (89/111)	0.675 (27/40)	0.500 (20/40)	–	**23,613**
	X1	0.622 (69/111)	0.775 (31/40)	–	–	2,570
	X2	0.793 (88/111)	0.975 (39/40)	0.550 (22/40)	–	3,196

Table 12: Results at every cutoff. Rates are followed by their counts. Disclosure is the number of assessed later facts stated or implied over $|F_t|$ and is undefined at $t = 5$; citation sufficiency has no value for X1, which cites nothing; assertions is the size of the cutoff's assertion inventory, counting a statement once for every page that prints it. A precision denominator below 40 reflects excluded frame errors. B4 (the graph baseline, which reuses NARRATIVEWIKI's graph) is unhighlighted; the treated-text cells are in Table 13. Bold: highest rates and assertion count at each cutoff among B1, B2, B3 and B5, and each non-zero disclosure count. Cyan, magenta, and violet highlight performance maxima, disclosures, and assertion counts, respectively.

screen hit on X2 exposed an unmatched phrase ("to marry"); after correcting the matcher, 144 sentences were reassessed. Of 224 eligible claims passed through the same search, 14 became candidates for a later fact about the same character. Retrieval alone does not count as disclosure.

The annotator model wrote and checked each fixture for its subject, event and object, independently of whether search retrieved it. An earlier fixture version, in which many implications did not establish their target fact, had retrieval of approximately one quarter and is released alongside the validated version. No second reader validated the fixtures. The search-independent checks of Section 6.2 also have limits: lexical screening detects only new names, while zero later-volume assertions in a sample of approximately 160 atoms gives limited evidence about rare disclosures. Overlapping cutoff wikis and clustering within pages reduce the effective sample size.

Matched and whole-artifact populations. Table 18 separates sampled assertions by page type. The comparison systems publish only cast character pages; on this matched population, 31 of NARRATIVEWIKI's 35 sampled atoms are supported. This small subsample does not support a precise comparison with the baselines, and no equivalence test is reported. Of NARRATIVEWIKI's 42 page-

Metric	t	B1	B2	B3	B4	B5	X1	X2
Recall	1	**0.957 (22/23)**	0.870 (20/23)	0.348 (8/23)	0.652 (15/23)	0.870 (20/23)	0.000 (0/23)	0.826 (19/23)
	2	**0.891 (41/46)**	0.783 (36/46)	0.457 (21/46)	0.674 (31/46)	0.826 (38/46)	0.000 (0/46)	0.696 (32/46)
	3	0.818 (54/66)	0.727 (48/66)	0.500 (33/66)	0.758 (50/66)	**0.848 (56/66)**	0.015 (1/66)	0.667 (44/66)
	4	0.764 (68/89)	0.775 (69/89)	0.539 (48/89)	0.663 (59/89)	**0.843 (75/89)**	0.000 (0/89)	0.764 (68/89)
	5	0.793 (88/111)	0.766 (85/111)	0.505 (56/111)	0.775 (86/111)	**0.829 (92/111)**	0.000 (0/111)	0.766 (85/111)
Precision	1	**0.975 (39/40)**	**0.975 (39/40)**	0.925 (37/40)	0.925 (37/40)	0.750 (30/40)	0.200 (8/40)	0.825 (33/40)
	2	0.950 (38/40)	**0.975 (39/40)**	0.949 (37/39)	**0.975 (39/40)**	0.700 (28/40)	0.150 (6/40)	0.825 (33/40)
	3	0.900 (36/40)	0.925 (37/40)	**1.000 (40/40)**	0.975 (39/40)	0.875 (35/40)	0.125 (5/40)	0.825 (33/40)
	4	0.825 (33/40)	0.950 (38/40)	**1.000 (40/40)**	0.923 (36/39)	0.750 (30/40)	0.150 (6/40)	0.825 (33/40)
	5	**0.950 (38/40)**	0.925 (37/40)	0.925 (37/40)	0.900 (36/40)	0.775 (31/40)	0.100 (4/40)	0.925 (37/40)
Citation suff.	1	0.875 (35/40)	**0.925 (37/40)**	0.625 (25/40)	0.850 (34/40)	0.500 (20/40)	–	0.575 (23/40)
	2	0.675 (27/40)	**0.900 (36/40)**	0.769 (30/39)	**0.900 (36/40)**	0.700 (28/40)	–	0.550 (22/40)
	3	0.425 (17/40)	0.825 (33/40)	**0.875 (35/40)**	**0.875 (35/40)**	0.575 (23/40)	–	0.475 (19/40)
	4	0.300 (12/40)	0.925 (37/40)	0.800 (32/40)	**0.949 (37/39)**	0.700 (28/40)	–	0.400 (16/40)
	5	0.375 (15/40)	0.850 (34/40)	**0.900 (36/40)**	**0.900 (36/40)**	0.650 (26/40)	–	0.375 (15/40)
Disclosed	1	0/88	0/88	0/88	0/88	0/88	0/88	**8/88**
	2	0/65	0/65	0/65	**2/65**	**2/65**	0/65	**8/65**
	3	0/45	0/45	0/45	0/45	**2/45**	0/45	**6/45**
	4	0/22	0/22	0/22	0/22	0/22	0/22	**1/22**
	5	–	–	–	–	–	–	–

Table 13: Every cell of the treated text behind Table 1: recall over the eligible gold facts, precision and citation sufficiency over 40 sampled statements, and stated or implied disclosures over the later gold facts (none are left to disclose at $t = 5$). The closed-book generator cites nothing. Bold cyan cells mark the highest rates among B1–B5 at each cutoff; bold magenta cells mark non-zero disclosure counts.

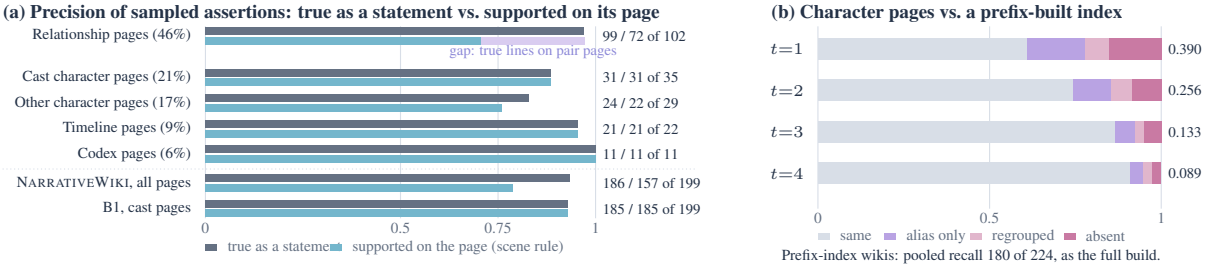


Figure 8: Context collapse (RQ4) and identity dependence (RQ5), original text. (a) On relationship pages nearly every line is true, but a quarter sit under a pair that shares no episode; on cast pages the two measures agree. (b) NarrativeWiki's character pages against an entity index discovered from volumes $1..t$; the printed rate is the share that differ.

context negatives, 30 occur on relationship pages; 27 of those 30 are supported as statements but placed under a pair that shares no episode of the scene.

For every NarrativeWiki page type, observed citation sufficiency is below the overall values of B1 and B2. Cited paragraphs establish 14 of 35 atoms on cast pages, 60 of 102 on relationship pages and 11 of 22 on the timeline. Timeline lines can carry up to four paragraphs selected to support a chapter-level summary.

The graph baseline (B4). Table 19 reports B4 at every cutoff, using the reported NarrativeWiki graph. Mean recall is 0.761, versus 0.802 for NarrativeWiki: lower at four cutoffs and equal at one. Mean precision is 0.965. No assessed later fact was disclosed (0 of 220); B4 publishes 1,701 distinct assertions per cutoff and has 1 lexical-screen hit, the same expanded page title found in the prefix baselines.

B4 generates cast character pages from a budgeted selection of retrieved graph facts; NarrativeWiki renders all eligible facts and also publishes relationship, codex and timeline pages. B4's higher precision, together with the 30 relationship-page negatives among NarrativeWiki's 42 page-context negatives, suggests that publication context contributes to the gap in Table 11. This comparison does not isolate context from selection and generation.

B4's cited paragraphs suffice for 0.915 of sam-

Recall ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	**0.848**	0.834	0.536	0.761	0.802	0.683	0.814
SD	0.022	0.063	0.133	0.049	0.026	0.082	0.043
Median	0.843	0.854	0.595	0.758	0.803	0.652	0.803
Min	0.818	0.758	0.304	0.696	0.761	0.622	0.761
Max	0.870	0.913	0.630	0.826	0.826	0.826	0.870

Precision ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	0.929	0.960	0.950	**0.965**	0.789	0.814	0.910
SD	0.028	0.029	0.018	0.014	0.087	0.071	0.058
Median	0.950	0.950	0.950	0.975	0.825	0.800	0.900
Min	0.897	0.925	0.925	0.950	0.675	0.725	0.825
Max	0.950	1.000	0.975	0.975	0.875	0.897	0.975

Citation sufficiency ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	0.658	0.774	0.765	**0.915**	0.512	–	0.625
SD	0.098	0.088	0.118	0.029	0.035	–	0.064
Median	0.650	0.800	0.775	0.925	0.500	–	0.625
Min	0.550	0.650	0.575	0.875	0.487	–	0.550
Max	0.800	0.850	0.900	0.950	0.575	–	0.700

Disclosure rate ($n = 4$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	0.000	0.000	0.000	0.000	**0.006**	**0.029**	**0.046**
SD	0.000	0.000	0.000	0.000	0.011	0.042	0.053
Median	0.000	0.000	0.000	0.000	0.000	0.013	0.045
Min	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Max	0.000	0.000	0.000	0.000	0.022	0.091	0.092

Assertions ($n = 5$)

Statistic	B1	B2	B3	B4	B5	X1	X2
Mean	1,918	1,805	1,466	1,796	**13,943**	1,637	1,930
SD	924	781	698	849	7,315	679	904
Median	1,823	1,730	1,331	1,704	13,947	1,511	1,740
Min	837	848	622	784	4,606	807	862
Max	3,234	2,892	2,440	2,981	23,613	2,570	3,196

Table 14: Distribution over cutoffs of every per-cutoff cell: unweighted mean, sample standard deviation, median, lowest and highest value over the n cutoffs measured ($t = 1..5$; disclosure $t = 1..4$). Disclosure rate is stated or implied disclosures over $|F_t|$. Rate means equal the point estimates of Table 11; the assertion panel counts every assertion as it occurs on a page, not the distinct assertions of Table 11. Bold: highest performance and assertion-count means among B1–B5, and each non-zero mean disclosure rate. Cyan highlights performance maxima, magenta non-zero disclosure means, and violet assertion count.

Metric	B5−	Mean	SD	Median	Min	Max
Recall	B1	**−0.045**	0.037	−0.036	−0.109	−0.015
	B2	−0.032	0.050	−0.034	−0.087	0.045
	B3	**0.266**	0.150	0.213	0.130	0.522
	X1	0.120	0.072	0.152	0.000	0.180
	X2	−0.011	0.021	0.000	−0.043	0.009
Disclosure	X1	−0.024	0.048	−0.013	−0.091	0.022
	X2	**−0.040**	0.060	−0.045	−0.092	0.022

Table 15: Paired differences of NARRATIVEWIKI (B5) from each system, mean over cutoffs of the per-cutoff difference, each system scored on its whole artifact; SD, median, lowest and highest value are over the per-cutoff differences (five cutoffs for recall, four for disclosure). Recall minus X1 compares source-conditioned output with the closed-book diagnostic. Bold cyan cells mark the largest mean difference in each direction.

	Cutoffs built	M1, M3, M4	M8
B1, B2, NARRATIVEWIKI	1–5	labeled	labeled
on prefix indexes	1–4	labeled	labeled
B3, B4, X2	1–5	labeled	labeled
X1 (cites nothing)	1–5	labeled	–
NARRATIVEWIKI, first build	1–5	labeled	labeled
from a prefix	2, 3	labeled	–

Table 16: What was built and labeled. Every label is by an LLM annotator (Section 5).

pled assertions, and none of its 200 sampled citations is judged wholly unrelated to the assertion. The corresponding whole-artifact value for NARRATIVEWIKI is 0.512; on character pages it is 29 of 64, versus 183 of 200 for B4. These observations show that evidence stored in the common graph can support generated assertions, while NARRATIVEWIKI's evidence selection or attachment is less reliable.

Identity dependence. Table 21 reports the measure defined in Section 6.5; its mean is 0.217. Every prefix-built character has a counterpart in the full build, although counterpart existence does not imply identical alias grouping.

Some differences arise deterministically during candidate mining. Of the full build's candidates first occurring by t, 109, 111, 78 and 44 are absent from prefix mining ($t = 1..4$). Most, 91, 87, 60 and 32, occur fewer than three times in the prefix and reach the miner's frequency threshold only with later occurrences. The remaining 18, 24, 18 and

Seeded sentence	Retrieved	Rate
Gold claim verbatim	220	1.000
Paraphrase, names the character	186	0.845
Paraphrase, pronoun, own page	169	0.768
Paraphrase, pronoun, other page	68	**0.309**
Implying, names the character	157	0.714
Implying, pronoun, own page	137	0.623
Implying, pronoun, other page	78	**0.355**

Table 17: Recall of the disclosure candidate search on seeded disclosures: 220 fact-by-cutoff units ($t = 1..4$), one paraphrase and one implying sentence per later fact, each checked to establish its fact. Bold magenta cells mark the lowest rate of each kind.

	Share	n	Page	Atom	Suff.
Character pages, cast	0.206	35	31	31	14
Character pages, other	0.171	29	22	24	15
Relationship pages	**0.463**	102	**72**	**99**	60
Codex pages	0.057	11	11	11	2
Timeline	0.090	22	21	21	11
Index	0.012	0	–	–	–
NARRATIVEWIKI, all pages	1.000	199	157	186	102
B1 (cast pages only)	1.000	199	185	185	131
B2 (cast pages only)	1.000	199	191	191	154

Table 18: The precision samples by kind of page, pooled over cutoffs. Share: of the system's assertions on that kind of page; n: sampled atoms; Page and Atom: supported on the page that prints them (relationship pages under the pair rule of Appendix B), and as statements; Suff.: established by their own cited paragraphs. Lavender marks the largest kind of page; magenta its page-context support, which falls short of its assertions true as statements.

12 fail a separate rule: a single capitalized word must stand alone in at least three quarters of its occurrences in some volume. Bare *Gilbert* appears alone in 62 of 113 book-one occurrences; the other 51 are in *Gilbert Blythe*. Prefix mining therefore omits the alias, while later volumes allow the full build to retain it. Disabling this rule admits 352 additional full-build candidates, mostly sentence-initial words. We retain the reported rule because changing it would alter the evaluated index.

Pages absent from the prefix index at $t = 1$ include `walter`, `jerry` and `marshall`, discussed below, and Anne's mother, whose full name occurs once in book one. The four prefix indexes use 2.61M input and 0.26M output tokens, including 0.37M and 0.04M in calls absent from the full build's cache. The measure captures entity-index sensitivity to construction scope; wiki outcomes on these indexes appear in Table 22.

Wikis built on the prefix indexes. Table 22 evaluates the complete reported pipeline on each prefix $t = 1..4$, using that prefix's entity index. Pooled recall is 180 of 224 gold-fact assessments for the prefix builds and 180 of 224 for the full build. Per-cutoff differences comprise one to three facts in either direction: prefix builds lose a minor character's trait and gain an office or prize omitted by the full build. Pooled precision is 134 of 159, versus 130 of 159, on different samples. Citation sufficiency is 91 of 159, versus 82 of 159, using different samples with labels finalized by Claude.

Neither build states a later fact. Both contain the same implied disclosure at $t = 3$, the bridesmaid relation inverted from eligible text. Prefix builds publish fewer character pages at early cutoffs: `walter` and `jerry`, retained by the full build at every cutoff because of book-five characters, never appear in prefix builds; `marshall`, from book four, appears only at $t = 4$. Prefix-specific discovery avoids using later text in entity construction while retaining the observed pooled gold recall. This is not evidence of equivalence on all outcomes: the approach requires a build per cutoff, and the four prefix builds together use more tokens than the single full build.

Construction scope. Table 20 compares prefix and full graphs made with first-build code; this complete graph comparison was not repeated for the reported build. The pooled recall difference is -0.027 (93 vs 96 of 112), with seven discordant facts; the precision difference is $+0.038$ (58 vs 55 of 78), using separate samples. These comparisons cannot isolate construction-scope effects from differences in entity indexes and model runs.

At $t = 2$, later-anchored pages include `walter` (Walter Blythe of book five, 127 of the entity's 130 mentions, but an early page describing Anne's father); `di` (Di Blythe, with an early duplicate page for Diana Barry under a nickname used once in the first two books); `mary` (Mary Vance, showing the twins' mother); `jerry` (Jerry Meredith, merging a book-one hired boy with a book-two agent); and `marshall`, `baxter` and `sam`. An entity-merge audit flags 42 entities in the full build and 2 in a volume-one build.

Shared pages have identical titles and no full-build alias absent from $D_{\leq t}$. Relation lines differ in both directions: at $t = 2$, 54 occur in both builds, 93 only in the full build and 68 only in the prefix build. Separate runs and entity lists prevent attribut-

t	Recall	Precision	Citation	Disclosed	Assertions
1	**0.826 (19/23)**	0.975 (39/40)	0.925 (37/40)	0/88	784
2	**0.696 (32/46)**	0.975 (39/40)	0.875 (35/40)	0/65	1,284
3	0.758 (50/66)	0.950 (38/40)	0.950 (38/40)	0/45	1,704
4	0.787 (70/89)	0.950 (38/40)	0.900 (36/40)	0/22	2,227
5	0.739 (82/111)	0.975 (39/40)	0.925 (37/40)	–	2,981

Table 19: B4, the graph baseline (the full graph, filtered at retrieval time), at every cutoff. Columns as in Table 12; Claude finalized the citation labels at every cutoff. Bold marks the highest and lowest recall, highlighted in cyan and pink, respectively.

	$t = 2$	$t = 3$
Pages, full build	133	173
Pages, prefix build	107	151
In both builds	100	145
same title	100	145
same alias list	80	128
alias not in $D_{\leq t}$ (full)	0	0
In one build only	40	34
later-anchored	**7**	**9**
second page, same person	9	9
same person, new address	9	5
minor entity	15	11
stating a later fact	0	0
Recall, prefix build	0.804 (37/46)	0.848 (56/66)
Recall, full build	**0.848 (39/46)**	**0.864 (57/66)**
Precision, prefix build	**0.769 (30/39)**	**0.718 (28/39)**
Precision, full build	0.718 (28/39)	0.692 (27/39)
Later facts disclosed	0/65	0/45
Prefix build tokens (M)	12.03 / 0.68	17.94 / 1.01

Table 20: Construction scope (character pages), first build of NARRATIVEWIKI: built from volumes $1..t$ alone (prefix) against the same cutoff of its five-volume build (full; 33.14M input and 1.81M output tokens once, for all cutoffs). Precision is under the pair rule of Appendix B. Disclosure counts are identical in the two builds; tokens are input / output, in millions. Bold magenta cells mark later-anchored pages; bold cyan cells mark the higher of the two builds on recall and precision.

ing these differences solely to later volumes. Prose on shared pages was not compared.

Later-anchored entities in the reported build. The reported build adjudicates bare names against fuller forms first appearing later, and assigns shared first names volume by volume to the locally dominant full name. Its `walter` entity now contains only Anne's father (three mentions), with Walter Blythe appearing separately from book five.

A separate screen flags character-page entities with at least ten mentions and at least four fifths occurring after the cutoff: 6 of 117 at $t = 2$ and 3 of 158 at $t = 3$. At $t = 2$, these are `susan` (3 of 204 mentions by book two; an empty page dominated by Susan Baker of books four and five); `sam` (2 of 10, with prose saying "Sam does not appear in the story provided"); `joseph` (5 of 54, an early pupil's uncle under an entity later dominated by others); `jerry-corcoran` (6 of 120, correctly described at $t = 2$ but later absorbing Jerry Meredith); and `stella-maynard` and `billy-andrews`, correctly resolved people introduced early. This screen identifies entities dominated by later mentions. It neither establishes dependence for every flagged page nor determines what a prefix build would publish instead.

Cost. Table 23 breaks down model use by stage; extraction dominates. Prose and shared-page summaries are generated once per distinct filtered fact set, across all five cutoffs, and included in the total. Verification includes rechecks after one re-extraction, so its count slightly exceeds a cold build from an empty cache. The first build used 33.14M input and 1.81M output tokens under the same accounting. B4 adds 1.11 / 0.36 million tokens (input / output) of page generation to the graph it shares with NARRATIVEWIKI; latency, hardware and the local evaluation models are not costed for any system.

Long-context input grows with the prefix (2.2M, 7.0M, 10.8M, 19.6M and 29.7M tokens at $t = 1..5$), which is sent once per page. The all-cutoff NARRATIVEWIKI build uses 33.1M input tokens. LightRAG's 5.08 / 1.01 million tokens (input / output) comprise 2.36M input and 0.72M output tokens for five incremental indexes and 2.72M and 0.29M for page generation. Prefix retrieval indexes use local embeddings and BM25, so its 3.42 / 0.37 million tokens cover page generation. The closed-book diagnostic uses 0.06 / 0.15.

Verification. At audit, the verifier had marked 21.2% of 2,535 facts unsupported by their cited evidence; these facts are withheld. Table 23 counts verification calls rather than distinct facts. On a

t	Pages	Same	Alias	Regrouped	Absent	Identity dependence	Regrouped or absent	Mentions
1	59	36	10	4	9	**0.390 (23/59)**	**0.220 (13/59)**	**0.943**
2	117	87	13	7	10	0.256 (30/117)	0.145 (17/117)	0.967
3	158	137	9	4	8	0.133 (21/158)	0.076 (12/158)	0.982
4	191	174	7	5	5	0.089 (17/191)	0.052 (10/191)	0.994

Table 21: Identity dependence of the reported NARRATIVEWIKI on the original text. Each character page of the five-volume build at cutoff t is compared with the entity index built from volumes $1..t$ alone. *Same*: a prefix-built character has exactly its set of names seen by t. *Alias*: one counterpart, with a name linked in one build only. *Regrouped*: names that one build gives one character the other gives two. *Absent*: no prefix-built entity carries any of its names. *Mentions*: character mentions in volumes $1..t$ under the prefix index, as a share of the full build's. One run per index, so model variance is inside the differences. Bold magenta cells mark the cutoff where identity dependence is largest.

	Build	$t = 1$	$t = 2$	$t = 3$	$t = 4$
Character pages	full	59	117	158	191
	prefix	48	108	155	185
Recall	full	0.826 (19/23)	0.761 (35/46)	0.803 (53/66)	**0.820 (73/89)**
	prefix	0.826 (19/23)	0.761 (35/46)	**0.818 (54/66)**	0.809 (72/89)
Precision	full	0.825 (33/40)	0.718 (28/39)	0.875 (35/40)	**0.850 (34/40)**
	prefix	**0.950 (38/40)**	**0.750 (30/40)**	0.875 (35/40)	0.795 (31/39)
Citation sufficiency	full	0.575 (23/40)	**0.487 (19/39)**	0.500 (20/40)	0.500 (20/40)
	prefix	**0.650 (26/40)**	0.425 (17/40)	**0.600 (24/40)**	**0.615 (24/39)**
Later facts disclosed	full	0/88	0/65	**1/45**	0/22
	prefix	0/88	0/65	**1/45**	0/22
Assertions	full	4,606	9,695	13,947	17,853
	prefix	3,977	8,732	13,432	17,723
Tokens, in / out (M)	prefix	5.68 / 0.31	11.95 / 0.65	18.04 / 0.99	24.04 / 1.37

Table 22: The reported NARRATIVEWIKI built on entity indexes discovered from the prefix (*prefix*: one build per cutoff, every stage on volumes $1..t$) against the same cutoff of its five-volume build (*full*). Metrics as in Table 12; precision is on a different sample of 40 assertions in each build. The prefix builds received a single LLM labeling pass; the full build's labels were finalized by Claude Opus 5.5 (Appendix B). The full build's tokens, for all five cutoffs at once, are in Table 23. Bold cyan cells mark the higher rate where the builds differ; bold magenta cells mark non-zero disclosure counts.

challenge set of 81 items created by minimal substitutions in source-verified facts, the verifier accepted 1 of 46 labeled unsupported candidates and rejected 0 of 35 labeled supported ones (Appendix K). This synthetic challenge does not establish performance on naturally occurring extraction errors.

Facts NARRATIVEWIKI misses. At $t = 5$, NARRATIVEWIKI misses 22 of the 111 gold facts; B1 or B2 conveys 18 of those omissions. Examples include Gilbert's teaching appointment at White Sands and Cooper Prize, Diana's role as improvement-society treasurer, Rilla's position as the youngest Blythe child, Leslie's mother's pressure to marry, Mr. Phillips's interest in Prissy Andrews, and Paul Irving's place of origin. Conversely, four facts are conveyed by NARRATIVEWIKI but neither B1 nor B2: Anne's fictional heroine, her mother's letters, Gilbert's boat rescue of Anne and Walter's hatred of a schoolmate. Four facts are conveyed by none of these three systems.

Typical errors. NARRATIVEWIKI still merges namesakes. Priscilla Grant's page lists the book-one schoolgirl "Prissy Andrews" as an alias. The `james-a` page joins Mr. Harrison of book two, who also has a separate page, with a deceased James of book three and a book-four sailor. A duplicate page for Anne, `shirley`, also absorbs her son Shirley at $t = 5$ ("the little brown boy").

Other errors include a remark by Mrs. Barry attributed to Marilla, an inverted relationship (Section 6.2), intervals persisting beyond their validity and an exhibition prize labeled "Fought in". Generated commentary about the input can reach pages: one subject "does not appear in the story provided", and 5 of the 341 non-timeline pages at $t = 4$ retain source-line keys ("[R3, R4, S1, S4]"; Table 33). Two first-build errors were not observed in the reported build: Anne's mother's death specified as

Stage	Calls	Input	Output
Entity discovery	2,771	1.39	0.14
Chapter pass	452	2.74	**0.97**
Mention pass	**2,932**	**14.04**	0.43
Long-context pass	980	7.10	0.11
Arbitration	64	0.02	0.00
Verification	2,816	6.21	0.04
Prose	953	0.89	0.15
Shared-page summaries	386	0.73	0.02
Total, reported runs	11,354	33.13	1.86
Rebuild from cache	0	0	0

Table 23: Size of the reported NARRATIVEWIKI build for all five cutoffs, including verification rechecks after one re-extraction: model calls, and input and output tokens in millions. Every call of the build is counted, whether or not the provider served it at a discount, so the token column of Table 11 is comparable across systems. A rebuild from the call cache sends nothing. Ingestion, assembly and linking make no model call. Bold violet cells identify the stage consuming the most resources in each column.

System	Original	Treated
B1	0/88	0/88
B2	0/88	0/88
B3	0/88	0/88
B4	0/88	2/88
B5 (NARRATIVEWIKI)	1/88	2/88
X1	3/88	0/88
X2	8/88	17/88

Table 24: Confirmed unique subject facts disclosed, over unique assessed subject facts. The unit is (character, claim, reveal volume): repeated cutoffs collapse, while reciprocal character facts remain distinct. These are not independent event or error counts.

"of yellow fever", beyond the text, and a merger of two men named George.

B1 and X2 also misattribute biographies: at $t = 5$, both place one man's courtship and marriage on another man's page. The closed-book diagnostic imports material outside the five-book corpus, describing Anne's years at "Windy Willows".

Uncertain disclosure verdicts. System-level uncertain counts at $t = 1..4$ are 3 (B1), 0 (B2), 1 (B3), 0 (B4), 4 (B5), 2 (X1), 6 (X2). For NARRATIVEWIKI, four verdicts arise from one relation line between Anne and Gilbert, "Romantically involved with — harbors romantic feelings for her". The interval opens in book one from a friend's report that Gilbert kept a rose Anne dropped. Assessed at $t = 1$ and $t = 2$ against two later facts about the couple, the line yields four uncertain judgments because it could suggest their book-four marriage.

	First	Reported
Recall	**0.855**	0.802
Precision, scene rule	0.751	**0.789**
Precision, line rule	0.604	**0.628**
True as statements	**0.949**	0.934
Citation sufficiency	0.112	**0.512**
With a paragraph citation (all)	0.198	**0.930**
Later facts disclosed	**0 of 220**	1 of 220
Lexical screen hits	4	**1**
Assertions per cutoff	20,823	13,943
distinct	8,808	6,970
Pair lines sharing an episode	76 of 116	**75 of 102**
Participants per scene record	9.0	5.9
Tokens, in / out (M)	33.14 / 1.81	33.13 / 1.86

Table 25: The two builds of NARRATIVEWIKI, means over cutoffs $t = 1..5$. Every relationship line of both samples was read by Claude Opus 5.5 under one rule; all other labels come from each build's own labeling round, the first build's before the revision and the reported build's after. Bold marks the better of the two builds where a direction exists; cyan emphasizes selected improvements, and magenta marks the observed disclosure.

Unique facts and error mechanisms. Table 24 collapses repeated fact–cutoff assessments to unique subject facts. On the original text, NARRATIVEWIKI's confirmed disclosure concerns one fact from one inverted-bridesmaid error. On the treated text, its four confirmed assessments cover two reciprocal marriage facts at two cutoffs, arising from a single prediction-to-marriage failure mechanism. B4's two treated assessments concern those same reciprocal facts. These witness-based descriptions do not establish that every output error has been found or that error mechanisms are independent. Grouping systems by information available to page generation includes B4 alongside B1–B3 and B5: together they have 1 of 1,100 original-text assessments, or 9 of 1,100 including uncertain verdicts.

The two builds and the pair rule. Table 25 compares the first and reported builds. The revision (Appendix D) lowered recall and raised citation sufficiency, merged two of the protagonist's three entities, removed three of four lexical-screen hits and introduced the implied disclosure. Precision is 0.751 and 0.789 under the main scene-sharing rule; 0.604 and 0.628 when the pair must participate in the line's own event; and 0.949 and 0.934 when assertions are assessed without page context. Rule choice changes the observed estimate more than

the revision does. At 40 atoms per cutoff, these estimates do not establish equal precision across builds.

Two earlier estimates are superseded because their judgment rules differ: 0.822, allowing both pair members anywhere in a chapter, and 0.748, an initial reported-build pass. The provisional and finalized passes agree on 90 of 99 pair verdicts. Eight of nine disagreements involve lines accepted in the finalized pass; four occur on Anne's duplicate page, whose identity the provisional pass rejects. Substituting the provisional pair verdicts while retaining all other labels yields precision 0.753.

The page-context error has a structural mechanism: scene records summarize chapters (median 38 paragraphs) and list their participants; pair pages print every line of every record naming both members. The revision requested participants rather than merely mentioned characters, reducing participants per record from 9.0 to 5.9 and implied pairs from 10,216 to 4,604. The share of sampled pair lines whose characters share an episode rises from 76 of 116 to 75 of 102. Episode-level records are a possible remedy that has not been tested.

H Where Accumulation Helps

Section 6.3 reports three situations in which NARRATIVEWIKI is ahead of the systems that rewrite each page per cutoff. This appendix gives the data behind them. Nothing here required a model call: all values come from the stored recall labels and the built wikis. On the treated text, characters are named by their original names.

Central characters. The three central characters are those with the most gold fact checks: Anne Shirley (81 of 335), Gilbert Blythe (39) and Diana Barry (21). They are selected by that count alone, which does not depend on any system's output; the decision to report recall by this split was made after the results were known. Table 26 gives the per-cutoff counts. On treated text NARRATIVEWIKI's recall on them rises from 86% at $t = 1$ to 95% at $t = 4$ and $t = 5$, while long context falls from 100% to 73–76% and prefix retrieval from 86% to 61–68%. On the original text the lead comes from the protagonist alone; on the treated text NARRATIVEWIKI leads on all three characters (Table 26).

Why pages diverge in length. Comparison systems write one page per character in one call under a shared template with sentence limits per section, and the retrieval systems pass at most 20,000 tokens of passages (Appendix E). Their pages therefore stay roughly the same length however much the story says about a character. NARRATIVEWIKI renders every eligible claim. Table 27 gives the length of Anne Shirley's page and of each wiki, in words with citations removed. Length is not monotone: Anne Shirley's NARRATIVEWIKI page has 3,719 words at $t = 4$ and 3,439 at $t = 5$.

		$t=1$ (14)	2 (21)	3 (31)	4 (37)	5 (38)	Pooled (141)
original	B1	11	**19**	25	30	33	118
	B2	**12**	15	22	29	32	110
	B3	3	13	12	19	20	67
	B4	**12**	15	22	29	28	106
	NARRATIVEWIKI	**12**	18	**27**	**34**	**35**	**126**
	X2	12	16	25	31	31	115
treated	B1	**14**	**19**	23	27	29	112
	B2	12	14	19	23	26	94
	B3	5	9	12	16	14	56
	B4	9	11	21	21	24	86
	NARRATIVEWIKI	12	**19**	**29**	**35**	**36**	**131**
	X2	12	14	16	26	27	95

Character	Text	B5	B1	B2	B3
Anne (81)	original	**78**	64	60	41
	treated	**72**	61	53	41
Gilbert (39)	original	31	33	**34**	12
	treated	**38**	31	30	8
Diana (21)	original	17	**21**	16	14
	treated	**21**	20	11	7

Table 26: Gold facts conveyed about the three central characters (Anne Shirley, Gilbert Blythe, Diana Barry). Top: at each cutoff, eligible fact checks in parentheses; bold, best of B1–B5 per column and text. Bottom: per character, pooled over $t = 1..5$ (fact checks in parentheses); B5 is NARRATIVEWIKI; bold, best per row.

Facts kept across cutoffs. A comparison system writes each cutoff's page afresh, so a fact present at t can be absent at $t + 1$. For NARRATIVEWIKI, every claim eligible at t is still eligible at $t + 1$; a value superseded by a later one is shown as a closed interval rather than removed. Table 28 counts the transitions. B4, which reads NARRATIVEWIKI's graph but writes each page in one call, loses 12 and 14, within the range of the prefix-bounded systems, which points to rendering rather than the graph as a source of the stability.

What is published. At $t = 1..5$ NARRATIVEWIKI publishes 59, 117, 158, 191 and 236 character pages, 34, 71, 111, 147 and 196 relationship pages (one per pair of characters), one timeline page per volume and three shared pages

Measure	System	$t=1$	$t=2$	$t=3$	$t=4$	$t=5$
Pages published	NARRATIVEWIKI	97	193	275	345	440
	B1, B2, B3	14	21	25	33	43
Words in the wiki	NARRATIVEWIKI	38,910	80,165	113,891	148,154	194,562
	B1	6,058	10,585	14,018	18,170	24,625
	B2	6,627	10,762	13,217	17,352	22,989
	B3	4,736	8,384	10,386	14,273	18,951
Words on Anne Shirley's page	NARRATIVEWIKI	1,813	2,363	2,688	3,719	3,439
	B1	864	852	913	984	1,157
	B2	699	553	708	711	688
	B3	620	731	653	554	410
Related characters on Anne Shirley's page	NARRATIVEWIKI	18	26	31	39	42
	B1	10	10	13	12	15
	B2	8	5	9	10	10
	B3	8	8	8	5	5

Table 27: Size of what each system publishes at each cutoff, original text. Words exclude front matter and citation blocks. NARRATIVEWIKI repeats a relation line on both characters' pages and on the pair's page, so its word count overstates distinct content; distinct assertions per cutoff (6,970 for NARRATIVEWIKI against 1,821, 1,718 and 1,404 for B1–B3) are the deduplicated measure: assertions identical after lower-casing and folding punctuation count once, so paraphrases, reciprocal phrasings and alias variants still count separately. Related characters: relationship blocks on the page, each linked to that character's page.

	Central three		Other 40		Facts lost	
System	orig.	treat.	orig.	treat.	orig.	treat.
B1	0.837	0.794	0.845	0.830	10	14
B2	0.780	0.667	**0.871**	**0.845**	4	24
B3	0.475	0.397	0.644	0.567	19	13
B4	0.752	0.610	0.758	0.799	12	14
B5	**0.894**	**0.929**	0.737	0.773	**1**	**1**
X2	0.816	0.674	0.804	0.789	7	11

Table 28: Where accumulation helps. Recall pooled over $t = 1..5$ on the three characters with the most gold assessments (Anne, Gilbert, Diana: 141 of 335) and on the other 40 (194), on original and treated text. Facts lost: gold facts conveyed at t and not at $t+1$, summed over $t = 1..4$ (lower is better). B5 is NARRATIVEWIKI. Cyan: best of B1–B5. Characters are split by their number of gold assessments (Appendix H).

for groups, places and objects. The comparison systems publish the character pages of the evaluated cast. Relations keep their history as intervals: at $t = 4$ the Anne–Gilbert block on Anne's page holds seven typed lines with paragraph citations (rival; romantically involved with; saved by; enemy, ended v1; comrade, since v2; friend, since v2; spouse, since v4), where long context prints one line, "Spouse of — Husband, former schoolmate and college rival". The cutoff-2 block lacks the spouse line, whose evidence is in book four (Appendix M).

	Input tokens	Pages	Input per page	Distinct per cutoff	Input per distinct
B5	33.13M	1,350	**24,541**	**6,970**	951
B1	69.23M	136	509,044	1,821	7,604
B2	**3.42M**	136	25,147	1,718	**398**
B3	5.08M	136	37,353	1,404	724

Table 29: Input tokens for all five cutoffs relative to what is published (B5: NARRATIVEWIKI; pages delivered over all cutoffs; distinct assertions as a mean per cutoff, and input per distinct assertion over their sum across the five cutoffs). Derived from Table 11 totals and the page counts of Table 27; not a separate run. Pages differ in length and kind across systems.

Cost for the amount published. One NARRATIVEWIKI build serves every cutoff, while long context resends the whole prefix for every page (Table 29). Per distinct assertion NARRATIVEWIKI uses one eighth of long context's input. Prefix retrieval remains the cheapest: its 9.7-fold advantage in total input tokens becomes 2.4-fold per distinct assertion and about equal per page.

Against LightRAG. On the 335 common fact checks, NARRATIVEWIKI alone conveys 94 and LightRAG alone 17 on the original text (both 175, neither 49); on the treated text 127 and 12. LightRAG remains ahead on precision, citation sufficiency and input tokens (Table 1).

Facts only NARRATIVEWIKI conveys. Gold facts that NARRATIVEWIKI conveys while B1 and

B2 both miss them number 7 (15) on the original text (facts, with fact checks in parentheses): Anne's braids and changeable eye color, her mother's old letters, her longing for puffed sleeves, her story heroine Averil Lester, Gilbert rowing to her when her flat sinks, Rachel Lynde's "red as carrots", and Walter's hatred of Dan Reese. On the treated text they number 13 (20): six of the above and, among others, Anne realizing she loves Gilbert when he falls ill, Gilbert giving up the Avonlea school for her, and his refused proposal. Most are early-book details about main characters that a later, budget-limited page no longer has room for. The reverse, facts B1 and B2 both convey and NARRATIVEWIKI misses, numbers 13 (31) on the original text and 10 (15) on the treated text.

The other 40 characters. NARRATIVEWIKI trails on the other 40 characters. Much of that deficit lies in characters about whom it conveys no gold fact at any cutoff on the original text (Mr. Phillips, 0 of 10 fact checks; Charlotta, 0 of 4; Reverend Jo, 0 of 3; Roy Gardner, 0 of 3). Facts lost across cutoffs compare labels at adjacent cutoffs, so a labeling inconsistency between them also counts as a loss.

I A Formal Account of the Trade-off

The results of Section 6 are consistent with one account: a claim, not its passage, is the generator's input. This appendix states three of its consequences precisely. The statements are elementary; their use is to separate differences that follow from the representation from those that depend on a particular build. They are not tested by ablation.

Notation. A stored claim is $c = (s, p, o, \mathrm{vol}(c), \mu(c))$: subject, predicate, object or value, the volume of its first evidence (`first_vol`) and a polarity $\mu(c) \in \{\textit{asserted}, \textit{presumed}\}$. The implemented schema stores no speaker and no finer modality (prediction, hypothesis, belief). For an entity e, let $I(e)$ be its verifier-accepted intervals and $I_t(e) = \{i \in I(e) : \mathrm{start}(i) \le t\}$ those eligible at cutoff t; the volume filter is $\phi_t(c) = \mathbf{1}[\mathrm{vol}(c) \le t]$. $G_t(e)$ denotes the gold facts about e eligible at t (Section 3).

Proposition 1 (stored claims are never evicted). For every entity e and cutoff $t < N$, $I_t(e) \subseteq I_{t+1}(e)$. Hence every structured line rendered for e at t is rendered at $t+1$, possibly with a closing volume added.

Proof. $\mathrm{start}(i) \le t$ implies $\mathrm{start}(i) \le t+1$. Structured fields are rendered deterministically from $I_t(e)$, and an interval that closes after t is shown as current at t and as a closed interval ("previously") once its end is reached, so it is never removed. □

Consequence. Under a fixed entity index, a gold fact conveyed by a structured line at t is conveyed at $t+1$. Losses can come only from prose regenerated per cutoff and from labelling differences between adjacent cutoffs. A page rewritten in one call per cutoff has no such guarantee: whatever a call omits is lost at that cutoff. NARRATIVEWIKI loses 1 conveyed fact per text (Table 28). The guarantee is conditional on the entity index: with an index rebuilt per cutoff (Section 6.5) a page's entity can change between cutoffs.

Proposition 2 (a volume filter is blind to modality). Let the renderer print (s, p, o) for every claim with $\phi_t(c) = 1$, whatever $\mu(c)$. If some claim has $\mathrm{vol}(c) \le t$, $\mu(c) \ne \textit{asserted}$, and (s, p, o) states a later gold fact $g \in F_t$, then the page at t discloses g.

Proof. $\phi_t(c) = 1$, so c is rendered, and the rendering states g. □

Consequence. Filtering by evidence volume is sufficient against disclosure only if every eligible non-asserted claim is either withheld from deterministic fields or rendered with its modality and speaker. On the treated text, a neighbour's prediction in book two was stored as a presumed marriage and printed as a spouse relation at $t = 2$ and $t = 3$ (4 of 220; Section 6.2). Prefix-bounded prompting keeps the passage, with speaker and mood, in the generator's input, and disclosed no assessed later fact.

Proposition 3 (routing by co-mention loses support). Let a scene record k with participants P_k produce a summary line a that is routed to every pair page $\{u, v\} \subseteq P_k$, and let a line be supported on a pair page only when u and v share an episode of the scene (the pair rule of Appendix B). If the episodes of the scene group the participants into disjoint sets $Q_1, \ldots, Q_m$, then of the $\binom{|P_k|}{2}$ pair pages that receive a, at most $\sum_j \binom{|Q_j|}{2}$ can support it.

Proof. A pair that shares an episode lies inside one Q_j. □

Consequence. A line can be true as a statement and unsupported on most of the pages it reaches,

and the unsupported share grows with the participants per record and the episodes per chapter. In the reported build, scene records list a mean of 5.9 participants (first build: 9.0); on relationship pages 0.971 (99/102) of sampled lines are true as statements but 0.706 (72/102) are supported in page context (Table 18). Statement truth and page support should therefore be scored separately (Appendix J).

J Reporting Recommendations

The audits behind this paper suggest six practices for evaluating cutoff-bounded or spoiler-sensitive generation over published narratives. Each is followed by the result of this paper that motivates it.

K Validation of Automatic Judges

Claude Opus 5.5 finalized the reported recall, precision, citation-sufficiency and disclosure labels after reading the source text (Section 5). We evaluated two types of automatic component: the verifier used by NARRATIVEWIKI and potential screens for evaluation.

The verifier inside NARRATIVEWIKI. Negatives were produced by minimal word substitutions in source-verified facts, following Jeong et al. (2025), each confirmed false against the text and stratified by type (entity, attribute, negation, reversed relation). On these 81 items the verifier reached a balanced accuracy of 0.989 (1 of 46 unsupported admitted, 0 of 35 supported rejected). A second, harder set of 12 items pairs a quoted lie, a hypothetical, and a character's embedded story with true counterparts from the same passages. The verifier's balanced accuracy on this set is 0.750; it admitted every claim drawn from the embedded story. These measurements concern the reported verifier. Its logged `APPEARANCE` call contains a definition error (Appendix D); no corrected verifier was evaluated.

Screens considered for evaluation. On the same minimal-edit set, MiniCheck (Tang et al., 2024) reached a balanced accuracy of 0.718, rejecting nearly every negative but also more than half of the true facts, and a local instruction model, Qwen2.5-14B-Instruct (Qwen Team, 2024) (4-bit, served by Ollama), reached 0.867. On the hard set they reached 0.500 and 0.667. Consistent with the finding of Subbiah et al. (2024) that entailment classifiers transfer poorly to narrative, neither screen assigns evaluation labels. The same local model splits prose into atomic assertions, and string search retrieves candidate disclosure sentences. Claude finalized the sampled-atom and candidate labels against the source text.

L Disclosure Casebook

Table 31 illustrates disclosure channels observed in the reported builds and the first build of NARRATIVEWIKI, including lexical-screen hits that did not disclose later facts. Table 32 documents defects found and corrected during development. The examples reveal plot points of the evaluated series.

M Qualitative Example: One Page at Two Cutoffs

Figures 10 and 11 show the reported NARRATIVEWIKI site for the Anne series. Readers select a cutoff before opening a wiki (Figure 10a), whose pages use claims with recorded evidence volumes at or below that cutoff. Gilbert Blythe's page is shown at cutoffs 2 and 4; volume 4 first establishes his marriage to Anne. The screenshots themselves were not annotated. Precision labels in Section 6 assess sampled atoms in the underlying pages.

The overview. Structured fields are rendered from graph rows with paragraph citations. At cutoff 2 (Figure 10b) Gilbert is listed as a teacher and holds the title "president" (since v2), and the page has no family line. At cutoff 4 (Figure 10c) the role is "country doctor (since v4)", the earlier roles are kept as closed intervals ("previously: teacher (v1–v2)"), the titles of his college years and "doctor" are added with the volumes that open them, and a family line names Anne Shirley as his spouse. These later roles, titles and the spouse field are absent from the cutoff-2 overview.

The relationship. The relationship block (Figure 11) contains the intervals illustrated in Figure 9. At cutoff 2 it lists rivalry, the enmity that ended in volume 1, comradeship and friendship, and its short prose ends with the two as close friends. At cutoff 4 the same block adds "Spouse of — husband and wife (since v4)", the other lines cite more paragraphs, and the prose, written from the cutoff-4 claims only, ends with the marriage. Each block's prose was generated independently from its cutoff's filtered claim set.

Recommendation	Practice	Why: evidence from this paper
R1 Pair every score with a paraphrase control	Report a closed-book floor and an aligned renamed-and-paraphrased condition beside any score on published fiction, with a recognition check of the treatment.	Renaming alone leaves the author identifiable in 0.625 of paragraphs; with paraphrase, closed-book recall falls from 0.683 to 0.003, and the treatment exposes X2's disclosures (8/88 to 17/88 unique later facts) and B1's citation decline (0.875 at t=1 to 0.300 at t=4).
R2 Add a single-call baseline on the same graph	Pair a multi-surface graph renderer with a baseline that writes pages in one call from the same graph and evidence.	B4, reading NARRATIVEWIKI's graph, reaches precision 0.965 and citation sufficiency 0.915 against NARRATIVEWIKI's 0.789 and 0.512: without it the gap would be attributed to extraction rather than to publication.
R3 Score statement truth and page support separately	Label sampled assertions both as standalone statements and in the page that prints them, and report both, by kind of page.	On relationship pages 0.971 (99/102) of NARRATIVEWIKI's sampled lines are true as statements but 0.706 (72/102) are supported on the page; page-context precision alone conflates misplacement with invention (Proposition 3).
R4 Audit construction scope against the prefix	Compare entities resolved over the whole series with indexes discovered from volumes 1..t at every cutoff before N.	Filtering claims leaves construction dependence intact: 0.390 of character pages at t=1 differ from a prefix-built index, 0.220 regrouped or absent (Table 21).
R5 Report retention across cutoffs, not only means	Count facts conveyed at t and lost at $t+1$, and stratify recall by how much the text says about a character.	Unweighted means (B1 0.848, NARRATIVEWIKI 0.802) hide that budgeted rewrites lose 4–24 conveyed facts across cutoffs while NARRATIVEWIKI loses 1 per text (Proposition 1).
R6 Validate the disclosure search and bound zero counts	Measure the candidate search on seeded paraphrases and pronoun references, and report character-clustered upper bounds with every zero.	The search retrieves 0.845 of named paraphrases but 0.309–0.355 when a pronoun on another character's page refers to the character (Table 17); a zero count at t=1 is compatible with a character-level rate up to 0.072 (0.221 at t=4).

Table 30: Six reporting recommendations for cutoff-bounded generation over published narratives, each with the result that motivates it.

Visible failure modes. The example contains four defects. (i) The line "Romantically involved with — harbors romantic feelings for her" is open from volume 1 at both cutoffs. Its relation label is stronger than its qualifier and evidence, producing NARRATIVEWIKI's uncertain disclosure verdicts (Section 6.2). (ii) The cutoff-4 prose prints the keys by which the generator names the lines it rests on ("[R3, R4, S1, S4]"). The renderer resolves those keys into the *Sources* line but fails to remove them from the prose; this occurs on 5 of the 341 non-timeline pages at cutoff 4. (iii) The age "nearly fourteen", stated in volume 1, still stands at cutoff 4 because an attribute with no later extracted value remains open. (iv) Gilbert's page carries a second block, "Shirley" (Figure 11c). The reported build merged the two entities the first build kept for "Anne" and "Anne Shirley", so the marriage is now on Anne Shirley's block and on her page; a third entity under her surname remains, and holds a rivalry line belonging to her. The screenshot shows the duplicate-entity failure discussed in Appendix G.

The same page as a structured comparison. Table 33 transcribes the fields of Figures 10 and 11 side by side, separating the intended interval behaviour from the four defects.

N Contamination Diagnostics and the Treated Text

A generator may recognize published novels (Chang et al., 2023). We compare aligned original, names-only and renamed-and-paraphrased text, assessing recognition separately from later-fact disclosure and cutoff compliance. Renaming follows prior practice (Gupta, 2025; Zhang et al., 2024); cross-condition differences motivate contamination diagnostics (Dekoninck et al., 2024). Because a treatment can damage the narrative (Sun et al., 2025), we examine gold-fact fidelity and document errors. The procedure aims to preserve facts, attribution, reveal order and cross-volume coreference, but has no independent validation of complete fidelity. Reduced recognition does not rule out memorization (Cheng et al., 2025).

System	t	Surface	Delivered text	Source and reveal volume	Verdict
X2	2	character page (Priscilla Grant)	"She entered Redmond College in Kingsport, taking up the Second Year work alongside Anne."	Anne studies at Redmond: `v03:c01:p0013`, reveal 3	stated; the later text was in the input with an instruction not to use it
X2	2	character page (Jane Andrews)	"Later, Jane accepted a teaching position in Winnipeg, West Canada, where she became engaged to a wealthy English millionaire."	Jane leaves for the West: `v03:c14:p0030`, reveal 3	stated
X2	4	character page (Susan Baker)	"Role: Housekeeper at Ingleside"	Anne lives at Ingleside: `v05:c01:p0001`, reveal 5	stated
X1	1	character page (Billy Andrews)	"Too bashful to propose to her himself, he sends his sister Jane to propose marriage to Anne on his behalf."	`v03:c08:p0012`, reveal 3	stated; no text in the input, so the source is the generator's memory
X1	4	character page (Little Jem)	"When Gilbert's medical practice expands, the Blythe family moves to Ingleside in Glen St. Mary."	`v05:c01:p0001`, reveal 5	stated (memory)
NARRATIVEWIKI	3	relation summary on a character page (Phil Gordon)	"Phil later serves as a bridesmaid for Anne."	Anne marries Gilbert: `v04:c04:p0000`, reveal 4	implied: a reader infers Anne's wedding, and by the end of book three she is engaged to Gilbert (`v03:c41:p0036`). Written from eligible text and wrong: in `v03:c39:p0039` Anne is Phil's bridesmaid
NARRATIVEWIKI	1, 2	relation line on two character pages and a pair page	"Anne Shirley — Romantically involved with — Gilbert Blythe — harbors romantic feelings for her · since v1", citing `v01:c25:p0055`	Anne marries Gilbert: `v04:c04:p0000`, reveal 4	uncertain: in the cited passage a friend reports that Gilbert kept a rose Anne dropped; the label reads towards the later romance
B1	2	character page (Rachel Lynde)	"…allowing Anne to attend Redmond College"	decision in book two, attendance `v03:c01:p0013`	uncertain
B4†	2	quote on a character page (Ruby Gillis)	"Oh, she does look really dead." (`v01:c28:p0013`, said of Anne playing a corpse)	Ruby Gillis dies: `v03:c14:p0033`, reveal 3	not a disclosure; an entailment screen scored it 0.52
B1–B4	3	title of a character page	"Reverend Jo", the name under which the evaluated cast lists the character	book three has "the Rev. Jo" (`v03:c39:p0039`); spelled out first in book four	expanded name form supplied by the evaluation's page list; no fact revealed
NARRATIVEWIKI †	2	existence and title of a character page (`walter`)	a page "Walter" describing Anne's father, wrongly	the entity is Walter Blythe of book five (127 of its 130 mentions)	later-anchored page; no fact revealed; absent from the prefix build and corrected in the reported build
NARRATIVEWIKI	2	existence of a character page (`susan`)	an empty page titled "Susan"	the entity is Susan Baker of books four and five (3 of its 204 mentions by book two)	later-anchored page; no fact revealed
NARRATIVEWIKI	1	codex page (groups)	"Conservatives"	book one has the singular	plural form; lexical screen hit, no fact revealed

Table 31: Cases from the reported builds and the first build of NARRATIVEWIKI, by disclosure channel. Each row gives the delivered text, the source and reveal volume of the fact concerned, and the verdict. †: the first build of NARRATIVEWIKI, or B4 on its graph.

System	t	Surface	Delivered text	Cause or correction	Verdict
baselines	1	list of pages and their titles	pages for eleven characters whose names do not occur in book one	the page-selection rule matched any single word of a name	disclosure by page title; rule changed to "a gold fact by t or the full name in $D_{\leq t}$"
NARRATIVEWIKI	3	page title and address	a page for Leslie Moore, who first appears in book four	a book-three mention "Will Leslie" was resolved to her	disclosure by title; resolution fixed
NARRATIVEWIKI	1	relation qualifier	a qualifier written from book-two evidence on a relation opened in book one	qualifier now taken from evidence no later than the interval's first volume	premature detail; fixed
NARRATIVEWIKI	all	shared navigation	names of later characters in a menu shared by every cutoff	the menu listed all pages of the site	disclosure by menu; shared navigation now names no character

Table 32: Development defects and the corrections applied before generating the reported evaluation cells.

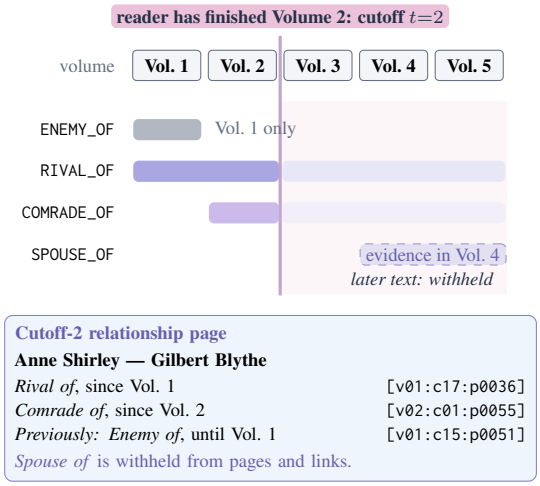


Figure 9: Reading cutoffs for one character pair. Relation intervals open at their first-evidence volume. The cutoff-2 wiki renders intervals opened by volume 2 and withholds the marriage evidenced in volume 4. Citations use `v<volume>:c<chapter>:p<paragraph>`.

Treatment. The treatment operates paragraph by paragraph, retaining identifiers, chapter boundaries and scene breaks so citations remain aligned.

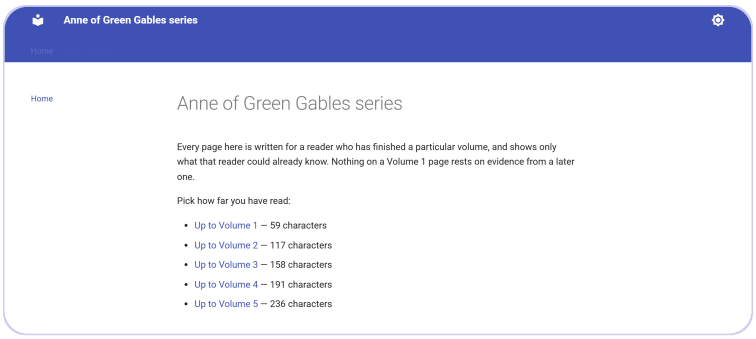
Step 1, names. A reviewed map of 239 entries replaces character, family and local place names with invented forms. Longer forms precede shorter ones in whole-word replacement, with consistent full names, surnames and nicknames. Mechanical checks require stand-ins to be absent from the source names and every part of a mapped full name to be mapped. An initial map failed both checks after one volume: a main character's stand-in matched two minor characters' first names, and a mapped full name left its 148 bare-first-name occurrences unchanged. We rebuilt the map and discarded that treated volume. Recurring minor characters and local places are renamed; countries and distant cities remain. A surname also used as a compass direction is replaced only in person-name contexts.

Step 2, paraphrase. The generator rewrites each renamed paragraph under instructions to preserve facts, events, numbers, places, ordering, names and dialogue speakers, with no additions or summarization. Reapplying the map replaces any restored source names. Nonempty rewrites between half and twice the source length are accepted; otherwise the renamed paragraph is retained and counted as unparaphrased. Residual source names are counted separately. In an initial treated volume, approximately one paragraph in 25, containing one word in 12, retained at least four fifths of its four-word sequences. We therefore request one retry when at least half the source's four-word sequences remain. A second near-copy is retained but counted as verbatim. Failed model calls stop the run, preventing silent fallback.

Step 3, gold. The same map transforms the original gold facts and cast; facts are neither added nor removed, and reveal volumes are unchanged. All systems in Section 5 are rebuilt with unchanged code, configuration and prompts. A names-only variant omits Step 2 and is used for recognition probes, rather than complete system rebuilding.

What the condition can and cannot show. If systems mainly use supplied evidence, much of their original-text coverage may survive the treatment. Reduced recognition may instead reduce closed-book coverage, where no source text is supplied. These are diagnostic expectations, not an identification strategy for each system's memory contribution: rewriting also changes wording, entity resolution and sometimes meaning.

A prefix-bounded system can disclose later facts



(a) Home page: the reader picks a cutoff before seeing anything.

Cutoff 2

Overview

- **Age:** nearly fourteen v1 CHAPTER XV. A Tempest in the School Teapot P24
- **Gender:** male v1 CHAPTER XV. A Tempest in the School Teapot P28, v2 XIX Just a Happy Day P51, v2 XXIV A Prophet in His Own Country P4, v2 XXX A Wedding at the Stone House P37
- **Role:** teacher v1 CHAPTER XXXVI. The Glory and the Dream P24, v1 CHAPTER XXXVII. The Reaper Whose Name Is Death P35, v1 CHAPTER XXXVIII. The Bend in the Road P38, v2 I An Irate Neighbor P55, v2 IV Different Opinions P0, v2 IV Different Opinions P12, +3 more
- **Status:** alive (since v2) v2 XXV An Avonlea Scandal P70
- **Titles:** president (since v2) v2 VI All Sorts and Conditions of Men . . . and women P8

(b) Gilbert Blythe, overview, cutoff 2.

Cutoff 4

Overview

- **Age:** nearly fourteen v1 CHAPTER XV. A Tempest in the School Teapot P24
- **Gender:** male v1 CHAPTER XV. A Tempest in the School Teapot P28, v2 XIX Just a Happy Day P51, v2 XXIV A Prophet in His Own Country P4, v2 XXX A Wedding at the Stone House P37, v3 ch.Change P27, v3 ch.Autumn P7, +4 more
- **Role:** country doctor (since v4) v4 CHAPTER 14 NOVEMBER DAYS P8
  previously: teacher (v1–v2) v1 CHAPTER XXXVI. The Glory and the Dream P24, v1 CHAPTER XXXVII. The Reaper Whose Name Is Death P35, v1 CHAPTER XXXVIII. The Bend in the Road P38, v2 I An Irate Neighbor P55, v2 IV Different Opinions P0, v2 IV Different Opinions P12, +3 more
  previously: Daily News employee (v3) v3 ch.Speaks P23
- **Origin:** Island (since v4) v4 CHAPTER 2 THE HOUSE OF DREAMS P7
- **Status:** alive (since v2) v2 XXV An Avonlea Scandal P70, v3 ch.Revelation P25
- **Titles:** president (since v2) v2 VI All Sorts and Conditions of Men . . . and women P8; Captain of the Freshman football team (since v3) v3 ch.Park P6; president of the Freshman Class (since v3) v3 ch.Home P0; doctor (since v4) v4 CHAPTER 14 NOVEMBER DAYS P3
- **Family:** Anne Shirley (spouse)

(c) The same section, cutoff 4.

Figure 10: The delivered site at two cutoffs (screenshots of the NARRATIVEWIKI wiki, Anne series). Fields carry paragraph citations; at cutoff 4 the role, title and family lines open intervals evidenced in volumes 3 and 4 and keep the earlier roles as closed intervals.

Cutoff 2

Anne Shirley · the relationship

Gilbert and Anne began as bitter enemies after he teased her on his return to school, causing Anne to hold a long-standing grudge despite his attempts to apologize and even saving her when her boat leaked. They were fierce academic rivals competing for top marks in school, but Anne eventually forgave him. By the second volume, they have become close friends who write newspaper notes together as literary collaborators, with Gilbert harboring romantic feelings for her.

Sources: v1 CHAPTER XV. A Tempest in the School Teapot P16, v1 CHAPTER XV. A Tempest in the School Teapot P33, v1 CHAPTER XV. A Tempest in the School Teapot P38, +34 more

- Rival of — academic rivals competing for top marks in school v1 CHAPTER XVII. A New Interest in Life P36, v1 CHAPTER XIX. A Concert, a Catastrophe, and a Confession P32, v1 CHAPTER XX. A Good Imagination Gone Wrong P4, +6 more
- Romantically involved with — harbors romantic feelings for her v1 CHAPTER XXV. Matthew Insists on Puffed Sleeves P55, v2 XIX Just a Happy Day P51, v2 XXVIII The Prince Comes Back to the Enchanted Palace P5
- Saved v1 CHAPTER XXVIII. An Unfortunate Lily Maid P26
- Enemy of — persistent grudge and refusal to speak after insult *(ended v1)* v1 CHAPTER XV. A Tempest in the School Teapot P47, v1 CHAPTER XV. A Tempest in the School Teapot P51, v1 CHAPTER XV. A Tempest in the School Teapot P93, +1 more
- Comrade of — literary collaborators writing newspaper notes together (since v2) v2 I An Irate Neighbor P55, v2 VI All Sorts and Conditions of Men . . . and women P8, v2 XV The Beginning of Vacation P40, +2 more
- Friend of — close friend (since v2) v2 VII The Pointing of Duty P1, v2 XIX Just a Happy Day P51, v2 XXVI Around the Bend P46, +1 more

(a) Gilbert Blythe, relationship with Anne Shirley, cutoff 2.

Cutoff 4

Anne Shirley · the relationship

Gilbert and Anne began as bitter enemies in school after Gilbert teased her, prompting Anne to smash a slate over his head and hold a persistent grudge despite his attempts to apologize and save her from a leaking boat [R3, R4, S1, S4]. They gradually transitioned into intense academic rivals, literary collaborators, and close friends as Anne eventually forgave him [R1, R5, R6, S5]. Though Anne initially rejected his marriage proposal, their relationship deepened until they ultimately married, settling down together as husband and wife [R2, R7, S41, S67].

Sources: v1 CHAPTER XV. A Tempest in the School Teapot P16, v1 CHAPTER XV. A Tempest in the School Teapot P33, v1 CHAPTER XV. A Tempest in the School Teapot P38, +80 more

- Rival of — academic rivals competing for top marks in school v1 CHAPTER XVII. A New Interest in Life P36, v1 CHAPTER XIX. A Concert, a Catastrophe, and a Confession P32, v1 CHAPTER XX. A Good Imagination Gone Wrong P4, +7 more
- Romantically involved with — harbors romantic feelings for her v1 CHAPTER XXV. Matthew Insists on Puffed Sleeves P55, v2 XIX Just a Happy Day P51, v2 XXVIII The Prince Comes Back to the Enchanted Palace P5, +10 more
- Saved v1 CHAPTER XXVIII. An Unfortunate Lily Maid P26
- Enemy of — persistent grudge and refusal to speak after insult *(ended v1)* v1 CHAPTER XV. A Tempest in the School Teapot P47, v1 CHAPTER XV. A Tempest in the School Teapot P51, v1 CHAPTER XV. A Tempest in the School Teapot P93, +1 more
- Comrade of — literary collaborators writing newspaper notes together (since v2) v2 I An Irate Neighbor P55, v2 VI All Sorts and Conditions of Men . . . and women P8, v2 XV The Beginning of Vacation P40, +5 more
- Friend of — close friend (since v2) v2 VII The Pointing of Duty P1, v2 XIX Just a Happy Day P51, v2 XXVI Around the Bend P46, +7 more
- Spouse of — husband and wife (since v4) v4 CHAPTER 1 IN THE GARRET OF GREEN GABLES P5, v4 CHAPTER 2 THE HOUSE OF DREAMS P7, v4 CHAPTER 3 THE LAND OF DREAMS AMONG P29, +26 more

(b) The same block, cutoff 4: “Spouse of” opens in volume 4.

Shirley · the relationship

Gilbert and Shirley maintained a fierce academic rivalry during their time as school classmates [R10, S13]. Their intense competition for top marks continued up through their final examinations at school [R10, S13, S14].

Sources: v1 CHAPTER XVII. A New Interest in Life P0, v1 CHAPTER XVII. A New Interest in Life P16, v1 CHAPTER XVII. A New Interest in Life P28, +5 more

- Rival of — academic rivals in schoolwork v1 CHAPTER XVII. A New Interest in Life P36

(c) The block “Shirley” on the same page at cutoff 4: a second entity for Anne, under her surname, holding one of her relations.

Figure 11: One relationship at two cutoffs. Each block’s prose is generated from its cutoff’s eligible claims; the cutoff-2 text therefore contains no marriage statement. The bracketed keys in (b) and the block in (c) are defects of the reported build.

Field	Cutoff 2 (*Anne of Avonlea*)	Cutoff 4 (*Anne's House of Dreams*)	Mechanism
Intended behaviour			
Role	teacher [v1 ch. XXXVI P24, v2 ch. I P55, +7]	country doctor (since v4) [v4 ch. 14 P8]; *previously:* teacher (v1–v2); Daily News employee (v3)	Supersession: an earlier value becomes a closed interval when a later one opens.
Titles	president (since v2) [v2 ch. VI P8]	adds Captain of the Freshman football team (v3), president of the Freshman Class (v3), doctor (v4)	Accumulation: volume-dated titles are appended; none is evicted.
Family	(no line)	Anne Shirley (spouse)	Withholding: no family claim has evidence by volume 2.
Anne–Gilbert block	rival; romantically involved with; saved; enemy (ended v1); comrade (since v2); friend (since v2)	the same six lines, with more citations, and *Spouse of* — husband and wife (since v4) [v4 ch. 1 P5, ch. 2 P7, ch. 3 P29, +26]	History is kept as intervals; long context prints one line ("Spouse of — husband, former schoolmate and college rival").
Block prose	ends with the two as close friends	ends with the marriage	Each cutoff's prose is generated from its own filtered claims.
Defects of the reported build			
(i) Relation label	*Romantically involved with* — harbors romantic feelings for her, open from v1 [v1 ch. XXV P55]	unchanged	The label is stronger than its qualifier and evidence: NARRATIVEWIKI's uncertain disclosure verdicts (Section 6.2).
(ii) Line keys in prose	none	prose prints "[R3, R4, S1, S4]" and "[R2, R7, S41, S67]"	The renderer resolves keys into the *Sources* line but leaves them in the prose: 5 of 341 non-timeline pages at $t{=}4$.
(iii) Age	nearly fourteen [v1 ch. XV P24]	nearly fourteen, beside the marriage	An attribute with no later extracted value stays open; elapsed years are not inferred.
(iv) Second entity	—	a block "Shirley" holds *Rival of* [v1 ch. XVII P36]	The build merged two of three Anne entities; the bare surname remains separate (Appendix G).

Table 33: Gilbert Blythe's page in the reported NARRATIVEWIKI site at cutoffs 2 and 4, transcribed from the delivered pages (screenshots are released with the code). Citations are printed as the site prints them (volume, chapter, paragraph; "+n": further citations). Cyan: intended interval behaviour; magenta: defects.

through extraction or inference errors, generation or residual recognition. A treated-text disclosure by the full-text diagnostic still demonstrates a failure to obey the cutoff while later evidence is available. A model can recognize a plot without its original names, so the treatment cannot establish the absence of memorization.

The treated text. The treated series contains 8,489 paragraphs and 428,122 and 467,246 words before and after treatment. Of these paragraphs, 8,468 (99.8%) were paraphrased, with the remainder renamed only; 1 remained near-copies after retrying. Paraphrase lengthens the series overall, although each accepted rewrite stays within the length bounds of Step 2. Whole-name checks find no mapped source name unchanged. A broader search finds 27 inflected or derived source-name occurrences in 26 paragraphs: chiefly family-name plurals ("the Gardners"), plus a college-derived adjective and a sentence explaining a name's origin. Recognition probes include these residual forms.

Some residuals reach generated pages. Across 1,597 character pages from seven systems, the name explained in the source sentence appears in 25 pages of six systems. A family name retained in a plural and a chapter title ("The Gardners' Call") appears on 9 pages of four systems. Three other family names and the college name, each from a residual form, appear on seven pages. A further closed-book match is a common surname in an invented story.

Another source name has no literal residual: Six pages from prefix retrieval and the full-text diagnostic print "Prince Edward Island", absent from the treated paragraphs and chapter titles. The renamed island's neighboring provinces remain, permitting geographic inference. This output therefore does not distinguish inference from residual recognition.

Fidelity. An unblinded LLM annotator checked all 111 gold facts against their treated evidence

Text condition	Title	Author
Original	0.820	0.865
Names remapped	0.325	0.625
Remapped and paraphrased	0.110	0.195

Table 34: Recognition ablation on 200 aligned paragraphs. Values are the share for which the frozen generator identifies an Anne title or its author. Renaming and paraphrasing both reduce recognition, while residual recognition remains. Cyan highlights the combined treatment. This probe does not establish that a wiki fact was memorized.

paragraphs. It judged all 111 still expressed, one with altered wording: a sleeve style is renamed in its evidence paragraph but given its source name in the next paragraph. This asks whether the treated passage retains the fact. The stricter paragraph-alone audit from Appendix B, yielding 85, 23 and 3 for the original text, was not repeated.

Errors also occur outside gold evidence. One paraphrase changes two children's ages into an interval of minutes, after which prefix retrieval reports twins born six minutes apart. The statement counts as supported against the supplied treated text. Treatment errors found during analysis are documented here and left in place.

Recognition. On 200 aligned paragraphs spanning five volumes, the generator restores a masked character name with accuracy 0.165 on original text and 0.000 on treated text. Title identification is 0.820, 0.325 and 0.110 for original, names-only and combined treatment, respectively; author identification is 0.865, 0.625 and 0.195. Renaming reduces title recognition more than author recognition. Adding paraphrase reduces both further, to approximately one third of their names-only levels, but roughly one treated paragraph in nine is still identified as belonging to the series. Only recognition was probed for names-only text; complete systems were rebuilt on original and combined-treatment text.

Systems on the treated text. Table 1 compares original and treated systems; Table 13 reports every cutoff. Gold facts, reveal volumes and labeling rules are transferred from the original; inverse name mapping allows the same per-fact tests. Claude Opus 5.5 finalized the AI labels for all systems on both texts, including precision and citation sufficiency.

Each condition has one build. Differences in entity resolution and paraphrase fidelity preclude interpreting small cross-condition changes as isolated memory effects. The closed-book diagnostic has a much larger recall reduction, from 0.683 to 0.003, with 1 of 335 matching fact–cutoff assessments remaining: a death also present in its invented story. This is consistent with recognition contributing substantially to original-text closed-book recall.

What the closed-book generator writes. Given stand-in names without text, the generator still writes full pages. Of 200 sampled assertions, 162 are unsupported. The stand-in for a shy book-one farmer becomes a sea captain founding a cult; a heroine's friend becomes a duelist at a magical academy; later cutoffs introduce a street gang and a starship. Supported judgments concern generic statements about gender, status, siblings or traits. No assertion carries a citation.

The disclosure by NARRATIVEWIKI. At $t = 2$ and $t = 3$, NARRATIVEWIKI lists both leads as spouses, a book-four fact (4 of 220, counting the marriage for both characters at both cutoffs). These are two reciprocal subject facts from one prediction-to-marriage failure mechanism, rather than four independent errors (Table 24). Another 4 verdicts are uncertain, involving the same courtship line as in the original. The marriage citations point to a book-two neighbor's prediction. Extraction stored a presumed marriage at volume two, and rendering turned presumed spouses into spouses. B4, using the same graph, prints "anticipated future marriage" at $t = 2$, labeled implied disclosure (2 of 220).

The original build did not extract this relation from the corresponding passage. The treated passage is cutoff-eligible, but the claim loses the distinction between a character's prediction and an established event. Volume filtering cannot repair that attribution and modality error.

Long-context prompting on the treated text. Mean recall is 0.845 on treated text versus 0.848 on original text; precision is 0.920 versus 0.929. Both treated estimates are lowest at $t = 4$ (Table 13). The close means do not establish equivalence from single builds.

Citation sufficiency is 0.875 and 0.675 for one and two volumes, then falls to 0.425, 0.300 and 0.375, versus 0.700, 0.590 and 0.550 on the original. Most insufficient citations accompany supported assertions. Many retain the paragraph num-

ber but lose the volume or chapter, pointing, for example, from a volume-one fact to the corresponding identifier in volume three. Such identifiers pass the harness's existence check.

The cause is untested. Familiarity with the original might help locate evidence, or paraphrased passages might be harder to distinguish. Neither explanation is isolated by this comparison.

The full-text diagnostic on the treated text. Despite the cutoff instruction, X2 discloses 23 of 220 assessed later facts on treated text, versus 8 of 220 on the original. Per-cutoff counts at $t = 1..4$ are 8/88, 8/65, 6/45 and 1/22; original counts are 0/88, 6/65, 0/45 and 2/22. At $t = 1$, the guardian's page describes the neighbor's widowhood and move, the heroine's college attendance and her marriage "years later". It cites volumes two to four, whose identifiers the harness accepts because all volumes are supplied. Reduced recognition therefore does not make the instruction reliable; the treated-text excess is largest at the first cutoff (8/88 against 0/88 on the original). Familiarity with the original's book boundaries is one untested explanation.

Treated recall is lower at every cutoff (0.744 versus 0.814); mean precision is also lower (0.845 versus 0.910). Most unsupported statements are misattributed biographies, and citation errors also omit volumes. The heroine's $t = 1$ page returned malformed JSON three times, closing a list with the wrong bracket. We corrected that bracket in the recorded first response without another model call (Appendix P).

Namesakes made by the name map. Whole-word renaming creates two collisions observed in the systems receiving the most text, B1 and X2. First, one stand-in replaces a shared first name across characters, making a book-one minor character a namesake of a schoolmate; passages about one are then cited for the other. Second, a book-two twin's stand-in is a short form of the stand-in for the heroine's romantic name, also borne by her friend's daughter in book four. With book four in context, the twin's page describes the friend (B1 at $t = 4$; X2 at $t = 2$, 4 and 5) or the heroine (X2 at $t = 3$), whereas original-text builds describe the twin.

This page contributes unsupported assertions and two stated X2 disclosures at $t = 3$. Neither mechanical map check detects these collisions. They are treatment defects retained in the reported system scores.

O Development Observations

Choice of generator. We sent stored claim-extraction prompts from a commercial development series to three Gemini Flash models, then checked the outputs of two of them, 3.1 Flash-Lite and 3.6 Flash, against their cited paragraphs. No numerical results from that series are reported (Appendix C). Qualitatively, Gemini 3.1 Flash-Lite returned more facts than Gemini 3.6 Flash, with a lower share fully supported and some unsupported claims; no unsupported claims were observed in the reviewed 3.6 Flash outputs. We selected 3.6 Flash for all stages and comparison systems. Model choice also affects entity discovery: from the same volume-one Anne candidate list, Flash-Lite retained 83 (50 characters) entities and 3.6 Flash retained 92 (54 characters). This illustrates an effect on the entity inventory before page generation.

Stories within the story. Characters tell stories, invent companions and make false confessions. Entity discovery admitted figures from embedded stories, including a story-club heroine and a child's imagined cave lady, and verification accepted claims about them (Appendix K). Local evidence can therefore be insufficient to distinguish the main narrative from an embedded or imagined account.

Reveal volumes need an earliest-statement check. Disclosure assessment found 2 of 90 draft facts assigned reveal volumes later than their earliest supporting statements. Pages containing these facts were consequently eligible. Reveal-volume annotation requires searching for the earliest support, rather than retaining the first passage encountered by the annotator.

Implicit temporal progression. Gilbert's age, *nearly fourteen*, is extracted from volume one and remains on his cutoff-4 page alongside the later marriage (Table 33). No extracted age supersedes it, so its interval stays open. The graph records explicit changes; it does not advance ages with narrative time. Reveal volumes order evidence rather than elapsed story years. A historical age should therefore be marked as an observation at a particular story time unless a supported update is available.

Cross-claim inference. Some updates require combining claims rather than extracting one ex-

plicit statement. As an illustrative example, suppose Anne is two years younger than Gilbert and marries him at age 25. The premises imply that Gilbert is 27 at their marriage. These ages are hypothetical, not verified annotations of the displayed example. The current pipeline does not systematically derive such updates. An extension would need to retain the supporting premises, align their narrative times and expose a derived claim only after all required premises become available at the reader's cutoff.

Coreference across extraction windows. Pronouns and other referring expressions can link a character or fact across sentences and scenes without repeating a name. Extraction may miss these links or attribute a claim to the wrong person when the relevant context lies outside its window. Both backward and forward references require discourse context, while a later identity revelation must remain unavailable at an earlier cutoff. These are qualitative failure modes; no error rate for them is measured here.

P Reproducibility and Release

Freeze. Before each block of runs, the source code, configuration, and scripts are hashed into a freeze record that every runner checks before making a call. Any change requires a new record and an account of which results it invalidates. Changes affecting a reported number are logged with the previous and revised values and the cells rebuilt or relabeled, including the two label audits in Section 5. Sampling seeds are fixed (assertion samples: seed 0).

Provenance. A shared client caches every model response under a hash of its provider, model, prompt, and parameters; each run writes a manifest and call log. Every number in the paper maps to a measurement-ledger row identifying the producing command and run. All generative calls of the systems and of the treatment used Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0), between 28 September and 7 October 2026. The provider's model name exposes no finer revision. Cached responses reproduce the released artifacts exactly; fresh calls need not produce identical generations. On the treated text, one page of the full-text diagnostic returned a list with an incorrect closing bracket in three attempts. The runner corrected the bracket in the recorded first response and applied its normal validation and rendering; no page was regenerated. Appendix Q presents one logged exchange for each generative stage, including the system prompt, input, and output. Claude Opus 5.5 finalized all reported AI annotation labels after evidence review. The prompt and response defects documented in Appendix Q remain part of the archived run configuration. The reported results are not measurements of a repaired prompt builder or citation validator.

Commands. With the corpus parsed, the reported artifacts are produced by:

- the reported NARRATIVEWIKI build of all five cutoffs:
  `run_series.py anne@v2 --upto 5` (the first build is the series `anne`);
- a comparison cell at cutoff t:
  `run_baseline.py anne {X1|B1|B2|B4|X2} --upto t`, with `--chunk-words 150 --budget-tokens 20000` for B2 and `--graph anne@v2 --budget-tokens 20000` for B4, and `b3_lightrag.py anne --upto t --mode mix`;
- the assertion inventory and sample of a cell:
  `assertion_inventory.py S --upto t` and `assertion_precision.py S --upto t --n 40 --population all`;
- disclosure candidates:
  `m4_tree.py S --upto t`;
- the lexical screen:
  `artifact_exposure.py S`;
- the pair rule and the timeline citations applied to a cell's labels:
  `apply_pair_rule.py S B5`;
- table values:
  `paper_numbers.py`; their spread over cutoffs:
  `summary_stats.py`;
- cost of the cells on disk:
  `run_cost.py S --final`.

Release. An anonymized archive provides the submission materials; the repository address is https://anonymous.4open.science/r/NarrativeWiki-Anon-341E/. It contains the code, configuration, dependency specification and freeze records; prompt templates and logged exchange excerpts; gold facts with evidence identifiers and reveal volumes; and AI evaluation label files (recall, annotated precision samples, citation sufficiency, disclosure verdicts with the

sentences assessed, and audit corrections). It also includes per-run cost summaries, every system's generated wikis at every cutoff, both builds of NARRATIVEWIKI, and the human-review package (554 items with source context: every gold fact and samples of the recall, precision and citation labels), plus the two reviewers' label files, accompanying records, provenance manifest, aligned decision audit and recomputed agreement. The reviewer labels are entirely human-produced. The source texts are public-domain and are fetched from Project Gutenberg by identifier.

Q Prompts and Model Exchanges

Every generative stage calls the generator through one client that records the system prompt, the user prompt and the raw response. Table 35 lists the stages whose first successful call from the run behind the reported results is exhibited in the release; all used Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). It first lists the five defects that the logged prompts and responses carry; the complete logged exchange of every stage follows (Appendix Q.2). Every call log is released (Appendix P); the excerpts are verbatim, with elisions marked. Evaluation labels were not produced by these stages (Appendix B).

Stage	Role	Tokens in / out	Run
NARRATIVEWIKI *build*			
Entity typing	Types each mined name candidate (gazetteer) (*"Anne", v1*)	518 / 39	`20261004T193719-gazetteer-v1-5`
Alias adjudication	Decides whether two surface forms are one entity (gazetteer) (*Charlie vs. Carrie Sloane*)	548 / 50	`20261004T193719-gazetteer-v1-5`
Claim extraction	Extracts dated, quoted claims about one character from one evidence window (*Anne Shirley, v5*)	8,884 / 734	`20261005T070204-extract-v5-5`
Scene extraction	Records scenes, participants and events over a chapter span (*v5, chapter 2*)	5,729 / 202	`20261004T195945-scenes-v5-5`
Infobox pass	Long-context pass for gender, age and origin (*Anne Shirley, v1*)	93,403 / 490	`20261005T070216-infobox-v1-5`
Backstory pass	Long-context pass for background facts (*Marilla Cuthbert, v1*)	135,267 / 309	`20261005T070229-backstory-v1-5`
Arbitration	Resolves a same-volume conflict between two values (*Anne Shirley, AGE, v1*)	258 / 55	`20261005T070243-graph_build-v1-5`
Verification	Checks a stored fact against its own cited evidence (*Anne Shirley, APPEARANCE*)	2,833 / 5	`20261005T053635-verify-upto5`
NARRATIVEWIKI *publication*			
Page prose (appearance)	Writes a physical description from the cutoff-filtered claims (*Anne Shirley,* t=1)	1,045 / 174	`20261005T070347-synthesize-upto5`
Codex summary	Writes the one-sentence summary of a shared codex entry (*Aid Society,* t=1)	1,974 / 69	`20261005T070527-site_build-upto5`
Comparison systems and diagnostics			
B1 page	Long context: one character page from the whole prefix (*Anne Shirley,* t=4)	559,824 / 5,688	`20261001T043209-baseline-b1-v4`
B2 page	Prefix retrieval: one character page from retrieved passages (*Marilla Cuthbert,* t=1)	24,083 / 2,477	`20261002T095544-baseline-b2-v1`
B3 index	LightRAG's own entity and relation extraction while building its index	3,243 / 1,022	`20260929T225346-baseline-b3-v5`
B3 page	LightRAG: one character page from LightRAG context (*Anne Shirley,* t=5)	17,438 / 2,191	`20261001T142328-baseline-b3-v5`
X1 page	Closed book: title and character name only (*Matthew Cuthbert,* t=5)	451 / 1,087	`20260930T214444-baseline-x1-v5`
X2 page	Full text: all five volumes and the cutoff instruction (*Gilbert Blythe,* t=1)	689,967 / 2,449	`20260930T222548-baseline-x2-v1`
Treated text and recognition			
Paraphrase	Paraphrases one renamed paragraph	385 / 209	`20261003T165238-decontaminate-v5`
Memorization probe	Name cloze or title identification on one paragraph	227 / 9	`20261003T165318-probe-memorization-v1-5`

Table 35: Every exhibited model stage: the first successful call of the run behind the reported results, with the exhibited character or entry and volume or cutoff in italics. Runs are under `anne@v2/` (build and pages) or `anne-decon/` (treatment and probe) in the release. B4's page prompt, the other prose sections and the shared-page summary check are in the release only.

Q.1 Defects carried by the logged prompts and responses

The five defects below are part of the reported runs. They are documented, not repaired; Section 7 and Appendix I relate each to the results.

D1 Implementation notes used as field definitions (extraction and verification)

The extraction prompt defines STATUS and APPEARANCE with internal revision labels and code paths, and the verifier is told to judge an APPEARANCE fact by that same text:

Verification user prompt (excerpt)

```
Fact: APPEARANCE = "little pale face with seven freckles on the nose" -- evidence: [v02:c27:p0041] [...]
Definition of APPEARANCE: subsume (Phase 22 A2): drop a value that is redundant next to a longer sibling ('hair' next to
   'long brown hair') or a bare single-word noun standing alone -- see graph/canonicalize.py::subsume_values.
  Deterministic, no embed call.
```

The exhibited claim is accepted; the example alone does not measure the defect's effect.

D2 Responses depart from prompt rules the code does not enforce (infobox pass)

The volume-1 infobox response gives a `GENDER` quote that does not contain Anne's name, and two `AGE` values (11 and 13). The code applies the evidence gates of Section 4, not these two prompt rules; the two ages go to arbitration (Appendix Q.2).

D3 Early-volume prompts carry later aliases

Alias lists come from the whole-series entity index, so the volume-1 infobox call already names Anne by her married names:

Infobox user prompt, volume 1 (excerpt)

```
Character: "Anne Shirley" (aliases: Anne, Anne-girl, Mistress Blythe, Mrs. Blythe, Mrs. Doctor, Mrs. Doctor, dear, Mrs.
  Dr. dear, Queen Anne, the Anne-girl, young Mrs. Doctor)
Volume 1.
```

This is construction dependence reaching the generator's input (Section 6.5); the comparisons of this paper do not isolate it.

D4 Residue of one prompt serving every corpus

The typing prompt lists a type unused here (`TECH`) and, as logged, repeats one example sentence three times; the extraction prompt's JSON example uses a character from another series ("Shin"); the paraphrase rule keeps "personal and ship" names, which has no effect on this series.

D5 A response cites a paragraph ID where a line key was requested (codex summary)

The codex prompt asks the model to cite input-line keys such as `W1`; the response returns `v01:c27:p0000`, a paragraph ID. The parser normalizes returned paragraph IDs during citation resolution, which maps an identifier to a paragraph but does not check that the paragraph supports the sentence.

Q.2 Complete logged exchanges

Every generative stage uses a shared client that records the system prompt, user prompt, and raw response. For each stage, we show the first successful call from the latest run underlying the reported results that sent calls to the provider. The exhibits cover each NARRATIVEWIKI stage (prose by its appearance section), the B1, B2, B3, X1 and X2 pages, the treatment and the recognition probe; B4's page prompt, the other prose sections and the shared-page summary check are in the release only. The exchanges retain the logged wording, with middle elisions that state the number of omitted characters and typographic characters folded to ASCII for typesetting. All calls shown used Gemini 3.6 Flash (Vertex AI, temperature 0.4, `thinking_budget` 0). The release contains the complete prompts and call logs (Appendix P). Evaluation labels were not produced by these stages; the annotation workflow is described in Appendix B.

The excerpts retain defects in the prompts and responses used for the reported runs. The notes below identify these defects without rewriting the logged exchanges or claiming that the pipeline has been repaired (Appendix D).

Entity typing

Types each mined name candidate (NarrativeWiki, gazetteer). Model `gemini-3.6-flash`; 518 input and 39 output tokens; run `anne@v2/20261004T193719-gazetteer-v1-5`.

Note on this log. One prompt serves every series the pipeline processes, so the type list includes types unused here (`TECH`). The example sentences are logged as sent: one sentence appears three times.

System prompt

```
You are typing candidate names mined from a novel. Each candidate is a capitalised phrase that
  occurs in the text; most are real entities but some are sentence fragments, honorific-only
  phrases, or common words that happened to be capitalised.

Classify the candidate into exactly one of these types, or say it is not an entity at all:
- CHARACTER
- FACTION
- LOCATION
- EVENT
- TECH

If the candidate is not a named entity -- a stray common word, part of a longer name cut short, a
   rank or honorific with no name attached -- respond with entity_type "NOT_ENTITY". This is a
  correct and expected answer for a large share of candidates; do not force a fit.

Respond with JSON only: {"entity_type": "...", "confidence": 0.0-1.0, "notes": "short reason"}
```

User prompt (input)

```
Candidate surface: "Anne"
Seen 3904 time(s), first in volume 1.
Mining signals: capitalised_token, honorific, vocative
Example sentences it appears in:
- ""Anne Shirley," reluctantly faltered forth the owner of that name, "but, oh, please do call me
   Cordelia. It can't matter much to you what you call me if I'm only going to be here a little
  while, can it? And Anne is such "
- ""Unromantic fiddlesticks!" said the unsympathetic Marilla. "Anne is a real good plain sensible
   name. You've no need to be ashamed of it.""
- ""Oh, I'm not ashamed of it," explained Anne, "only I like Cordelia better. I've always
  imagined that my name was Cordelia--at least, I always have of late years. When I was young I
  used to imagine it was Geraldine, but I "
- ""Oh, I'm not ashamed of it," explained Anne, "only I like Cordelia better. I've always
  imagined that my name was Cordelia--at least, I always have of late years. When I was young I
  used to imagine it was Geraldine, but I "
- ""Oh, I'm not ashamed of it," explained Anne, "only I like Cordelia better. I've always
  imagined that my name was Cordelia--at least, I always have of late years. When I was young I
  used to imagine it was Geraldine, but I "

What is this?
```

Model output

```
{"entity_type": "CHARACTER", "confidence": 1.0, "notes": "Anne (Anne Shirley) is the main
  character of the novel Anne of Green Gables."}
```

Alias adjudication

Decides whether two surface forms are one entity (NARRATIVEWIKI, gazetteer). Model `gemini-3.6-flash`; 548 input and 50 output tokens; run `anne@v2/20261004T193719-gazetteer-v1-5`.

System prompt

```
You are deciding whether two surface forms mined from a novel refer to the SAME entity (e.g. a
  full name and a nickname, a call sign and a real name, a title and a bare surname) or to two
  DIFFERENT entities that merely look similar.

Leaving two different surface forms unmerged is the SAFE error -- a human reviewer can merge them
   in five seconds. Merging two different people is the COSTLY error -- it silently mixes their
  facts on one page and is hard to catch later. When genuinely unsure, say they are different.

Decide from the quoted passages alone. Anything you may know about this book or its series from
  elsewhere is not evidence: the reader of this page knows only the text so far, and a merge the
  passages do not show can reveal a later twist or join two different people.

Respond with JSON only: {"same_entity": true/false, "confidence": 0.0-1.0, "notes": "short reason
  "}
```

User prompt (input)

```
Form A: "Charlie Sloane" (CHARACTER, seen 38 time(s), first vol 1):
  - "in half a dozen take-notices. "It's only meant as a joke. And don't you be too sure your
  name won't ever be written up. Charlie Sloane is dead gone on you. He told his mother--his
  mother, mind you--that you were the smartest girl in school. That's better than being good-
  looking.""
  - ""No, it isn't," said Anne, feminine to the core. "I'd rather be pretty than clever. And I
  hate Charlie Sloane, I can't bear a boy with goggle eyes. If anyone wrote my name up with his I
  'd never get over it, Diana Barry. But it is nice to keep head of your class.""

Form B: "Carrie Sloane" (CHARACTER, seen 10 time(s), first vol 1):
  - "der brothers and sisters who were going to take part. Everybody in school over nine years of
   age expected to go, except Carrie Sloane, whose father shared Marilla's opinions about small
  girls going out to night concerts. Carrie Sloane cried into her grammar all the afternoon and
  felt that life was n"
  - "a model pupil like Minnie Andrews. She hadn't anything on her conscience. The girls cried
  all the way home from school. Carrie Sloane kept saying every few minutes, 'The time has come
  for us to part,' and that would start us off again whenever we were in any danger of cheering
  up. I do feel dreadful"

Are these the same entity?
```

Model output

```
{"same_entity": false, "confidence": 1.0, "notes": "Charlie Sloane (a male classmate described as
   a boy) and Carrie Sloane (a female classmate) are two distinct individuals/siblings within the
   same family."}
```

Claim extraction

Extracts dated, quoted claims about one character from one evidence window (NARRATIVEWIKI). Model `gemini-3.6-flash`; 8,884 input and 734 output tokens; run `anne@v2/20261005T070204-extract-v5-5`.

Note on this log. STATUS and APPEARANCE include internal revision labels, code paths, and implementation notes in the prompt's field descriptions. These entries are preserved as run evidence. Implementation metadata should be separated from semantic definitions in the prompt builder before future runs.

System prompt

```
You are extracting facts about ONE named character from passages of a novel. You will be told the
   character's name; extract facts ONLY about that character, never about anyone else named
  alongside them.

Three kinds of fact exist:

ATTRIBUTE (a scalar fact about the character):
- AGE
- GENDER
- STATUS -- alive / deceased / missing in action / captured. Use polarity=presumed when the text
  only reports a belief. `default` (Phase 22 A5, U6): materialized by synth/assemble.py when zero
   STATUS claims exist -- 'alive' is the reader's safe default at any cutoff (a later death is
  itself a claim that only appears once the reader reaches it), marked inferred: true so it never
   masquerades as evidence.
- RANK -- A formal grade inside a named organisation or hierarchy: a military rank, a guild or
  adventurer grade, a noble or church rank. Always a category, never a number. NOT a power level,
   a game level or any other statistic, NOT a job (OCCUPATION), NOT an honorific or epithet (
  TITLE).
- TITLE -- An honorific, epithet or office the text calls the character by (Supreme Being, the
  Wisewolf of Yoitsu, head of the branch). A grade in a hierarchy is RANK; a job is OCCUPATION.
- NICKNAME -- Includes call signs and code names. first_vol matters: an alias revealed in V3 must
   not appear on a V1 page.
- ORIGIN -- Where the character comes from: a birthplace, homeland, village or people. Never a
  job or a past occupation (that is OCCUPATION or BACKGROUND).
- OCCUPATION -- What the character does in the world: their job, trade or standing role (merchant
  , maid, village chief, floor guardian). NOT a combat class or skill, NOT a species, NOT a rank.
- APPEARANCE -- subsume (Phase 22 A2): drop a value that is redundant next to a longer sibling ('
  hair' next to 'long brown hair') or a bare single-word noun standing alone -- see graph/
  canonicalize.py::subsume_values. Deterministic, no embed call.

RELATION (a link to another character or entity actually named in the passage; "object" must be
  copied verbatim from a name that appears in the passage, not paraphrased or described):
- PARENT_OF -- An actual stated parent/child relationship -- never inferred from a shared surname
  , a title like 'son of', or
[... 9,813 characters elided ...]
BLING_OF, SPOUSE_OF, STUDENT_OF), not just the subject.

Respond with JSON only, shaped exactly like this worked example (a RELATION fact about a
character named "Shin", using null for the fields a TRAIT or ATTRIBUTE fact would instead need):

{"facts": [
  {"predicate": "AFFILIATED_WITH", "kind": "relation", "object": "Spearhead Squadron", "value":
  null, "qualifier": "captain", "para_id": "v01:c00:p0031", "quote": "the Spearhead Squadron's
  captain, code name Undertaker", "confidence": 0.92, "polarity": "asserted"}
]}

Return {"facts": []} if this passage says nothing new about this character.
```

User prompt (input)

```
Character: "Anne Shirley" (aliases: Anne, Anne-girl, Mistress Blythe, Mrs. Blythe, Mrs. Doctor,
  Mrs. Doctor, dear, Mrs. Dr. dear, Queen Anne, the Anne-girl, young Mrs. Doctor)
Volume 5.
```

User prompt (input) (continued)

```
Other people/things named in these passages: Aunt Martha (also: old Martha), Avonlea (also: The
  Avonlea), Blythe, Carl, Cornelia Bryant (also: Bryant, Cornelia, Cornelia Elliott, Miss Bryant,
   Miss Cornelia, Mrs. Elliott, Mrs. Marshall Elliott, fair Cornelia), Europe, Faith (also: Faith
   Meredith, Red Rose), Four Winds, Four Winds Harbor (also: Four Winds Harbour), Glen, Green
  Gables, Harbour Head, Harrison, Ingleside, James A (also: James), Jem (also: Jem Blythe, Little
   Jem), Jerry Corcoran (also: Jerry), London, Maria, Marilla Cuthbert (also: Aunt Marilla,
  Cuthbert, Marilla, Miss Cuthbert), Mary Vance (also: Mary, Mary-creature), Methodist (also:
  Methodists, The Methodists), Paul Irving (also: Paul), Presbyterian (also: Presbyterians),
  Priscilla Grant (also: Grant, Pris, Priscilla, Prissy, Prissy Andrews), Rachel Lynde (also:
  Lynde, Mrs. Rachel, Rachel), Rainbow Valley, Rilla Blythe (also: Rilla), Shirley, Susan (also:
  Susan Baker, old Susan, old Susan Baker), Taylor, Thomas Lynde (also: Rachel Lynde's husband,
  Thomas), Una Meredith (also: Una), Walter Blythe (also: Walter)

Passages (each paragraph labeled with its id; a passage break means the text on either side comes
   from a different part of the volume and does not continue):
[v05:c01:p0000] It was a clear, apple-green evening in May, and Four Winds Harbour was mirroring
  back the clouds of the golden west between its softly dark shores. The sea moaned eerily on the
   sand-bar, sorrowful even in spring, but a sly, jovial wind came piping down the red harbour
  road along which Miss Cornelia's comfortable, matronly figure was making its wa
[... 18,642 characters elided ...]
ained untinged of charity. Susan smiled slyly.
[v05:c02:p0042] "They do say, Mrs. Marshall Elliott, that the Methodists and Presbyterians are
  talking of uniting," she said.
[v05:c02:p0043] "Well, all I hope is that I'll be under the sod if that ever comes to pass,"
  retorted Miss Cornelia. "I shall never have truck or trade with Methodists, and Mr. Meredith
  will find that he'd better steer clear of them, too. He is entirely too sociable with them,
  believe me. Why, he went to the Jacob Drews' silver-wedding supper and got into a nice scrape
  as a result."

Extract facts about "Anne Shirley" only.
```

Model output


```
{"facts": [
  {"predicate": "SPOUSE_OF", "kind": "relation", "object": "Blythe", "value": null, "qualifier":
  "husband", "para_id": "v05:c01:p0001", "quote": "Miss Cornelia was going up to Ingleside to see
   Dr. and Mrs. Blythe, who were just home from Europe.", "confidence": 0.95, "polarity": "
  asserted"},
  {"predicate": "PARENT_OF", "kind": "relation", "object": "Rilla Blythe", "value": null, "
  qualifier": "mother", "para_id": "v05:c01:p0003", "quote": "Behind her, in the hammock, Rilla
  Blythe was curled up, a fat, roly-poly little creature of six years, the youngest of the
  Ingleside children.", "confidence": 0.95, "polarity": "asserted"},
  {"predicate": "PARENT_OF", "kind": "relation", "object": "Shirley", "value": null, "qualifier":
   "mother", "para_id": "v05:c01:p0004", "quote": "After his birth Anne had been very ill for a
  long time, and Susan "mothered" the baby with a passionate tenderness which none of the other
  children, dear as they were to her, had ever called out.", "confidence": 0.95, "polarity": "
  asserted"},
  {"predicate": "PARENT_OF", "kind": "relation", "object": "Jem", "value": null, "qualifier": "
  mother", "para_id": "v05:c02:p0001", "quote": ""Shirley is in bed and Jem and Walter and the
  twins are down in their beloved Rainbow Valley," said Anne.", "confidence": 0.95, "polarity": "
  asserted"},
  {"predicate": "PARENT_OF", "kind": "relation", "object": "Walter Blythe", "value": null, "
  qualifier": "mother", "para_id": "v05:c02:p0001", "quote": ""Shirley is in bed and Jem and
  Walter and the twins are down in their beloved Rainbow Valley," said Anne.", "confidence":
  0.95, "polarity": "asserted"},
  {"predicate": "APPEARANCE", "kind": "attribute", "object": null, "value": "beautiful gray-green
   eyes", "qualifier": null, "para_id": "v05:c01:p0003", "quote": "and the beautiful gray-green
  eyes, gazing down the harbour road, were as full of unquenchable sparkle and dream as ever.", "
  confidence": 0.95, "polarity": "asserted"},
  {"predicate": "PERSONALITY", "kind": "trait", "object": null, "value": "loves gossip", "
  qualifier": null, "para_id": "v05:c01:p0009", "quote": "Do you know, Susan, I have a dreadful
  suspicion that I love gossip.", "confidence": 0.9, "polarity": "asserted"}
]}
```

Scene extraction

Records scenes, participants and events over a chapter span (NARRATIVEWIKI). Model `gemini-3.6-flash`; 5,729 input and 202 output tokens; run `anne@v2/20261004T195945-scenes-v5-5`.

System prompt

```
You are reading one FULL CHAPTER of a novel, span by span, to build a scene-level summary -- not
  fact-by-fact extraction. You are given a whole passage in reading order, possibly spanning
  multiple scenes or POV shifts (marked inline with a line reading "[SCENE BREAK]").

For this passage, report:

PARTICIPANTS: every character who takes part in this passage -- present in the scene, speaking,
  acting, or acted upon -- copy each name EXACTLY as it appears in "Known entities in this
  chapter" below, never invent a new name. A character who is only talked about, remembered,
  written to, or named in passing is NOT a participant: the wiki lists this scene on every
  participant's page.

LOCATION: where this passage takes place, if stated or clearly implied -- copy it EXACTLY from "
  Known entities in this chapter", or leave it null if not stated or not in that list.

BEAT_SUMMARY: 2-4 sentences paraphrasing what actually happens in this passage -- this is the one
   field in the whole scene-extraction pass that does NOT need a verbatim quote; summarize freely
  , but only describe what this passage itself shows.

SUMMARY_PARA_IDS: the ids of the 1-4 paragraphs that most directly show what your beat_summary
  says, copied exactly from their labels in the passage (e.g. "v01:c03:p0082"). A reader follows
  them to check the summary.

Rules:
- Only name a participant/location that is in "Known entities in this chapter" -- never a name
  invented or paraphrased from elsewhere.
- Omitting something this passage does not support is correct -- do not force a fit.

Respond with JSON only, shaped exactly like this worked example:

{{"participants": ["Shin", "Spearhead Squadron"], "location": null,
"beat_summary": "Shin briefs the squadron before the next sortie, reminding them of the plan.",
"summary_para_ids": ["v01:c03:p0082", "v01:c03:p0084"], "confidence": 0.85}}

Return {{"participants": [], "location": null, "beat_summary": "...", "summary_para_ids": [], "
  confidence": 0.5}} (with a short beat_summary even if nothing else was found) if this passage
  supports nothing more specific.
```

User prompt (input)

```
Volume 5, chapter 2.
Known entities in this chapter: Alec Davis (also: Kitty Alec), Anne Shirley (also: Anne, Anne-
  girl), Aunt Martha, Avonlea (also: The Avonlea), Canada, Carl, Cornelia Bryant (also: Bryant,
  Cornelia), Ellen West (also: Ellen), Episcopalians, Europe, Faith (also: Faith Meredith), Four
  Winds, Four Winds Harbor (also: Four Winds Harbour), Glen, Harbour Head, James A (also: James),
   Japan, Jem (also: Jem Blythe, Little Jem), Jerry Corcoran (also: Jerry), Joseph, Kenneth,
  Leslie Moore (also: Leslie), Lowbridge, Maria, Marilla Cuthbert (also: Aunt Marilla, Cuthbert,
  Marilla), Marshall Elliott (also: Marshall), Martin Crawford (also: Martin), Mary Vance (also:
  Mary), Merediths (also: The Meredith), Methodist (also: Methodists, The Methodists), Moses,
  Owen Ford, Paul Irving (also: Paul), Presbyterian (also: Presbyterians), Priscilla Grant (also:
   Grant, Pris, Priscilla, Prissy, Prissy Andrews), Rainbow Valley, Rilla Blythe (also: Rilla),
  Rosemary West (also: Rosemary), Shirley, Susan (also: Susan Baker), Thomas Lynde (also: Thomas)
  , Una Meredith (also: Una), Uncle Jim (also: Boyd, Jim, Jim Boyd), Walter Blythe (also: Walter)

Passage (each paragraph labeled with its id):
[v05:c02:p0000] "Where are the other children?" asked Miss Cornelia, when the first greetings--
  cordial on her side, rapturous on Anne's, and dignified on Susan's--were over.
[v05:c02:p0001] "Shirley is in bed and Jem and Walter and the twins are down in their beloved
  Rainbow Valley," said Anne. "They just came home this afternoon, you know, and they could
  hardly wait until supper was over before rushing down to the valley. They love it above every
  spot on earth. Even the maple grove doesn't rival it in their affections."
```

User prompt (input) (continued)

```
[v05:c02:p0002] "I am afraid they love it too well," said Susan gloomily. "L
[... 15,858 characters elided ...]
d see me soon. Tell them the doughnut jar is always full."
[v05:c02:p0061] "Oh, at supper they were planning a descent on you. They'll go soon; but they
  must settle down to school again now. And the twins are going to take music lessons."
[v05:c02:p0062] "Not from the Methodist minister's wife, I hope?" said Miss Cornelia anxiously.
[v05:c02:p0063] "No--from Rosemary West. I was up last evening to arrange it with her. What a
  pretty girl she is!"
[v05:c02:p0064] "Rosemary holds her own well. She isn't as young as she once was."

Report participants, location, and a beat summary for this passage.
```

Model output

```
{
  "participants": [
    "Anne Shirley",
    "Cornelia Bryant",
    "Susan"
  ],
  "location": null,
  "beat_summary": "Cornelia Bryant and Susan visit Anne to welcome her back and catch her up on
  the local news, particularly regarding the new minister, Mr. Meredith, and his motherless
  children. They discuss the minister's well-meaning but absent-minded nature, his unruly family,
   and various church gossip before Miss Cornelia takes her leave.",
  "summary_para_ids": [
    "v05:c02:p0007",
    "v05:c02:p0011",
    "v05:c02:p0023",
    "v05:c02:p0054"
  ],
  "confidence": 0.95
}
```

Infobox pass

Long-context pass for gender, age and origin (NARRATIVEWIKI). Model `gemini-3.6-flash`; 93,403 input and 490 output tokens; run `anne@v2/20261005T070216-infobox-v1-5`.

Note on this log. The response departs from two rules of its prompt: the `GENDER` quote does not contain Anne's name, and the `AGE` entries give two values (11 and 13). The code does not enforce these two prompt rules; it applies the evidence gates of Section 4. Two values of one attribute from one volume go to arbitration, shown next.

System prompt

```
You are filling in the infobox of ONE named character's wiki page, from passages of a novel. The
  passages are every part of one volume where this character appears, in reading order. Extract
  ONLY these fields, and only about the named character:

- GENDER
- AGE
- ORIGIN -- Where the character comes from: a birthplace, homeland, village or people. Never a
  job or a past occupation (that is OCCUPATION or BACKGROUND).

Rules:
- Each field needs a "para_id" copied EXACTLY from a bracketed label, and a "quote" copied
  VERBATIM from that single paragraph (no added quotation marks) that shows the value.
- GENDER: "female" or "male". The quote MUST contain both the character's name (or a listed alias
  ) AND a gendered word referring to them -- e.g. "Lena shook her head", "Shin said, and he
  laughed". Pick such a sentence even if a vaguer one ("said the girl") appears earlier; a quote
  without the name is discarded, and a quote without a gendered word does not show gender. Never
  infer gender from a name alone.
- AGE: a number or range only when the text states it or states something that fixes it.
```

System prompt (continued)

```
- ORIGIN: where the character was born or comes from (homeland, hometown). NOT where they
  currently live, work, trade or are visiting.
- If the passages do not support a field, leave it out. Omitting is correct; guessing is not.
- polarity is "asserted", or "presumed" when the text only reports a belief or a claim the
  passage marks as untrue or uncertain. confidence is 0.0-1.0.
- You may give up to 3 entries per field, each with a different quote, so that at least one meets
   the evidence requirements below. Every entry for a field must give the same value.

EVIDENCE REQUIREMENTS -- a fact that fails any of these is discarded:
- "quote" must be at least 6 words long. A shorter quote cannot support a fact on its own and is
  thrown away -- quote the whole clause, not two words of it.
- The character's own name or a known alias MUST occur in the cited paragraph or within 1
  paragraph(s) before or after it in the supplied passage. Cite and quote the paragraph that
  actually states the fact, even when it uses a pronoun; do not substitute a nearby paragraph
  merely because it names them. The antec
[... 402 characters elided ...]
of" is a label, not a fact.
- For these relations the cited paragraph must name BOTH people (CHILD_OF, COMMANDS, KILLED,
  MENTOR_OF, PARENT_OF, RELATIVE_OF, ROMANTIC_WITH, SAVED, SERVES_UNDER, SIBLING_OF, SPOUSE_OF,
  STUDENT_OF), not just the subject.

Respond with JSON only, shaped exactly like this:
{"facts": [
  {"predicate": "GENDER", "kind": "attribute", "object": null, "value": "female", "qualifier":
  null, "para_id": "v01:c02:p0040", "quote": "She pulled her hood lower over her ears", "
  confidence": 0.95, "polarity": "asserted"}
]}

Return {"facts": []} if none of these fields is supported.
```

User prompt (input)

```
Character: "Anne Shirley" (aliases: Anne, Anne-girl, Mistress Blythe, Mrs. Blythe, Mrs. Doctor,
  Mrs. Doctor, dear, Mrs. Dr. dear, Queen Anne, the Anne-girl, young Mrs. Doctor)
Volume 1.
Other people/things named in these passages: Adella, Aid Society (also: Aid, Aids), Allan,
  American (also: Americans), Anne of Green Gables, Avenue (also: White Way, White Way of Delight
  ), Avonlea (also: The Avonlea), Barry, Beechwood, Ben Hur, Bentley, Bertha Shirley, Billy
  Andrews (also: Billy, Billy Booth), Bingen, Birch Path, Blewett, Blythe, Bolingbroke, Bonny,
  Boston, Bright River, Camelot, Canada, Carmody, Carrie Sloane, Charlie Sloane (also: Goggle-
  eyes), Charlottetown, Chester Ross, Conservative (also: Conservatives, Tory), Cuthberts,
  Debating Club, Diana Barry (also: Di, Diana, Diana Wright), Dryad's Bubble (also: Bubble, Dryad
  ), Edward Island, Elaine, Emily (also: the Fair Unknown), Entrance, Evans, Exhibition, Faith (
  also: Faith Meredith, Red Rose), Flora Jane, Frank Stockley, Geraldine, Gertie Pye (also:
  Gertie), Gilbert Blythe (also: Gil, Gilbert), God, Green Gables, Grit (also: Grits), Guinevere,
   Hammond, Harmon Andrews (also: Harmon), Haunted Wood (also: The Haunted Wood), Idlewild,
  Island, Jane Andrews (also: Jane), Jerry, Josephine Barry (also: Aunt Josephine, Miss Barry,
  Old Miss Barry), Josie Pye (also: Josie), Julia Bell, Lake of Shining Waters (also: Shining
  Waters), Lancelot, Lavendar Lewis (also: Lavendar, Lewis), Liberal, Lover (also: Lover's Lane),
   Marilla Cuthbert (also: Aunt Marilla, Cuthbert, Marilla, Miss Cuthbert), Martin, Mary Joe (
  also: Aunt Mary, Mary, Young Mary Joe), Marysville, Matthew Cuthbert (also: Matthew), Merry
  Christmas (also: Christmas), Minnie, Minnie Andrews, Minnie May, Moody Spurgeon MacPherson (
  also: Moody Spurgeon), New Brunswick, Newbri
[... 351,617 characters elided ...]
he," answered Anne, vexed to find herself blushing. "I met him on Barry's hill."

--- new passage, not continuous with the text above ---

[v01:c38:p0061] Anne sat long at her window that night companioned by a glad content. The wind
  purred softly in the cherry boughs, and the mint breaths came up to her. The stars twinkled
  over the pointed firs in the hollow and Diana's light gleamed through the old gap.

--- new passage, not continuous with the text above ---

[v01:c38:p0063] "'God's in his heaven, all's right with the world,'" whispered Anne softly.

Extract facts about "Anne Shirley" only.
```

Model output


```
{"facts": [
  {"predicate": "GENDER", "kind": "attribute", "object": null, "value": "female", "qualifier":
  null, "para_id": "v01:c03:p0045", "quote": "With a sob she hastily discarded her garments, put
  on the skimpy nightgown and sprang into bed where she burrowed face downward into the pillow
  and pulled the clothes over her head.", "confidence": 1.0, "polarity": "asserted"},
  {"predicate": "AGE", "kind": "attribute", "object": null, "value": 11, "qualifier": null, "
  para_id": "v01:c05:p0011", "quote": ""I was eleven last March," said Anne, resigning herself to
   bald facts with a little sigh.", "confidence": 1.0, "polarity": "asserted"},
  {"predicate": "AGE", "kind": "attribute", "object": null, "value": 11, "qualifier": null, "
  para_id": "v01:c06:p0017", "quote": ""Anne Shirley," faltered the shrinking child, not daring
  to make any stipulations regarding the spelling thereof, "and I'm eleven years old."", "
  confidence": 1.0, "polarity": "asserted"},
  {"predicate": "AGE", "kind": "attribute", "object": null, "value": 13, "qualifier": null, "
  para_id": "v01:c26:p0004", "quote": ""Just think, Diana, I'm thirteen years old today,"
  remarked Anne in an awed voice.", "confidence": 1.0, "polarity": "asserted"},
  {"predicate": "ORIGIN", "kind": "attribute", "object": null, "value": "Bolingbroke, Nova Scotia
  ", "qualifier": null, "para_id": "v01:c05:p0011", "quote": ""And I was born in Bolingbroke,
  Nova Scotia.", "confidence": 1.0, "polarity": "asserted"}
]}
```


Backstory pass

Long-context pass for background facts (NARRATIVEWIKI). Model `gemini-3.6-flash`; 135,267 input and 309 output tokens; run `anne@v2/20261005T070229-backstory-v1-5`.

System prompt

```
You are writing the background of ONE named character's wiki page, from passages of a novel. The
  passages are every part of one volume where this character appears, with nearby context, in
  reading order. Extract ONLY this kind of fact, and only about the named character:

- BACKGROUND -- A biographical fact, not a narrative. 'Orphaned in the First Offensive' -- the
  prose stage joins these.

What counts: the character's life BEFORE the story's present, or lasting facts about their past
  that the text reveals -- upbringing, family, where they lived or grew up, earlier work, how
  they came to be where they are, a formative event. What does NOT count: things they do during
  the story's present, personality, appearance, or their current job.

Rules:
- "value" is a short phrase of at most 20 words, not a sentence, in the PAST tense, describing
  something that happened or was true before the story's present: "raised in an almshouse
  attached to an abbey", "apprenticed to a relative at twelve". If it cannot be phrased that way,
   it is not backstory -- leave it out.
- Never return: an age; a current job, title, employer or backer; who they work with now; health
  complaints; where they are from, as a bare place name (a separate field covers that).
- The character's own first-person account of their past counts; so does another character or the
   narrator stating it as fact. A rumour, a boast or a guess is "presumed", not "asserted".
- Each fact needs a "para_id" copied EXACTLY from a bracketed label and a "quote" copied VERBATIM
   from that single paragraph (no added quotation marks).
- Give each distinct fact once, at most 8 facts. Omitting is correct; guessing is not.

EVIDENCE REQUIREMENTS -- a fact that fails any of these is discarded:
- "quote" must be at least 6 words long. A shorter quote cannot support a fact on its own and is
  thrown away -- quote the whole clause, not two words of it.
- The character's own name or a known alias MUST occur in the cited paragraph or within 1
  paragraph(s) before or after it in the supplied passage. Cite and quote the paragraph that
  actually states the fact, even when it uses a pronoun; do not substitute a nearby paragraph
  merely because it names
[... 482 characters elided ...]
raph must name BOTH people (CHILD_OF, COMMANDS, KILLED, MENTOR_OF, PARENT_OF, RELATIVE_OF,
  ROMANTIC_WITH, SAVED, SERVES_UNDER, SIBLING_OF, SPOUSE_OF, STUDENT_OF), not just the subject.

Respond with JSON only, shaped exactly like this:
{"facts": [
  {"predicate": "BACKGROUND", "kind": "trait", "object": null, "value": "set out on his own at
  eighteen", "qualifier": null, "para_id": "v01:c01:p0006", "quote": "This year made the seventh
```

System prompt (continued)

```
  since he'd struck out on his own", "confidence": 0.9, "polarity": "asserted"}
]}

Return {"facts": []} if the passages reveal nothing about this character's past.
```

User prompt (input)

```
Character: "Marilla Cuthbert" (aliases: Aunt Marilla, Cuthbert, Marilla, Miss Cuthbert)
Volume 1.
Other people/things named in these passages: Adam, Adella, Aid Society (also: Aid, Aids),
  Alexander Spencer, Allan, Almighty, American (also: Americans), Anne Shirley (also: Anne, Anne-
  girl, Mistress Blythe, Mrs. Blythe, Mrs. Doctor, Mrs. Doctor, dear, Mrs. Dr. dear, Queen Anne,
  the Anne-girl, young Mrs. Doctor), Anne of Green Gables, Avenue (also: White Way, White Way of
  Delight), Avonlea (also: The Avonlea), Barry, Beechwood, Ben Hur, Bentley, Bertha Shirley,
  Bertram, Billy Andrews (also: Billy, Billy Booth), Bingen, Birch Path, Blewett, Blythe, Blythes
   (also: The Blythe), Bolingbroke, Bonny, Bright River, Camelot, Canada, Carmody, Carrie Sloane,
   Charlie Sloane (also: Goggle-eyes), Charlottetown, Chester Ross, Conservative (also:
  Conservatives, Tory), Cuthberts, Daniel, Debating Club, Diana Barry (also: Di, Diana, Diana
  Wright), Dryad's Bubble (also: Bubble, Dryad), Edward Island, Elaine, Emily (also: the Fair
  Unknown), England, Entrance, Europe, Evans, Exhibition, Faith (also: Faith Meredith, Red Rose),
   Flora Jane, Fourth, Frank Stockley, Geraldine, German, Germany, Gertie Pye (also: Gertie),
  Gilbert Blythe (also: Gil, Gilbert), God, Green Gables, Grit (also: Grits), Hammond, Harmon
  Andrews (also: Harmon), Harris, Haunted Wood (also: The Haunted Wood), Hopeton (also: Hopetown)
  , Idlewild, Island, James A (also: James), Jane Andrews (also: Jane), Jerry, Josephine Barry (
  also: Aunt Josephine, Miss Barry, Old Miss Barry), Josie Pye (also: Josie), Julia Bell, Katie
  Maurice, Lake of Shining Waters (also: Shining Waters), Lauretta, Liberal, London, Lover (also:
   Lover's Lane), Lynde's Hollow, Marshall, Martin, Mary Joe (also: Aunt Mary, Mary, Young Mary
  Joe), Marysville, Matthew
[... 488,087 characters elided ...]
med through the old gap.
[v01:c38:p0062] Anne's horizons had closed in since the night she had sat there after coming home
   from Queen's; but if the path set before her feet was to be narrow she knew that flowers of
  quiet happiness would bloom along it. The joy of sincere work and worthy aspiration and
  congenial friendship were to be hers; nothing could rob her of her birthright of fancy or her
  ideal world of dreams. And there was always the bend in the road!
[v01:c38:p0063] "'God's in his heaven, all's right with the world,'" whispered Anne softly.

Extract facts about "Marilla Cuthbert" only.
```

Model output

```
{"facts": [
  {"predicate": "BACKGROUND", "kind": "trait", "object": null, "value": "grown up when the new
  house was built", "qualifier": null, "para_id": "v01:c01:p0031", "quote": "Matthew and Marilla
  were grown up when the new house was built", "confidence": 0.9, "polarity": "asserted"},
  {"predicate": "BACKGROUND", "kind": "trait", "object": null, "value": "heard an aunt call her a
   dark, homely little thing", "qualifier": null, "para_id": "v01:c09:p0044", "quote": "She had
  been a very small child when she had heard one aunt say of her to another, "What a pity she is
  such a dark, homely little thing."", "confidence": 0.9, "polarity": "asserted"},
  {"predicate": "BACKGROUND", "kind": "trait", "object": null, "value": "quarreled with John
  Blythe and refused to forgive him", "qualifier": null, "para_id": "v01:c37:p0040", "quote": "We
   had a quarrel. I wouldn't forgive him when he asked me to.", "confidence": 0.9, "polarity": "
  asserted"}
]}
```

Arbitration

Resolves a same-volume conflict between two values (NARRATIVEWIKI). Model `gemini-3.6-flash`; 258 input and 55 output tokens; run `anne@v2/20261005T070243-graph_build-v1-5`.

Handling of flagged conflicts. A `flag` resolution keeps the higher-confidence value provisionally (here 11), so the interval sequence has no gap, and records the conflict in the build's conflict report; the page is rendered from the provisional value.

System prompt

```
You are arbitrating a contradiction found in facts extracted about a character from a novel. Two
  facts were extracted from the SAME volume and disagree, which means one is an extraction error
  -- the source text does not actually contradict itself within one volume. You will be shown
  both facts and the exact passage each was drawn from. Decide which is correct, or whether both
  are legitimately true at once (e.g. two compatible phrasings of the same fact), or whether
  neither passage clearly supports a value (flag for human review). Base your decision only on
  the quoted evidence, never on outside knowledge of the series.
```

User prompt (input)

```
Character: anne-shirley
Attribute: AGE
Volume: 1

Fact A: value="11", confidence=1.00, evidence: "I was eleven last March," said Anne"
Fact B: value="13", confidence=0.99, evidence: "It's such a solemn thing to be almost fourteen,
  Marilla."; ""Just think, Diana, I'm thirteen years old today," remarked Anne in an awed voice."

Which fact is correct? resolution must be one of keep_a, keep_b, keep_both, flag, plus a one-
  sentence rationale citing the evidence.
```

Model output

```
{
  "resolution": "flag",
  "rationale": "Fact A states Anne was eleven based on an earlier passage, while Fact B shows her
   turning thirteen later in the same volume, indicating a progression of time rather than a
  single static age."
}
```

Verification

Checks stored facts against their own cited evidence (NARRATIVEWIKI). Model `gemini-3.6-flash`; 2,833 input and 5 output tokens; run `anne@v2/20261005T053635-verify-upto5`. This run precedes the extraction runs shown above because those later runs replayed the call cache, producing the same claims, and the final verification run (`20261005T070323`) made no call: every fact of the final graph had a stored verdict, which is reused only while the fact's identity and cited evidence are unchanged.

Note on this log. The supplied definition of APPEARANCE is a canonicalization instruction, rather than a semantic definition of a character's physical features. Because the verifier is explicitly told to judge by that definition, this is a defect in the verification input, even though the exhibited claim is accepted. The example alone does not quantify the defect's impact.

System prompt

```
You are fact-checking one entry on a character's page on a wiki generated from a novel series.
  You will be shown a single fact extracted about this character, together with the exact quoted
  evidence it was extracted from. Three known failure patterns to watch for: (1) a fact that
  actually belongs to a DIFFERENT character whose name is similar or who appears in the same
  passage (two people conflated as one); (2) a relationship or membership claim the quoted
  evidence does not actually support (for example, evidence describing an enemy or a captor does
  not support "member of" or "friend of"); (3) a family relationship (parent/child/sibling/spouse
  /relative) whose "evidence" is really a boast, a joke, a false identity given to a stranger, or
```

System prompt (continued)

```
 someone invoking a respected name for effect (e.g. "I'm X, son of the great Y" as bravado or
  name-dropping, not a literal claim of parentage) -- treat this the same as pattern (2): the
  quoted line does not actually support the relationship. Judge the fact ONLY against its OWN
  quoted evidence: never reject it because it is surprising, or because you know the story from
  outside sources. Judge only the fact shown. The character may appear under any of the names
  listed after "also called": evidence about any of those names is evidence about this character,
   not about someone else. For a GENDER fact, a gendered pronoun or noun that clearly refers to
  the character ("she", "his", "the girl") is sufficient evidence. Nearby paragraphs, when
  supplied, are context for the quote, not additional cited evidence. Use them to distinguish
  literal statements from banter, figurative kinship, and false identities. Base your judgement
  solely on the supplied quote and context. When the fact comes with a definition of its label,
  judge it by that definition: the evidence must show that specific relationship or quality (a
  friendship, a mentorship, a membership), not merely a meeting, a business dealing, a single
  helpful act, being in the same scene, or being paid by someone. The description in parentheses
  must be supported too; if the label holds but the description does not, the fact is not
  supported.
```

User prompt (input)

```
Character: Anne Shirley (also called: Anne, Anne-girl, the Anne-girl, Queen Anne, Mistress Blythe
  , young Mrs. Doctor, Mrs. Doctor, dear, Mrs. Doctor)

Fact: APPEARANCE = "little pale face with seven freckles on the nose" -- evidence: [v02:c27:p0041
  ] "When she looked in the glass all she saw was a little pale face with seven freckles on the
  nose thereof."; [v02:c20:p0017] "There were those seven freckles on her nose, more rampant than
   ever, or so it seemed in the glare of light from the unshaded window."
Definition of APPEARANCE: subsume (Phase 22 A2): drop a value that is redundant next to a longer
  sibling ('hair' next to 'long brown hair') or a bare single-word noun standing alone -- see
  graph/canonicalize.py::subsume_values. Deterministic, no embed call.

Nearby paragraph context (interpretation only, not additional citations):
[v02:c27:p0037] Anne and Charlotta the Fourth accordingly betook themselves back to Mr. Kimball's
   pasture, a green remote place where the air was as soft as velvet and fragrant as a bed of
  violets and golden as amber.
[v02:c27:p0038] "Oh, isn't it sweet and fresh back here?" breathed Anne. "I just feel as if I
  were drinking in the sunshine."
[v02:c27:p0039] "Yes, ma'am, so do I. That's just exactly how I feel too, ma'am," agreed
  Charlotta the Fourth, who would have said precisely the same thing if Anne had remarked that
  she felt like a pelican of the wilderness. Always after Anne had visited Echo Lodge Charlotta
  the Fourth mounted to her little room over the kitchen and tried before her looking glass to
  speak and look and move like Anne. Charlotta could never flatter herself that she quite
  succeeded; but practice makes perfect, as Charlotta had learned at school, and she fondly hoped
   that in time she might catch the trick of that dainty uplift
[... 6,974 characters elided ...]
d with a veritable halo of feathers. At this auspicious moment a knock sounded at the kitchen
  door.
[v02:c20:p0021] "That must be Mr. Shearer," thought Anne. "I'm in a dreadful mess but I'll have
  to run down as I am, for he's always in a hurry."

Does the quoted evidence actually support this fact? If it does not, give a one-sentence
  rationale citing what the evidence actually shows.

Respond with JSON only, shaped exactly like this:
{"supported": false, "rationale": "The quote shows Shin hiring the man, not being related to him
  ."}

Return {"supported": true} if the evidence supports the fact.
```

Model output

```
{"supported": true}
```

Page prose: physical appearance

Writes a physical description from the cutoff-filtered claims (NARRATIVEWIKI). Model `gemini-3.6-flash`; 1,045 input and 174 output tokens; run `anne@v2/20261005T070347-synthesize-upto5`.

This logged prompt requests 3–4 sentences, while the intended character-page template specifies 2–4. The four-sentence response satisfies both; the prompts express different minimum lengths.

System prompt

```
You are writing a physical description of a character from a novel, for a wiki page a reader is
  viewing at one specific point in the story. You are given ONLY the attributes extracted so far,
   each with the exact passage it came from. Write 3-4 sentences, third person, encyclopedic tone
  , no meta-commentary. Do not mention any name, number, place, relationship or event that is not
   stated in the facts you were given -- the reader has not read past this point in the story,
  and a detail you add or infer from outside knowledge of the series would spoil it. If the facts
   describe a change over time, you may mention that in passing, but never state or imply what
  happens after the given facts.

Every line you are given starts with a key such as [F1] or [S2]. In "cited", list the keys of the
   lines your text actually rests on, and no others.

Respond with JSON only: {"text": "...", "cited": ["F1"]}
```

User prompt (input)

```
Character: Anne Shirley
As known through Volume 1.

[F1] Appearance: solemn gray eyes (since v1) -- "Her pointed freckled face and solemn gray eyes
  peered back at her."
[F2] Appearance: long red braids (since v1) -- "now Anne, with a toss of her long red braids,
  spelled him down"; "Gilbert reached across the aisle, picked up the end of Anne's long red
  braid, held it out at arm's length and said in a piercing whisper"; "Gilbert reached across the
   aisle, picked up the end of Anne's long red braid"
[F3] Appearance: auburn hair (since v1) -- "people are nice enough to tell me my hair is auburn
  now--all but Josie Pye."
[F4] Appearance: red hair (since v1) -- "Redheaded people can't wear pink, not even in
  imagination."; ""You'd find it easier to be bad than good if you had red hair," said Anne
  reproachfully."; "It's bad enough to have red hair myself, but I positively couldn't endure it
  in a bosom friend."; "My hair is red and I'm freckled and skinny and ugly."; "She should look
  at him, that red-haired Shirley girl with the little pointed chin and the big eyes that weren't
   like the eyes of any other girl in Avonlea school."; "I'm perfectly happy--yes, in spite of my
   red hair."; "A minister mightn't mind my red hair because he wouldn't be thinking of such
  worldly things."; "Josie says she thinks a red-haired fairy is just as ridiculous as a fat one
  "; "I thought nothing could be as bad as red hair."; "You'd find it easier to be bad than good
  if you had red hair," said Anne reproachfully. "People who haven't red hair don't know what
  trouble is. Mrs. Thomas told me that God made my hair red on purpose"; "Who is that girl on the
   platform with the splendid Titian hair?"
[F5] Appearance: thin legs (since v1) -- "below which her thin legs seemed ungracefully long."
[F6] Appearance: sho
[... 776 characters elided ...]
oroughly and it had been necessary to shingle the hair as closely as possible."
[F11] Appearance: cheeks faintly flushed with pink (since v1) -- "A series of staccato taps on
  the west window brought Anne flying in from the yard, eyes shining, cheeks faintly flushed with
   pink, unbraided hair streaming behind her in a torrent of brightness."
[F12] Appearance: pale, big-eyed style (since v1) -- "She's a real pretty girl got to be, though
  I can't say I'm overly partial to that pale, big-eyed style myself."
[F13] Appearance: taller than Miss Barry (since v1) -- "You're taller than I am, I declare."
```

Model output

```
{"text": "Anne is a skinny girl with a pointed freckled face, a little pointed chin, and solemn,
  big gray eyes. She has red hair that she wears in long braids, which Streams behind her in a
  bright torrent when unbraided, though some consider it auburn or Titian. At various points, her
   hair is dyed green, closely shingled, and eventually grows back into short, silky curls. She
```

Model output (continued)

```
  has ungracefully long, thin legs, stands taller than Miss Barry, and displays a pale style with
   cheeks that flush faintly pink.", "cited": ["F1", "F2", "F3", "F4", "F5", "F6", "F7", "F8", "
  F9", "F10", "F11", "F12", "F13"]}
```

Codex summary

Writes the one-sentence summary of a shared codex entry (NARRATIVEWIKI). Model `gemini-3.6-flash`; 1,974 input and 69 output tokens; run `anne@v2/20261005T070527-site_build-upto5`.

Paragraph-ID normalization. The prompt requests input-line keys such as `W1`, but the response returns `v01:c27:p0000`, a paragraph ID. The parser normalizes returned paragraph IDs during citation resolution, handling this departure from the requested format. The excerpt elides the input lines; the complete call is in the released log. Normalization maps an identifier to a paragraph; it does not check that the paragraph supports the sentence.

System prompt

```
You are writing a short encyclopedia definition of a faction or organization from a novel, for a
  wiki page a reader is viewing at one specific point in the story. You are given ONLY the
  passages that mention it so far. Write exactly 1 sentence, third person, encyclopedic tone, no
  meta-commentary. Do not mention any name, number, place or event that is not stated in the
  passages you were given -- the reader has not read past this point in the story.

Every line you are given starts with a key such as [F1] or [S2]. In "cited", list the keys of the
   lines your text actually rests on, and no others.

Respond with JSON only: {"text": "...", "cited": ["W1"]}
```

User prompt (input)

```
Faction: Aid Society
As known through Volume 1.

[W1] [v01:c26:p0023] "But we're so careful to put a moral into them all, Marilla," explained Anne
  . "I insist upon that. All the good people are rewarded and all the bad ones are suitably
  punished. I'm sure that must have a wholesome effect. The moral is the great thing. Mr. Allan
  says so. I read one of my stories to him and Mrs. Allan and they both agreed that the moral was
   excellent. Only they laughed in the wrong places. I like it better when people cry. Jane and
  Ruby almost always cry when I come to the pathetic parts. Diana wrote her Aunt Josephine about
  our club and her Aunt Josephine wrote back that we were to send her some of our stories. So we
  copied out four of our very best and sent them. Miss Josephine Barry wrote back that she had
  never read anything so amusing in her life. That kind of puzzled us because the stories were
  all very pathetic and almost everybody died. But I'm glad Miss Barry liked them. It shows our
  club is doing some good in the world. Mrs. Allan says that ought to be our object in everything
  . I do really try to make it my object but I forget so often when I'm having fun. I hope I
  shall be a little like Mrs. Allan when I grow up. Do you think there is any prospect of it,
  Marilla?"
[v01:c26:p0024] "I shouldn't say there was a great deal" was Marilla's encouraging answer. "I'm
  sure Mrs. Allan was never such a silly, forgetful little girl as you are."
[v01:c26:p0025] "No; but she wasn't always so good as she is now either," said Anne seriously. "
  She told me so herself--that is, she said she was a dreadful mischief when she was a girl and
  was always getting into scrapes. I felt so encouraged when I heard that. Is it very wicked of
  me, Marilla, to feel encouraged when I hear that other people have be
[... 4,833 characters elided ...]
But I'm bringing her up, not you."
[v01:c27:p0006] It was dark when supper was ready, and still no sign of Anne, coming hurriedly
  over the log bridge or up Lover's Lane, breathless and repentant with a sense of neglected
  duties. Marilla washed and put away the dishes grimly. Then, wanting a candle to light her way
  down the cellar, she went up to the east gable for the one that generally stood on Anne's table
  . Lighting it, she turned around to see Anne herself lying on the bed, face downward among the
  pillows.
[v01:c27:p0007] "Mercy on us," said astonished Marilla, "have you been asleep, Anne?"
```

Model output

```
{"text": "The Aid Society, also known as the Aids, is a community organization whose members
  assemble at regular meetings to attend to church affairs, such as providing a new carpet for
  the vestry room, and manage charitable initiatives like their missionary box.", "cited": ["v01:
  c27:p0000"]}
```

B1 page

Long-context baseline: one character page from the whole prefix. Model `gemini-3.6-flash`; 559,824 input and 5,688 output tokens; run `anne@b1/20261001T043209-baseline-b1-v4`.

System prompt

```
You write one character's page for a spoiler-safe wiki of a novel series. The reader has
read up to a stated volume and nothing later.

Return JSON with these keys (omit or leave empty what the passages do not support):
- age, gender, status, rank, origin, occupation: {"text": short value, "evidence": [paragraph ids
  ]}
- titles, nicknames: lists of {"text", "evidence"}
- appearance: {"text": 2-4 sentences, "evidence": [...]}
- personality: {"text": 2-6 sentences, "evidence": [...]}
- traits (personality traits and fears), abilities (skills), goals (motivations),
  background (facts about the character's past): lists of {"text": one short phrase, "evidence":
  [...]}
- history: {"text": 3-12 sentences, a biography of what the character did and experienced, "
  evidence": [...]}
- quotes: up to 5 of {"text": the character's own words copied exactly, "para_id": id}
- affiliations: list of {"name": group, "role": short, "evidence": [...]}
- relationships: list of {"name": another character, "predicate": one of PARENT_OF, CHILD_OF,
  SIBLING_OF, RELATIVE_OF, SPOUSE_OF, ROMANTIC_WITH, FRIEND_OF, RIVAL_OF, ENEMY_OF, MENTOR_OF,
  STUDENT_OF, COMMANDS, SERVES_UNDER, COMRADE_OF, "note": short, "evidence": [...]}

Rules:
- Use only the passages given. Every item cites the ids of the paragraphs that support it, copied
  exactly as they appear in brackets, e.g. v01:c06:p0003.
- State nothing a reader of the stated volumes could not know.
- Write in the third person, in plain encyclopaedic English. JSON only.
```

User prompt (input)

```
Passages from a novel series, volumes 1-4 (the reader has read exactly these volumes):

[v01:c01:p0000] MRS. Rachel Lynde lived just where the Avonlea main road dipped down into a
  little hollow, fringed with alders and ladies' eardrops and traversed by a brook that had its
  source away back in the woods of the old Cuthbert place; it was reputed to be an intricate,
  headlong brook in its earlier course through those woods, with dark secrets of pool and cascade
  ; but by the time it reached Lynde's Hollow it was a quiet, well-conducted little stream, for
  not even a brook could run past Mrs. Rachel Lynde's door without due regard for decency and
  decorum; it probably was conscious that Mrs. Rachel was sitting at her window, keeping a sharp
  eye on everything that passed, from brooks and children up, and that if she noticed anything
  odd or out of place she would never rest until she had ferreted out the whys and wherefores
  thereof.
[v01:c01:p0001] There are plenty of people, in Avonlea and out of it, who can attend closely to
  their neighbors' business by dint of neglecting their own; but Mrs. Rachel Lynde was one of
  those capable creatures who can manage their own concerns and those of other folks into the
  bargain. She was a notable housewife; her work was always done and well done; she "ran" the
  Sewing Circle, helped run the Sunday-school, and was the strongest prop of the Church Aid
  Society and Foreign Missions Auxiliary. Yet with all this Mrs. Rachel found abundant time to
  sit for hours at her kitchen window, knitting "cotton warp" quilts--she had knitted sixteen of
  them, as Avonlea housekeepers were wont to tell in awed voices--and keeping a sharp eye on the
  main road that crossed the hollow and wound up the steep red hill beyond. Since Avonlea
  occupied a little triangular peni
[... 1,993,961 characters elided ...]
ves would still call from the red rock-coves.
[v04:c40:p0062] "But we will be gone," said Anne through her tears.
```

User prompt (input) (continued)

```
[v04:c40:p0063] She went out, closing and locking the door behind her. Gilbert was waiting for
  her with a smile. The lighthouse star was gleaming northward. The little garden, where only
  marigolds still bloomed, was already hooding itself in shadows.
[v04:c40:p0064] Anne knelt down and kissed the worn old step which she had crossed as a bride.
[v04:c40:p0065] "Good-bye, dear little house of dreams," she said.

Write the page for the character "Anne Shirley" as known after volume 4.
```

Model output


```
{
  "age": {
    "text": "21 years old (turned 20 in March before starting her final year at Redmond, and
  turned 21 during her final year)",
    "evidence": [
      "v03:c01:p0000",
      "v03:c01:p0016",
      "v03:c19:p0000",
      "v03:c30:p0006"
    ]
  },
  "gender": {
    "text": "Female",
    "evidence": [
      "v01:c01:p0029",
      "v01:c03:p0019"
    ]
  },
  "status": {
    "text": "Married",
    "evidence": [
      "v04:c04:p0015",
      "v04:c05:p0008"
    ]
  },
  "rank": null,
  "origin": {
    "text": "Bolingbroke, Nova Scotia",
    "evidence": [
      "v01:c05:p0011",
      "v02:c04:p0034"
    ]
  },
  "occupation": {
    "text": "Housewife, former schoolteacher, and writer",
    "evidence": [
      "v02:c01:p0002",
      "v02:c06:p0003",
      "v03:c36:p0003",
      "v04:c05:p0008"
    ]
  },
  "titles": [
    {
      "text": "B.A. (Bachelor of Arts)",
      "evidence": [
        "v03:c34:p0027",
        "v03:c37:p0031"
      ]
    },
    {
      "text": "Teacher",
      "evidence": [
        "v02:c05:p0020",
        "v02:c15:p0004"
      ]
    }
```

Model output (continued)


```
  ],
  "nicknames": [
    {
      "text": "Cordelia",
      "evidence": [
        "v01:c03:p0015",
        "v01:c08:p0054"
      ]
    },
    {
      "text": "Anne of Green Gables",
      "evidence": [
        "v01:c08:p0056",
        "v01:c19:p0079"
      ]
    },
    {
      "text": "Carrots",
      "evidence": [
        "v01:c15:p0034",
        "v01:c15:p0041"
      ]
    },
    {
      "text": "Queen Anne",
      "evidence": [
        "v03:c05:p0006",
        "v03:c06:p0006"
      ]
    }
  ],
  "appearance": {
    "text": "Anne is a tall, slender woman with a milk-white, freckled complexion, luminous gray
  eyes that appear green in some lights, and thick hair that darkened from plain red in her youth
   to a
[... 11,273 characters elided ...]
ed spirit",
      "evidence": [
        "v02:c05:p0008",
        "v02:c11:p0038",
        "v04:c03:p0032"
      ]
    },
    {
      "name": "James Matthew Blythe",
      "predicate": "PARENT_OF",
      "note": "Son",
      "evidence": [
        "v04:c34:p0006",
        "v04:c34:p0021"
      ]
    },
    {
      "name": "Joyce Blythe",
      "predicate": "PARENT_OF",
      "note": "Deceased infant daughter",
      "evidence": [
        "v04:c19:p0010",
        "v04:c19:p0021"
      ]
    }
  ]
}
```

B2 page

Prefix-retrieval baseline: one character page from retrieved passages. Model `gemini-3.6-flash`; 24,083 input and 2,477 output tokens; run `anne@b2/20261002T095544-baseline-b2-v1`.

System prompt

```
You write one character's page for a spoiler-safe wiki of a novel series. The reader has
read up to a stated volume and nothing later.

Return JSON with these keys (omit or leave empty what the passages do not support):
- age, gender, status, rank, origin, occupation: {"text": short value, "evidence": [paragraph ids
  ]}
- titles, nicknames: lists of {"text", "evidence"}
- appearance: {"text": 2-4 sentences, "evidence": [...]}
- personality: {"text": 2-6 sentences, "evidence": [...]}
- traits (personality traits and fears), abilities (skills), goals (motivations),
  background (facts about the character's past): lists of {"text": one short phrase, "evidence":
  [...]}
- history: {"text": 3-12 sentences, a biography of what the character did and experienced, "
  evidence": [...]}
- quotes: up to 5 of {"text": the character's own words copied exactly, "para_id": id}
- affiliations: list of {"name": group, "role": short, "evidence": [...]}
- relationships: list of {"name": another character, "predicate": one of PARENT_OF, CHILD_OF,
  SIBLING_OF, RELATIVE_OF, SPOUSE_OF, ROMANTIC_WITH, FRIEND_OF, RIVAL_OF, ENEMY_OF, MENTOR_OF,
  STUDENT_OF, COMMANDS, SERVES_UNDER, COMRADE_OF, "note": short, "evidence": [...]}

Rules:
- Use only the passages given. Every item cites the ids of the paragraphs that support it, copied
  exactly as they appear in brackets, e.g. v01:c06:p0003.
- State nothing a reader of the stated volumes could not know.
- Write in the third person, in plain encyclopaedic English. JSON only.
```

User prompt (input)

```
Passages from a novel series, volumes 1-1 (the reader has read exactly these volumes):

[v01:c01:p0002] She was sitting there one afternoon in early June. The sun was coming in at the
  window warm and bright; the orchard on the slope below the house was in a bridal flush of pinky
  -white bloom, hummed over by a myriad of bees. Thomas Lynde--a meek little man whom Avonlea
  people called "Rachel Lynde's husband"--was sowing his late turnip seed on the hill field
  beyond the barn; and Matthew Cuthbert ought to have been sowing his on the big red brook field
  away over by Green Gables. Mrs. Rachel knew that he ought because she had heard him tell Peter
  Morrison the evening before in William J. Blair's store over at Carmody that he meant to sow
  his turnip seed the next afternoon. Peter had asked him, of course, for Matthew Cuthbert had
  never been known to volunteer information about anything in his whole life.
[v01:c01:p0003] And yet here was Matthew Cuthbert, at half-past three on the afternoon of a busy
  day, placidly driving over the hollow and up the hill; moreover, he wore a white collar and his
   best suit of clothes, which was plain proof that he was going out of Avonlea; and he had the
  buggy and the sorrel mare, which betokened that he was going a considerable distance. Now,
  where was Matthew Cuthbert going and why was he going there?
[v01:c01:p0005] "I'll just step over to Green Gables after tea and find out from Marilla where he
  's gone and why," the worthy woman finally concluded. "He doesn't generally go to town this
  time of year and he never visits; if he'd run out of turnip seed he wouldn't dress up and take
  the buggy to go for more; he wasn't driving fast enough to be going for a doctor. Yet something
   must have happened since last night to start him off. I'm clean puzzled
[... 82,377 characters elided ...]
like a deer, and disappeared in the firry shadows of the Haunted Wood. Mrs. Lynde looked after
  her indulgently.
[v01:c38:p0043] "There's a good deal of the child about her yet in some ways."
[v01:c38:p0044] "There's a good deal more of the woman about her in others," retorted Marilla,
  with a momentary return of her old crispness.
[v01:c38:p0045] But crispness was no longer Marilla's distinguishing characteristic. As Mrs.
  Lynde told her Thomas that night.
[v01:c38:p0046] "Marilla Cuthbert has got mellow. That's what."

Write the page for the character "Marilla Cuthbert" as known after volume 1.
```

Model output


```
{
  "age": null,
  "gender": {
    "text": "Female",
    "evidence": [
      "v01:c01:p0013"
    ]
  },
  "status": null,
  "rank": null,
  "origin": null,
  "occupation": {
    "text": "Homemaker and farmer",
    "evidence": [
      "v01:c01:p0009",
      "v01:c06:p0036",
      "v01:c14:p0018"
    ]
  },
  "titles": [
    {
      "text": "Miss Cuthbert",
      "evidence": [
        "v01:c06:p0004",
        "v01:c08:p0011"
      ]
    }
  ],
  "nicknames": [],
  "appearance": {
    "text": "Marilla is a tall, thin woman with angular lines and no soft curves. Her dark hair,
  which exhibits some gray streaks, is twisted up in a tight knot secured aggressively by two
  wire hairpins. She possesses a mouth that suggests a hidden sense of humor, though it remains
  largely undeveloped. As she ages, her hair grows grayer, but she retains a gaunt frame and
  angular stature, with her face softening in expression over time.",
    "evidence": [
      "v01:c01:p0013",
      "v01:c33:p0015"
    ]
  },
  "personality": {
    "text": "Marilla is a woman of narrow experience, rigid conscience, and firm, unsentimental
  principles who prefers practical duty over imagination. She is neat, precise, and highly
  disciplined, maintaining an exceptionally clean home and yard. Though initially strict, stern,
  and undemonstrative, she possesses a deep capacity for love and grows deeply attached to Anne.
  Over time, her hard exterior mellows, revealing a soft and compassionate heart despite her
  struggle to express affection openly.",
    "evidence": [
      "v01:c01:p0008",
      "v01:c01:p0009",
      "v01:c01:p0013",
      "v01:c05:p0021",
      "v01:c06:p0019",
      "v01:c08:p0013",
      "v01:c08:p0017",
      "v01:c30:p0003",
      "v01:c33:p0015",
      "v01:c38:p0046"
    ]
  },
  "traits": [

[... 4,359 characters elided ...]
01:c01:p0006"
      ]
    },
    {
```

Model output (continued)


```
      "name": "Anne Shirley",
      "predicate": "PARENT_OF",
      "note": "Adoptive guardian who raises her at Green Gables",
      "evidence": [
        "v01:c06:p0031",
        "v01:c06:p0037",
        "v01:c08:p0007",
        "v01:c19:p0089"
      ]
    },
    {
      "name": "Rachel Lynde",
      "predicate": "FRIEND_OF",
      "note": "Longtime neighbor and friend despite their differences",
      "evidence": [
        "v01:c01:p0012"
      ]
    }
  ]
}
```


B3 index (LightRAG)

LightRAG's own entity and relation extraction while building its index. Model `gemini-3.6-flash`; 3,243 input and 1,022 output tokens; run `anne@b3/20260929T225346-baseline-b3-v5`.

System prompt

```
---Role---
You are a Knowledge Graph Specialist responsible for extracting entities and relationships from
  the `---Input Text---` section of user prompt.

---Instructions---
1. **Entity Extraction:**
  - Identify clearly defined and meaningful entities only in the current user prompt's fenced
  `---Input Text---` section.
  - For each entity, extract:
    - `entity_name`: The name of the entity. If the entity name is case-insensitive, capitalize
  the first letter of each significant word (title case). Ensure **consistent naming** across the
   entire extraction process.
    - `entity_type`: Categorize the entity using the type guidance provided in the `---Entity
  Types---` section below. If none of the provided entity types apply, classify it as `Other`.
    - `entity_description`: Provide a concise yet comprehensive description of the entity's
  attributes and activities, based *solely* on the information present in the input text.

2. **Relationship Extraction:**
  - Identify direct, clearly stated, and meaningful relationships between previously extracted
  entities.
  - If a single statement describes a relationship involving more than two entities, decompose it
   into multiple binary relationships.
  - For each binary relationship, extract:
    - `source_entity`: The name of the source entity. Ensure **consistent naming** with entity
  extraction. Capitalize the first letter of each significant word (title case) if the name is
  case-insensitive.
    - `target_entity`: The name of the target entity. Ensure **consistent naming** with entity
  extraction. Capitalize the first letter of each significant word (title case) if the name is
  case-insensitive.
    - `relationship_keywords`: One or more high-level keywords summarizing the relationship.
  Multiple keywords within this field must be separated by a comma `,`. **DO NOT use `<|#|>` for
   separating multiple keywords within this field.**
    - `relationship_description`: A concise explanation of the nature of the relationship between
   the source and target entities.

3. **Record Types:**
  - `entity` is used only for entity rows and those rows always contain exactly 4 tuple parts
  total.
  - `relation` is used only for relationship
```

System prompt (continued)

```
[... 4,217 characters elided ...]
: Quantitative or structured information (statistics, datasets, measurements)
- Artifact: Physical or digital objects created by humans (tools, software, devices)
- NaturalObject: Natural non-living objects (minerals, celestial bodies, chemical compounds)

---Output Format Template---
The following content is an output format template only. It is not source text and must never be
  used as extraction content.

entity<|#|><entity_name><|#|><entity_type><|#|><entity_description>
relation<|#|><source_entity><|#|><target_entity><|#|><relationship_keywords><|#|><
  relationship_description>
<|COMPLETE|>
```

User prompt (input)

````
---Task---
Extract entities and relationships from the `---Input Text---` section below.

---Instructions---
1. **Strict Adherence to Format:** Strictly adhere to all format requirements for entity and
  relationship lists, including output order, field delimiters, and proper noun handling, as
  specified in the system prompt.
2. **Quantity Limits:** In this response, output at most 100 total rows and at most 40 entity
  rows. Output fewer rows if fewer high-value items are present. Only output relationship rows
  whose source and target entities are both included in this response.
3. **Output Content Only:** Output *only* the extracted list of entities and relationships. Do
  not include any introductory or concluding remarks, explanations, or additional text before or
  after the list.
4. **Completion Signal:** Output `<|COMPLETE|>` as the final line after all relevant entities and
   relationships have been extracted and presented. If the row limit is reached, output `<|
  COMPLETE|>` immediately after the last allowed row.
5. **Output Language:** Ensure the output language is English. Proper nouns (e.g., personal names
  , place names, organization names) must be kept in their original language and not translated.

---Input Text---
```
[v05:c02:p0000] "Where are the other children?" asked Miss Cornelia, when the first greetings--
  cordial on her side, rapturous on Anne's, and dignified on Susan's--were over.
[v05:c02:p0001] "Shirley is in bed and Jem and Walter and the twins are down in their beloved
  Rainbow Valley," said Anne. "They just came home this afternoon, you know, and they could
  hardly wait until supper was over before rushing down to the valley. They love it above every
  spot on earth. Even the maple grove doesn't rival it in their affections."
[v05:c02:p0002] "I am afraid they lo
[... 3,392 characters elided ...]
d about the worst candidating text there is in the Bible--'Curse ye Meroz.'"
[v05:c02:p0020] "Whenever he got stuck for an idea, he would bang the Bible and shout very
  bitterly, 'Curse ye Meroz.' Poor Meroz got thoroughly cursed that day, whoever he was, Mrs. Dr.
   dear," said Susan.
[v05:c02:p0021] "The minister who is candidating can't be too careful what text he chooses," said
   Miss Cornelia solemnly. "I believe Mr. Pierson would have got the call if he had picked a
  different text. But when he announced 'I will lift my eyes to the hills' he was done for. Every
   one grinned, for
```

---Output---
````

Model output

```
entity<|#|>Miss Cornelia<|#|>Person<|#|>A resident of Glen St. Mary who visits Anne and provides
  news about the local church and new minister candidates.
entity<|#|>Anne<|#|>Person<|#|>Also referred to as Mrs. Dr. dear, she is the mother of Shirley,
  Jem, Walter, and the twins, and has recently returned home.
entity<|#|>Susan<|#|>Person<|#|>A member of Anne's household who frequently speaks in tandem with
   Miss Cornelia, addressing Anne as "Mrs. Dr. dear."
entity<|#|>Shirley<|#|>Person<|#|>One of Anne's children, who is in bed after returning home.
```

Model output (continued)

```
entity<|#|>Jem<|#|>Person<|#|>Also known as Little Jem, one of Anne's children who loves Rainbow
  Valley and is doted on by Marilla.
entity<|#|>Walter<|#|>Person<|#|>One of Anne's children who enjoys spending time in Rainbow
  Valley.
entity<|#|>Rainbow Valley<|#|>Location<|#|>A beloved valley near the home of Anne's family that
  the children adore visiting.
entity<|#|>Avonlea<|#|>Location<|#|>The place where the children recently visited Marilla.
entity<|#|>Marilla Cuthbert<|#|>Person<|#|>An eighty-five-year-old lady, also called Miss
  Cuthbert, living in Avonlea who spoils Anne's children, especially Jem.
entity<|#|>Glen St. Mary<|#|>Location<|#|>The village where Miss Cornelia and Susan live, home to
   the local church.
entity<|#|>Glen St. Mary Church<|#|>Organization<|#|>The local church in Glen St. Mary that
  recently called a new minister.
entity<|#|>John Knox Meredith<|#|>Person<|#|>The newly settled minister at Glen St. Mary church,
  described as learned, spiritual, comely, and absent-minded.
entity<|#|>Mr. Folsom<|#|>Person<|#|>A minister candidate for Glen St. Mary church who was
  rejected for being too dark and sleek.
entity<|#|>Mr. Rogers<|#|>Person<|#|>A minister candidate whose sermon was disrupted by a stray
  sheep entering
[... 1,112 characters elided ...]
illa lives in Avonlea where the children visited her.
relation<|#|>John Knox Meredith<|#|>Glen St. Mary Church<|#|>employment, leadership<|#|>John Knox
   Meredith was called as the new minister for the Glen St. Mary church.
relation<|#|>Miss Cornelia<|#|>Glen St. Mary<|#|>residence<|#|>Miss Cornelia resides in Glen St.
  Mary and attends its church.
relation<|#|>Caleb Ramsay<|#|>Mr. Rogers<|#|>disruption<|#|>Caleb Ramsay's stray sheep
  interrupted Mr. Rogers's candidate sermon at church.
<|COMPLETE|>
```

B3 page

LightRAG baseline: one character page from LightRAG context. Model `gemini-3.6-flash`; 17,438 input and 2,191 output tokens; run `anne@b3/20261001T142328-baseline-b3-v5`.

System prompt

```
You write one character's page for a spoiler-safe wiki of a novel series. The reader has
read up to a stated volume and nothing later.

Return JSON with these keys (omit or leave empty what the passages do not support):
- age, gender, status, rank, origin, occupation: {"text": short value, "evidence": [paragraph ids
  ]}
- titles, nicknames: lists of {"text", "evidence"}
- appearance: {"text": 2-4 sentences, "evidence": [...]}
- personality: {"text": 2-6 sentences, "evidence": [...]}
- traits (personality traits and fears), abilities (skills), goals (motivations),
  background (facts about the character's past): lists of {"text": one short phrase, "evidence":
  [...]}
- history: {"text": 3-12 sentences, a biography of what the character did and experienced, "
  evidence": [...]}
- quotes: up to 5 of {"text": the character's own words copied exactly, "para_id": id}
- affiliations: list of {"name": group, "role": short, "evidence": [...]}
- relationships: list of {"name": another character, "predicate": one of PARENT_OF, CHILD_OF,
  SIBLING_OF, RELATIVE_OF, SPOUSE_OF, ROMANTIC_WITH, FRIEND_OF, RIVAL_OF, ENEMY_OF, MENTOR_OF,
  STUDENT_OF, COMMANDS, SERVES_UNDER, COMRADE_OF, "note": short, "evidence": [...]}

Rules:
- Use only the passages given. Every item cites the ids of the paragraphs that support it, copied
  exactly as they appear in brackets, e.g. v01:c06:p0003.
- State nothing a reader of the stated volumes could not know.
- Write in the third person, in plain encyclopaedic English. JSON only.
```

User prompt (input)

```
Passages from a novel series, volumes 1–5 (the reader has read exactly these volumes):

LightRAG graph context:
- Land Of The Maple: The country or region where Jem and thousands of young people like him live.
- Marilla: Marilla Cuthbert, frequently referred to simply as Marilla or Miss Cuthbert, is a
  central character, maternal figure, and the mistress of the Green Gables estate in Avonlea,
  Prince Edward Island, where she has resided for sixty years. A gaunt woman with graying hair,
  Marilla is defined by her practical outlook, firm adherence to traditional decorum, and strong
  moral principles. Though stern, strict, and prone to headaches, she possesses profound
  resilience, a dry sense of humor, and a deep emotional warmth. Over the nine years following
  Anne Shirley's arrival at Green Gables, Marilla grows noticeably softer, gentler, and more
  expressive. She takes immense pride in her immaculate household, her fine china, her culinary
  crafts--such as yellow plum preserves and Christmas delicacies--and domestic activities like
  making round, braided mats.

Originally living with her brother, Matthew Cuthbert, Marilla intended to adopt an orphan boy to
  help with farm work, but instead took in the imaginative Anne Shirley. Marilla serves as a
  steady, sensible foil to Anne, guiding her through domestic duties, funding her education, and
  ultimately forming an indelible bond with her. Throughout Anne's upbringing, Marilla offers
  practical guidance and manages household dynamics, including inquiring about local news,
  discussing neighborhood events like Miss Josephine Barry's passing alongside Mrs. Rachel Lynde,
   expressing concern when Anne returns tired from visiting friends like Ruby, and leaving Anne
  alone with Gilbert Blythe during his visits. When Marilla later faces failing
[... 68,594 characters elided ...]
ms. She really looks upon boys as good comrades, and she is pleased when she has dozens of them
  tagging round, simply because she likes to be popular and to be thought popular. Even Alex and
  Alonzo--I'll never be able to think of those two names separately after this--are to her just
  two playfellows who want her to play with them all their lives. I'm glad we met her, and I'm
  glad we went to Old St. John's. I believe I've put forth a tiny soul-root into Kingsport soil
  this afternoon. I hope so. I hate to feel transplanted."

Write the page for the character "Anne Shirley" as known after volume 5.
```

Model output


```
{
  "age": null,
  "gender": {
    "text": "Female",
    "evidence": [
      "v01:c35:p0001",
      "v01:c35:p0004",
      "v04:c01:p0000"
    ]
  },
  "status": null,
  "rank": null,
  "origin": null,
  "occupation": {
    "text": "Schoolteacher and student",
    "evidence": [
      "v03:c31:p0001",
      "v04:c01:p0000",
      "v04:c01:p0004"
    ]
  },
  "titles": [
    {
      "text": "Miss Shirley",
      "evidence": [
        "v03:c04:p0056"
      ]
    }
  ],
  "nicknames": [],
  "appearance": {
```

Model output (continued)


```
    "text": "Anne has gray eyes that are compared to a morning sky and bold red hair, which she
  considers auburn. She possesses a very nice nose.",
    "evidence": [
      "v03:c31:p0003",
      "v03:c31:p0004",
      "v03:c04:p0056",
      "v04:c01:p0000"
    ]
  },
  "personality": {
    "text": "Anne is a thoughtful, imaginative, and ambitious individual with a blithe,
  irresistible laugh. Possessing a natural genius for friendship, she values comradeship and
  views intellectual discussions with deep appreciation. Though prone to romanticism, she
  maintains a sense of humor and warmth that endears her to others.",
    "evidence": [
      "v01:c35:p0002",
      "v01:c35:p0003",
      "v01:c35:p0004",
      "v03:c04:p0060",
      "v04:c01:p0005",
      "v04:c01:p0008"
    ]
  },
  "traits": [
    {
      "text": "Thoughtful and imaginative",
      "evidence": [
        "v01:c35:p0004"
      ]
    },
    {
      "text": "Ambitious",
      "evidence": [
        "v01:c35:p0004"
      ]
    },
    {
      "text": "Genius for friendship",
      "evidence": [
        "v01:c35:p0003"
      ]
    },
    {
      "text": "Dislikes geometry",
      "evidence": [
        "v04:c01:p0000",
        "v04:c01:p0003"
      ]
    }
  ],
  "abilities": [
    {
      "text": "Teaching",
      "
[... 3,313 characters elided ...]
 Green Gables.",
      "evidence": [
        "v04:c01:p0005"
      ]
    },
    {
      "name": "Philippa Gordon",
      "predicate": "FRIEND_OF",
      "note": "Intimate college friend and correspondent.",
      "evidence": [
        "v03:c04:p0060",
        "v03:c31:p0000"
      ]
```

Model output (continued)

```
    },
    {
      "name": "Priscilla Grant",
      "predicate": "FRIEND_OF",
      "note": "Close friend and fellow student.",
      "evidence": [
        "v01:c35:p0004",
        "v03:c04:p0068"
      ]
    }
  ]
}
```

X1 page

Closed-book diagnostic: title and character name only. Model `gemini-3.6-flash`; 451 input and 1,087 output tokens; run `anne@x1/20260930T214444-baseline-x1-v5`.

System prompt

```
You write one character's page for a spoiler-safe wiki of a novel series. The reader has
read up to a stated volume and nothing later.

Return JSON with these keys (omit or leave empty what the passages do not support):
- age, gender, status, rank, origin, occupation: {"text": short value, "evidence": [paragraph ids
  ]}
- titles, nicknames: lists of {"text", "evidence"}
- appearance: {"text": 2-4 sentences, "evidence": [...]}
- personality: {"text": 2-6 sentences, "evidence": [...]}
- traits (personality traits and fears), abilities (skills), goals (motivations),
  background (facts about the character's past): lists of {"text": one short phrase, "evidence":
  [...]}
- history: {"text": 3-12 sentences, a biography of what the character did and experienced, "
  evidence": [...]}
- quotes: up to 5 of {"text": the character's own words copied exactly, "para_id": id}
- affiliations: list of {"name": group, "role": short, "evidence": [...]}
- relationships: list of {"name": another character, "predicate": one of PARENT_OF, CHILD_OF,
  SIBLING_OF, RELATIVE_OF, SPOUSE_OF, ROMANTIC_WITH, FRIEND_OF, RIVAL_OF, ENEMY_OF, MENTOR_OF,
  STUDENT_OF, COMMANDS, SERVES_UNDER, COMRADE_OF, "note": short, "evidence": [...]}

Rules:
- No passages are given: write what you know of this character from the novels up to the stated
  volume. Leave every evidence list empty.
- State nothing a reader of the stated volumes could not know.
- Write in the third person, in plain encyclopaedic English. JSON only.
```

User prompt (input)

```
The novels: Anne of Green Gables series, volumes 1-5 (the reader has read exactly these volumes).

Write the page for the character "Matthew Cuthbert" as known after volume 5.
```

Model output

```
{
  "age": {
    "text": "Sixty years old at the beginning of the series; died at sixty-eight.",
    "evidence": []
  },
  "gender": {
    "text": "Male",
    "evidence": []
  },
  "status": {
    "text": "Deceased",
```

Model output (continued)


```
    "evidence": []
  },
  "rank": null,
  "origin": {
    "text": "Avonlea, Prince Edward Island, Canada",
    "evidence": []
  },
  "occupation": {
    "text": "Farmer",
    "evidence": []
  },
  "titles": [],
  "nicknames": [],
  "appearance": {
    "text": "Matthew was an awkwardly shaped man with stooped shoulders and dark hair turning
  gray. He wore a full, soft brown beard that he had kept since his twenties. He dressed plainly
  in farm clothes and was known for his shy, diffident posture, especially in the presence of
  women.",
    "evidence": []
  },
  "personality": {
    "text": "Matthew was extraordinarily shy, quiet, and reclusive, famously terrified of women
  and young girls other than his sister and Anne. Beneath his silent demeanor lay a gentle,
  deeply affectionate heart and an intuitive understanding of Anne's imaginative nature. He was
  generous, quiet in his devotion, and almost incapable of refusing Anne anything she set her
  heart on.",
    "evidence": []
  },
  "traits": [
    "Extremely shy",
    "Gentle and soft-hearted",
    "Fear of women and girls",
    "Quietly supportive",
    "Industrious"
  ],
  "abilities": [
    "Farming and livestock care",
    "Quiet listening and unspoken empathy"
  ],
  "goals": [
    "To provide a loving home for Anne and ensure her happiness",
    "To keep Green Gables running smoothly alongside his sister Marilla"
  ],
  "background": [
    "Lived his entire life at Green Gables in Avonlea with his family.",
    "Never married due to his intense shyness around women.",
    "Took over the farm duties alongsid
[... 1,820 characters elided ...]
 co-owner of Green Gables; she managed the household while he ran the farm.",
      "evidence": []
    },
    {
      "name": "Anne Shirley",
      "predicate": "PARENT_OF",
      "note": "Adoptive father figure who loved her unconditionally from their first meeting.",
      "evidence": []
    },
    {
      "name": "Rachel Lynde",
      "predicate": "FRIEND_OF",
      "note": "Neighbor and long-time family friend, though her blunt manner often intimidated
  him.",
      "evidence": []
    }
  ]
}
```

X2 page

Instructed full-text diagnostic: all five volumes and the cutoff instruction. Model `gemini-3.6-flash`; 689,967 input and 2,449 output tokens; run `anne@x2/20260930T222548-baseline-x2-v1`.

System prompt

```
You write one character's page for a spoiler-safe wiki of a novel series. The reader has
read up to a stated volume and nothing later.

Return JSON with these keys (omit or leave empty what the passages do not support):
- age, gender, status, rank, origin, occupation: {"text": short value, "evidence": [paragraph ids
  ]}
- titles, nicknames: lists of {"text", "evidence"}
- appearance: {"text": 2-4 sentences, "evidence": [...]}
- personality: {"text": 2-6 sentences, "evidence": [...]}
- traits (personality traits and fears), abilities (skills), goals (motivations),
  background (facts about the character's past): lists of {"text": one short phrase, "evidence":
  [...]}
- history: {"text": 3-12 sentences, a biography of what the character did and experienced, "
  evidence": [...]}
- quotes: up to 5 of {"text": the character's own words copied exactly, "para_id": id}
- affiliations: list of {"name": group, "role": short, "evidence": [...]}
- relationships: list of {"name": another character, "predicate": one of PARENT_OF, CHILD_OF,
  SIBLING_OF, RELATIVE_OF, SPOUSE_OF, ROMANTIC_WITH, FRIEND_OF, RIVAL_OF, ENEMY_OF, MENTOR_OF,
  STUDENT_OF, COMMANDS, SERVES_UNDER, COMRADE_OF, "note": short, "evidence": [...]}

Rules:
- Use only the passages given. Every item cites the ids of the paragraphs that support it, copied
  exactly as they appear in brackets, e.g. v01:c06:p0003.
- State nothing a reader of the stated volumes could not know.
- Write in the third person, in plain encyclopaedic English. JSON only.
```

User prompt (input)

```
Passages from a novel series, volumes 1-5. The reader has read ONLY volumes 1-1: write for that
  reader and do not reveal anything that happens after volume 1.

[v01:c01:p0000] MRS. Rachel Lynde lived just where the Avonlea main road dipped down into a
  little hollow, fringed with alders and ladies' eardrops and traversed by a brook that had its
  source away back in the woods of the old Cuthbert place; it was reputed to be an intricate,
  headlong brook in its earlier course through those woods, with dark secrets of pool and cascade
  ; but by the time it reached Lynde's Hollow it was a quiet, well-conducted little stream, for
  not even a brook could run past Mrs. Rachel Lynde's door without due regard for decency and
  decorum; it probably was conscious that Mrs. Rachel was sitting at her window, keeping a sharp
  eye on everything that passed, from brooks and children up, and that if she noticed anything
  odd or out of place she would never rest until she had ferreted out the whys and wherefores
  thereof.
[v01:c01:p0001] There are plenty of people, in Avonlea and out of it, who can attend closely to
  their neighbors' business by dint of neglecting their own; but Mrs. Rachel Lynde was one of
  those capable creatures who can manage their own concerns and those of other folks into the
  bargain. She was a notable housewife; her work was always done and well done; she "ran" the
  Sewing Circle, helped run the Sunday-school, and was the strongest prop of the Church Aid
  Society and Foreign Missions Auxiliary. Yet with all this Mrs. Rachel found abundant time to
  sit for hours at her kitchen window, knitting "cotton warp" quilts--she had knitted sixteen of
  them, as Avonlea housekeepers were wont to tell in awed voices--and keeping a sharp eye on the
  main road that crossed the hollow and wound up th
[... 2,458,388 characters elided ...]
05:c35:p0017] The girls shivered.
[v05:c35:p0018] "You know you're only pretending," protested Mary Vance, "and I wish you wouldn't
  . You make it too real. I hate that old Piper of yours."
[v05:c35:p0019] But Jem sprang up with a gay laugh. He stood up on a little hillock, tall and
  splendid, with his open brow and his fearless eyes. There were thousands like him all over the
  land of the maple.
[v05:c35:p0020] "Let the Piper come and welcome," he cried, waving his hand. "I'll follow him
  gladly round and round the world."

Write the page for the character "Gilbert Blythe" as known after volume 1.
```

Model output


```
{
  "age": {
    "text": "Nearly fourteen (in autumn of Volume 1) / Around fifteen to sixteen",
    "evidence": [
      "v01:c15:p0024"
    ]
  },
  "gender": {
    "text": "Male",
    "evidence": [
      "v01:c15:p0016"
    ]
  },
  "status": {
    "text": "Single",
    "evidence": [
      "v01:c15:p0028"
    ]
  },
  "rank": null,
  "origin": {
    "text": "Avonlea, Prince Edward Island",
    "evidence": [
      "v01:c15:p0016"
    ]
  },
  "occupation": {
    "text": "Student",
    "evidence": [
      "v01:c15:p0024",
      "v01:c17:p0037"
    ]
  },
  "titles": [],
  "nicknames": [
    {
      "text": "Gil",
      "evidence": [
        "v01:c15:p0024"
      ]
    }
  ],
  "appearance": {
    "text": "Gilbert is a tall boy with an ungainly, athletic build and hazel eyes. He has curly
  brown hair and a teasing, roguish smile. He is widely considered very handsome by his peers.",
    "evidence": [
      "v01:c15:p0016",
      "v01:c15:p0028"
    ]
  },
  "personality": {
    "text": "Gilbert is confident, mischievous, and fond of teasing girls. Despite his
  mischievous streak, he is stout-hearted, honorable, and quick to admit his mistakes when he
  realizes he has offended someone. He is academically ambitious, competitive, and good-natured
  in rivalry.",
    "evidence": [
      "v01:c15:p0016",
      "v01:c15:p0028",
      "v01:c15:p0042",
      "v01:c15:p0049",
      "v01:c17:p0036"
    ]
  },
  "traits": [
    {
      "text": "Handsome",
      "evidence": [
```

Model output (continued)


```
        "v01:c15:p0016"
      ]
    },
    {
      "text": "Mischievous and teasing",
      "evidence": [
        "v01:c15:p0016",
        "v01:c15:p0028"
      ]
    },
    {
      "text": "Contrite and honorable when wrong",
      "evidence": [
        "v01:c15:p0042",
        "v01:c15:p0049"
      ]
    },
    {

[... 4,248 characters elided ...]
. Phillips",
      "predicate": "SERVES_UNDER",
      "note": "Schoolmaster at Avonlea.",
      "evidence": [
        "v01:c15:p0027"
      ]
    },
    {
      "name": "Ruby Gillis",
      "predicate": "FRIEND_OF",
      "note": "Classmate at Avonlea School.",
      "evidence": [
        "v01:c15:p0028"
      ]
    },
    {
      "name": "Charlie Sloane",
      "predicate": "RIVAL_OF",
      "note": "Classmate and fellow student.",
      "evidence": [
        "v01:c17:p0036"
      ]
    }
  ]
}
```


Paraphrase for the treated text

Paraphrases one renamed paragraph. Model `gemini-3.6-flash`; 385 input and 209 output tokens; run `anne-decon/20261003T165238-decontaminate-v5`.

Note on this log. The same instructions serve every corpus the treatment is applied to; the rule to keep "personal and ship" names comes from another corpus and has no effect here.

System prompt

```
You rewrite passages of fiction in new words for a research corpus. You never add, remove or
  explain anything.
```

User prompt (input)

```
Rewrite the paragraph below in different words and different sentence structure.

Rules:
- Keep every fact, event, number, date and place, and keep them in the same order.
```

User prompt (input) (continued)

```
- Keep every personal and ship name exactly as written. Do not introduce any name that is not in
  the paragraph.
- Dialogue stays dialogue, spoken by the same person. Use the same quotation marks as the
  original (" " stay " "). Only spoken words go inside them; a speech tag such as "said I" stays
  outside.
- Keep the narrator's person (first person stays first person).
- Do not summarise, shorten substantially, or add commentary.

Return only the rewritten paragraph, nothing else.

Paragraph:
It was a clear, apple-green evening in May, and Seven Gulls Harbour was mirroring back the clouds
   of the golden west between its softly dark shores. The sea moaned eerily on the sand-bar,
  sorrowful even in spring, but a sly, jovial wind came piping down the red harbour road along
  which Miss Winifred's comfortable, matronly figure was making its way towards the village of
  Glen St. Agnes. Miss Winifred was rightfully Mrs. Ambrose Kerr, and had been Mrs. Ambrose Kerr
  for thirteen years, but even yet more people referred to her as Miss Winifred than as Mrs. Kerr
  . The old name was dear to her old friends, only one of them contemptuously dropped it. Tabitha
   Pratt, the gray and grim and faithful handmaiden of the Carrow family at Hollowmere, never
  lost an opportunity of calling her "Mrs. Ambrose Kerr," with the most killing and pointed
  emphasis, as if to say "You wanted to be Mrs. and Mrs. you shall be with a vengeance as far as
  I am concerned."
```

Model output

```
The May evening was transparent and apple-green, as Seven Gulls Harbour reflected the clouds from
   the golden west amidst its gently darkened banks. On the sand-bar, the ocean uttered an
  uncanny mournfulness, melancholic despite the spring, yet a cunning, merry breeze whistled down
   the crimson harbour road where the cozy, motherly form of Miss Winifred traveled toward the
  village of Glen St. Agnes. Legally, Miss Winifred was Mrs. Ambrose Kerr, a status she had held
  for thirteen years, though even now she was called Miss Winifred by more people than called her
   Mrs. Kerr. Her former name remained cherished by her longtime companions, with a single
  contemptuous exception. Tabitha Pratt, who served as the gray, austere, and loyal servant to
  the Carrow household at Hollowmere, seized every chance to address her as "Mrs. Ambrose Kerr,"
  using the most devastating and deliberate stress, as though implying "You desired to be a Mrs.,
   so a Mrs. you will certainly be to the absolute extreme where I am concerned."
```

Memorization probe

Name cloze / title identification on one paragraph. Model `gemini-3.6-flash`; 227 input and 9 output tokens; run `anne-decon/20261003T165318-probe-memorization-v1-5`.

User prompt (input)

```
You have seen the following passage in your training data. What is the proper name that fills in
  the [MASK] token in it? This name is exactly one word long, and is a proper name (not a pronoun
   or any other word). You must make a guess, even if you are uncertain.

Example:

Input: Stay gold, [MASK], stay gold.
Output: <name>Ponyboy</name>

Input: The door opened, and [MASK], dressed and hatted, entered with a cup of tea.
Output: <name>Gerty</name>

Input: "Pretty? Oh, pretty doesn't seem the right word to use. Nor beautiful, either. They don't
  go far enough. Oh, it was wonderful--wonderful. It's the first thing I ever saw that couldn't
  be improved upon by imagination. It just satisfied me here"--she put one hand on her breast--"
  it made a queer funny ache and yet it was a pleasant ache. Did you ever have an ache like that,
   Mr. [MASK]?"
Output:
```

Model output

```
<name>Cuthbert</name>
```